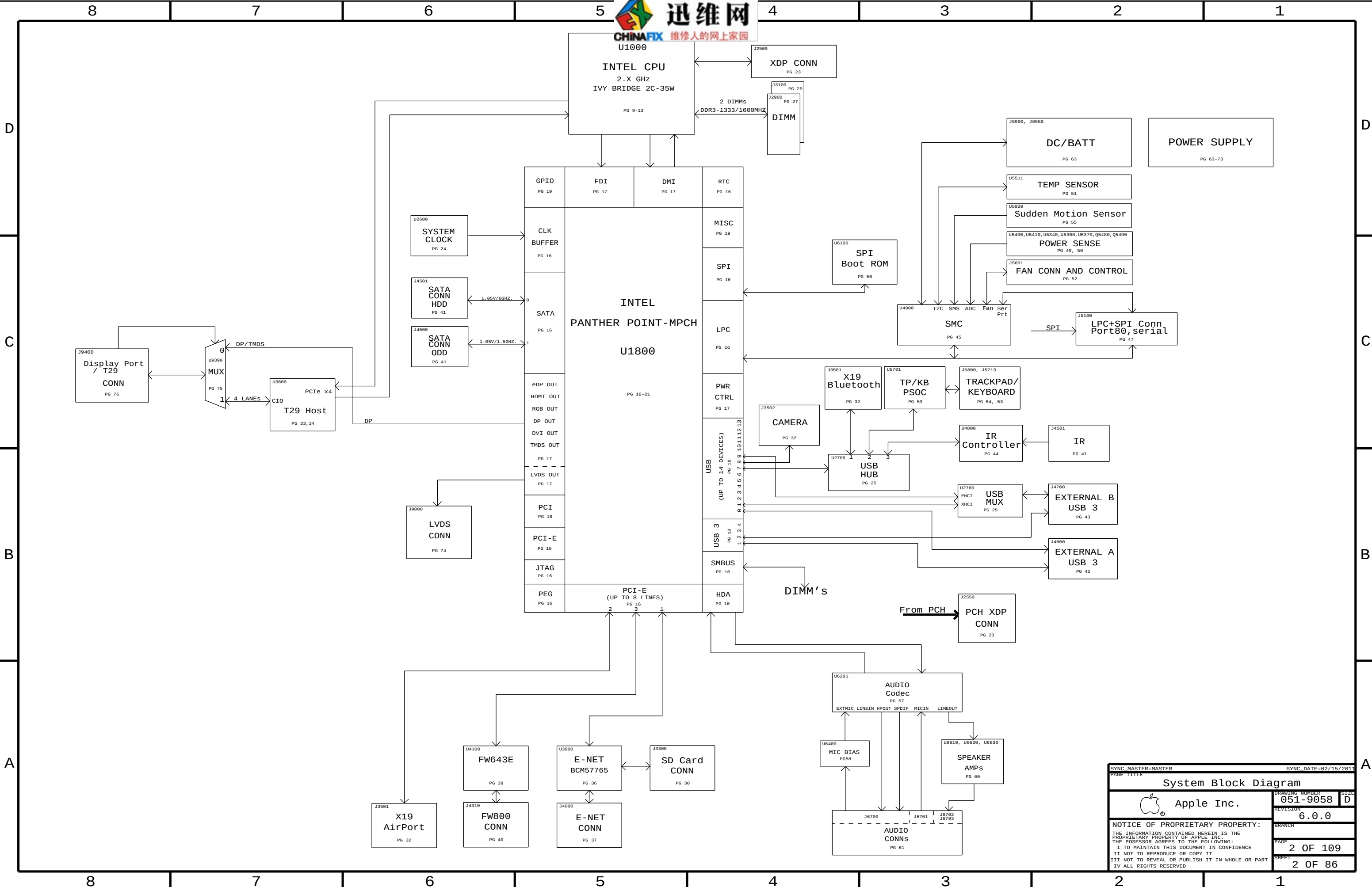
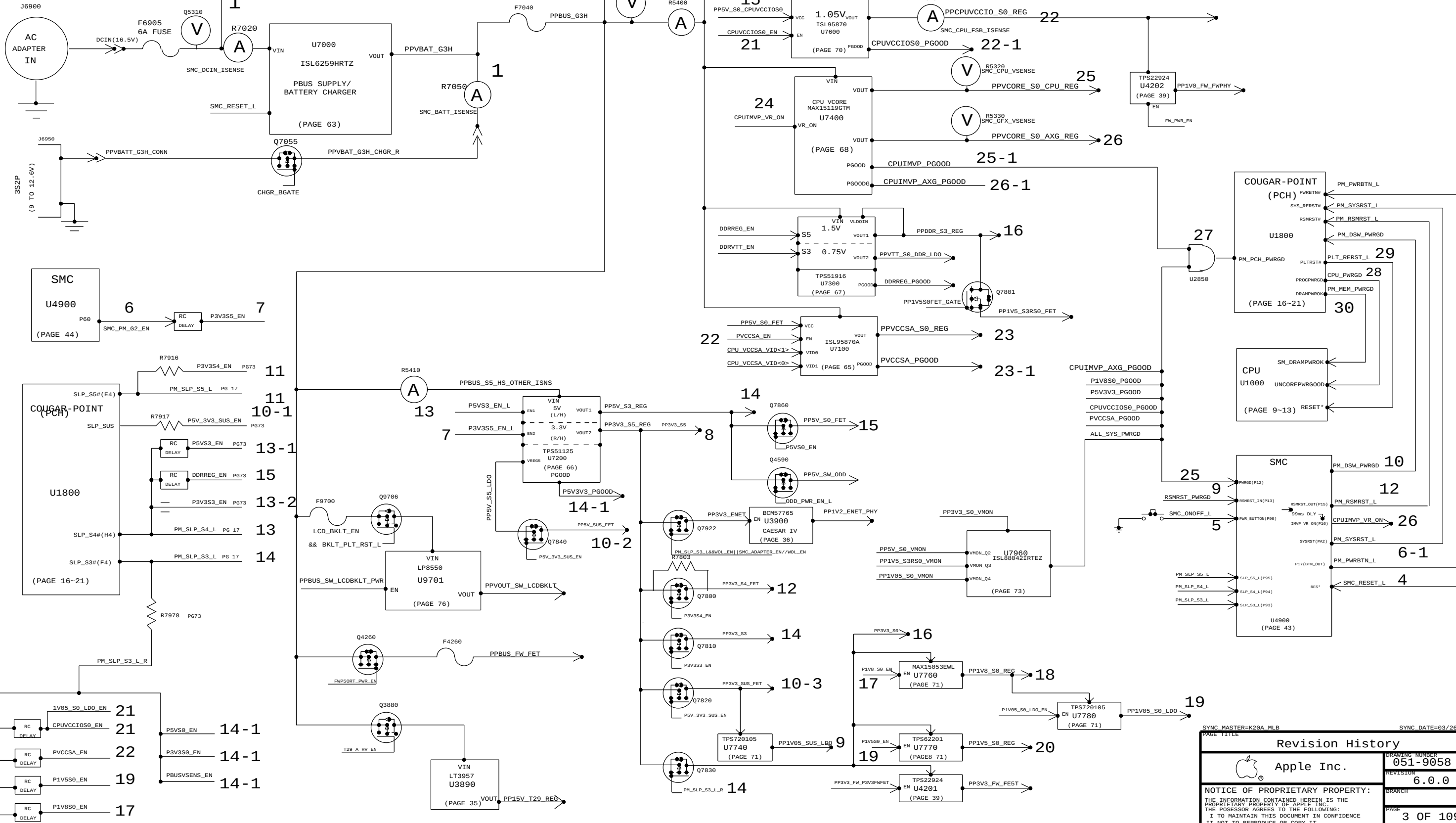


DRAWING TITLE		SCHEM, MLB, J30	
 Apple Inc.	DRAWING NUMBER		SIZE
	051-9058		D
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		PAGE	
		1 OF 109	
		SHEET	
		1 OF 86	

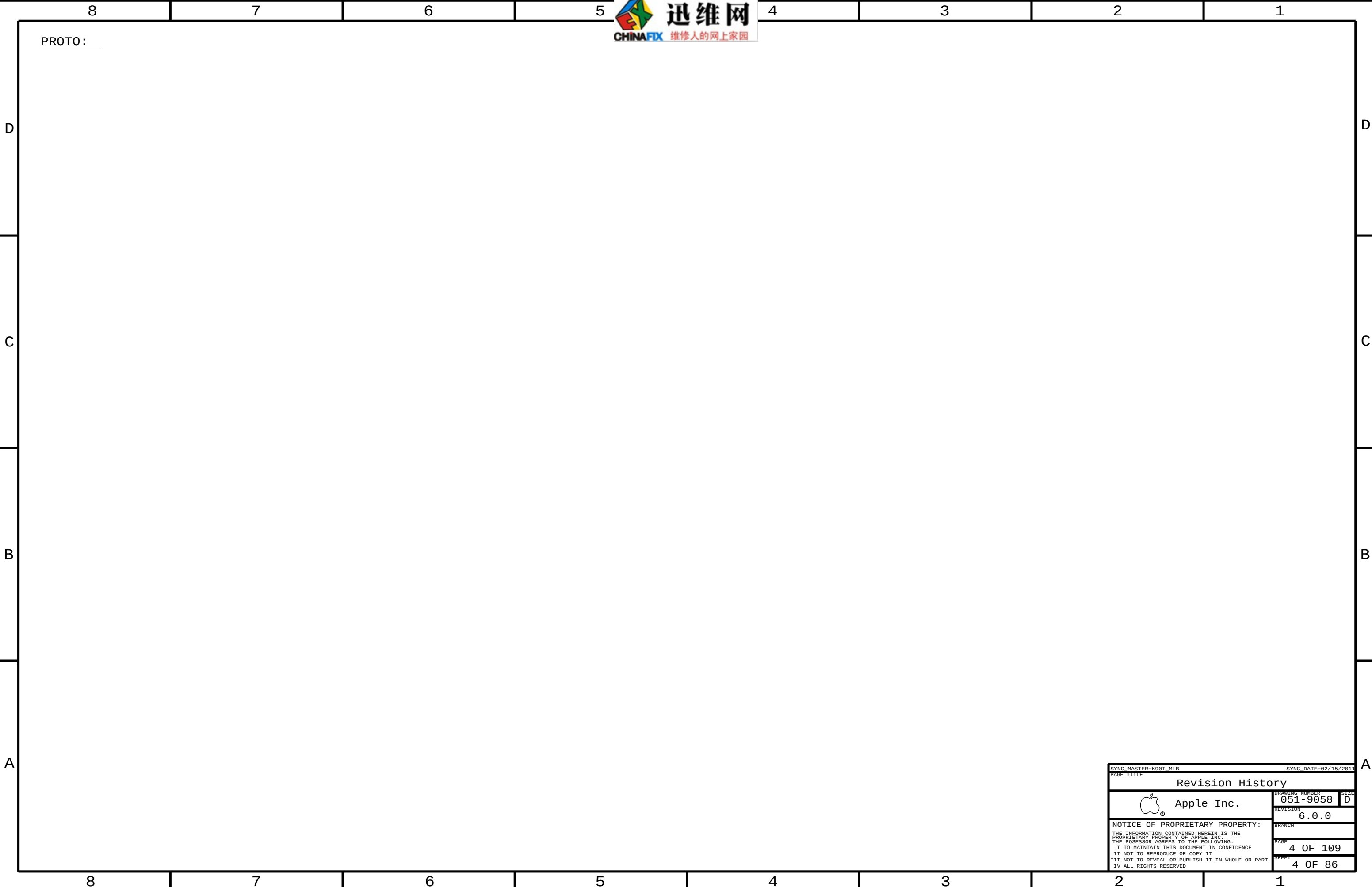
Schematic / PCB #'s

DRAWING
TITLE=MLB
ABBREV=DRAWING
LAST_MODIFIED= Tue Mar 13 14:00:17 2012





SYNC MASTER-K20A MLB		SYNC DATE=03/26/2009	
PAGE TITLE			
Revision History			
	Apple Inc.		DRAWING NUMBER 051-9058
			SIZE D
			REVISION 6.0.0
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		PAGE 3 OF 109	SHEET 3 OF 86



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SYNC MASTER=K98I.MLB		SYNC DATE=02/15/2011	
PAGE TITLE		Revision History	
 Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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		PAGE	4 OF 109
		SHEET	4 OF 86

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
607-8895	CMN PTS,PCBA,MLB,J30	J30_COMMON,FET_PAIR
085-3092	J30 MLB DEVELOPMENT BOM	J30_LEVEL:ENG
607-8721	POWER FETS PAIR,FAIRCHILD,DDR,J30	DDR_POWER_FET:FAIR
607-8722	POWER FETS PAIR,FAIRCHILD,5V_S3,J30	5V_S3_POWER_FET:FAIR
607-8723	POWER FETS PAIR,FAIRCHILD,PBUS_CHARGER,J30	CHARGER_POWER_FET:FAIR
607-9309	POWER FETS PAIR,RENESAS,DDR,J30	DDR_POWER_FET:REN
607-9310	POWER FETS PAIR,RENESAS,5V_S3,J30	5V_S3_POWER_FET:REN
607-9311	POWER FETS PAIR,RENESAS,PBUS_CHARGER,J30	CHARGER_POWER_FET:REN
639-3752	PCBA,MLB,MOL,2.9G,J30	J30_CMNPTS,CPU_2.9GHZ,SODIMM:MOLEX,EEEE_F1YK
639-3756	PCBA,MLB,HYB,2.9G,J30	J30_CMNPTS,CPU_2.9GHZ,SODIMM:HYBRID,EEEE_F1YH
639-3753	PCBA,MLB,FOX,2.5G,J30	J30_CMNPTS,CPU_2.5GHZ,SODIMM:FOXCONN,EEEE_F1YL
639-3755	PCBA,MLB,HYB,2.5G,J30	J30_CMNPTS,CPU_2.5GHZ,SODIMM:HYBRID,EEEE_F1YJ
639-3751	PCBA,MLB,MOL,2.5G,J30	J30_CMNPTS,CPU_2.5GHZ,SODIMM:MOLEX,EEEE_F1YM
639-3754	PCBA,MLB,FOX,2.9G,J30	J30_CMNPTS,CPU_2.9GHZ,SODIMM:FOXCONN,EEEE_F1YG

J30 BOM GROUPS

BOM GROUP	BOM OPTIONS
J30_COMMON	ALTERNATE, COMMON, J30_COMMON1, J30_COMMON2, J30_DEBUG:ENG, J30_PROGPARTS, T29BST:Y, TBTHV:P15V
J30_COMMON1	BATT_3S, CPUMEM_S0, USBHUB2513B, HUB_3NONREM, T29:YES, SDRV_PD, SDRV12C:MCU, AXG_PHASE1, BTPWR:S4, UV_GLUE_J30
J30_COMMON2	MIKEY, TPAD:Z2, RAMCFG_SL0T
J30_PROGPARTS	BOOTROM_PROG, SMC_PROG, TPAD_PROG, ENET_PROG, T29ROM:PROG, T29MCU:PROG
J30_DEVEL:ENG	BKLT:ENG, XDP_CONN, XDP_CPU:BPM, XDP_PCH, LPCPLUS_CONN:YES, LOADISNS:YES, D0RVREF_DAC, S0P600D_TSL
J30_DEVEL:PVT	LPCPLUS_CONN:YES, XDP_CONN
J30_DEBUG:ENG	DEVEL_BOM, MOJO:YES, XDP, LPCPLUS_R:YES, VREFDQ:M1_M3, VREFCA:L00_DAC
J30_DEBUG:PVT	DEVEL_BOM, BKLT:PROD, MOJO:YES, XDP, LPCPLUS_R:YES, VREFDQ:M1_M3, VREFCA:L00, USBHUB2514B
J30_DEBUG:PROD	BKLT:PROD, MOJO:YES, XDP, LPCPLUS_R:YES, LOADISNS:NO, VREFDQ:M1_M3, VREFCA:L00, USBHUB2513B

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
33754113	1	IC, IVB, 2C, 35W, 1023BGA	U1000	CRITICAL	CPU_IVB_2C
33754264	1	IVB, S R8N0, PRQ, L1, 2.5, 35W, 2+2, 1.1, 3M, BGA	U1000	CRITICAL	CPU_2_5GHZ
33754265	1	IVB, S R8MU, PRQ, L1, 2.9, 35W, 2+2, 1.25, 4M, BGA	U1000	CRITICAL	CPU_2_9GHZ
33754269	1	PANTHERPOINT, C1, SL3BC, PRQ, BDB2HM77	U1800	CRITICAL	
34350534		IC, BCM57765B0, ENET&SD, 8XB	U3900	CRITICAL	
33850753	1	IC, FW48E, 1394B PHY/ONC1 LINK/PCI-E, 12	U4100	CRITICAL	
33851072	1	IC, T29, PRQ, S L3JY, FCGBA, 15x15MM, C1	U3600	CRITICAL	T29: YES
35353055	1	IC, PI3VEDP212, X2 DISPLAYPORT 2:1 MUX, QFN	U9390	CRITICAL	
946-3827	1	300 KLB DYNAM ADHESIVE 29993-SC 0.48G	UV_GLUE_330	CRITICAL	UV_GLUE_330
51650806	1	CONN, 204P, SODIMM, SOCKET, DDR3, RAM, BGA, FOXCONN	J3100	CRITICAL	SODIMM: FOXCONN
516-0246		CONN, 204P, SODIMM, DDR3, P=0.6MM, FOXCONN	J2900	CRITICAL	SODIMM: FOXCONN
51650805	1	CONN, 204P, SODIMM, SOCKET, DDR3, RAM, BGA, MOLEX	J3100	CRITICAL	SODIMM: MOLEX
516-0245		CONN, 204P, SODIMM, DDR3, P=0.6MM, MOLEX	J2900	CRITICAL	SODIMM: MOLEX
51650805	1	CONN, 204P, SODIMM, SOCKET, DDR3, RAM, BGA, MOLEX	J3100	CRITICAL	SODIMM: HYBRID
516-0246	1	CONN, 204P, SODIMM, DDR3, P=0.6MM, FOXCONN	J2900	CRITICAL	SODIMM: HYBRID

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:F1YG]	CRITICAL	EEEE_F1YG
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:F1YH]	CRITICAL	EEEE_F1YH
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:F1YJ]	CRITICAL	EEEE_F1YJ
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:F1YK]	CRITICAL	EEEE_F1YK
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:F1YL]	CRITICAL	EEEE_F1YL
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE:F1YM]	CRITICAL	EEEE_F1YM

Programmable Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S0862	1	IC, FLASH, SERIAL, SPI, 1MBIT, 2V7, REV F	U3990	CRITICAL	ENET_BLANK
341S3096	1	IC ENET, 11MBITFLAN, CIV REV01, K9x	U3990	CRITICAL	ENET_PROG
335S0550	1	IC, EEPROM, SERIAL, SPI, 4Kx8, 1.8V, MLP8, LF	U3690	CRITICAL	T29ROM:BLANK
341S3430	1	IC, T29 EEPROM, LR, J30/ J31	U3690	CRITICAL	T29ROM:PROG
337S3997	1	IC, MCU, 32B, LPC1112A, 16KB/2KB, HVQFN25	U9330	CRITICAL	T29MCU:BLANK
341S3365	1	IC, PROGRAMM, T29, PORT MCU, K901A, K91A, K92A	U9330	CRITICAL	T29MCU:PROG
338S1098	1	IC, SMC12-A3, 40MHZ/560MIPS MCU, 9x9, 157BGA	U4900	CRITICAL	SMC_BLANK
341S3300	1	IC, SMC, EXTERNAL, FSB, A3, J30	U4900	CRITICAL	SMC_PROG
335S0807	1	IC, SPI SRL 50MHZ FLASH, 64MBT, B50P, FUSE=1	U6100	CRITICAL	BOOTROM_BLANK
335S0812	1	64 MBIT SPI SRL DUAL I/O FLASH, 501CB	U6100	CRITICAL	BOOTROM_BLANK
341S3558	1	IC, EFI, V90C7, J30/ J31	U6100	CRITICAL	BOOTROM_PROG
341S2384	1	IR, ENCORE II, CY7C53803-LQXC	U4800	CRITICAL	
341S3522	1	IC, P50C, TP/KB, J30/ J31	U5701	CRITICAL	TPAD_PROG

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
13850603	13850602		ALL	Murata alt to Samsung
15750058	15750084		ALL	Belix alt to WM Negotiation
12850303	12850323		ALL	Panasonic alt to Sanyo
13850676	13850691		ALL	Murata alt to Samsung
15250778	15250693		ALL	Cytec alt to Vishay
37650855	37651032		ALL	Diodes alt to Toshiba
37650977	37650859		ALL	Diodes alt to Toshiba
37650972	37651017		ALL	Rohe alt to Toshiba
37650937	37650845		ALL	Fairchild alt to Renesas
37650777	37650761		ALL	ADM alt to Siliconix
37650957	37650958		ALL	Fairchild alt to Fairchild
37650953	37650958		ALL	Fairchild alt to Renesas
37750107	37750126		ALL	Onsemi alt to Semtech
37150709	37150652		ALL	NOP alt to Infineon
514-8788	514-0671		ALL	Acorn(w/ litem) alt to Acorn
607-9310	607-8722		ALL	Renesas alternate to Fairchild
607-9311	607-8723		ALL	Renesas alternate to Fairchild

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
15251499	15250864		ALL	Collcraft alt to Murata
15251493	15251300		ALL	Collcraft alt to Murata
13850652	13850648		ALL	Samung/Murata alt to Taiyo
13850684	13850660		ALL	Murata alt to Taiyo
15251512	15251295		ALL	Cytron alt to NEC
15251019	15251271		ALL	Cytron alt to TMO
37651023	37650960		ALL	Siliconix alt to Renesas
35353312	35353055		ALL	MP alt to Pericom
35353238	35351428		ALL	Intersil alt to TI
35353519	35352179		ALL	Intersil alt to TI
15550578	15550367		ALL	Taiyo alt to Murata
13850681	13850638		ALL	Taiyo alt to Samsung
13850671	13850673		ALL	Taiyo alt to Murata
37650998	37650796		ALL	Fairchild alt to Vishay
37750124	37750057		ALL	Anotech alt to Tdk
34153492	34153096		ALL	Sumida alt to Amel (ENET ROM)
37651053	37650604		ALL	Diodes alt to Fairchild
37651076	37650634		ALL	Diodes alt to onsemi

Sub BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
085-3692	1	J30 MLB DEVELOPMENT	DEVEL	CRITICAL	DEVEL_BOM
607-8895	1	CMN PTS,PCBA,MLB,J30	CMNPTS	CRITICAL	J30_CMNPTS
607-8721	1	POWER_FETS PAIR,FAIRCHILD,DDR,J30	CSET1	CRITICAL	FET_PAIR
607-8722	1	POWER_FETS PAIR,FAIRCHILD,5V_S3,J30	CSET2	CRITICAL	FET_PAIR
607-8723	1	POWER_FETS PAIR,FAIRCHILD,PBUS_CHARGER,J30	CSET3	CRITICAL	FET_PAIR

SYNCH MASTER-K907 MLB		SYNCH DATE=02/15/2011	
PAGE TITLE			
BOM Configuration			
 Apple Inc.		DRAWING NUMBER 051-9058	SIZE D
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BRANCH		PAGE	5 OF 109
		SHEET	5 OF 86

Functional Test Points

D

Fan Connectors

612	TRUE	PP5V_S0	6 7
615	TRUE	FAN_RT_PWM	52
616	TRUE	FAN_RT_TACH	52
(NEED TO ADD 1 GND TP)			

MIC FUNC_TEST

6550	TRUE	BI_MIC_L0	61 62
6550	TRUE	BI_MIC_HI	61 62
6550	TRUE	BI_MIC_SHIELD	61 62
(NEED TO ADD 1 GND TP)			

SPEAKER FUNC_TEST

6699	TRUE	SPKRAMP_L_N_OUT	69 61 85
6700	TRUE	SPKRAMP_L_P_OUT	69 61 85
6700	TRUE	SPKRAMP_R_N_OUT	69 61 85
6700	TRUE	SPKRAMP_R_P_OUT	69 61 85
6700	TRUE	SPKRAMP_SUB_N_OUT	69 61 85
6700	TRUE	SPKRAMP_SUB_P_OUT	69 61 85

LVDS FUNC_TEST

6260	TRUE	PP3V3_LCDVDD_SW_F (NEED 2 TP)	6 74
6260	TRUE	PP3V3_S0_LCD_F	6 74
6260	TRUE	PPVOUT_SW_LCDBKLT (NEED 2 TP)	74 77
6260	TRUE	LVDS_DDC_CLK	8 74
6260	TRUE	LVDS_DDC_DATA	8 74
6260	TRUE	LVDS_IG_A_DATA_N<0>	17 74 89
6260	TRUE	LVDS_IG_A_DATA_P<0>	17 74 89
6260	TRUE	LVDS_IG_A_DATA_N<1>	17 74 89
6260	TRUE	LVDS_IG_A_DATA_P<1>	17 74 89
6260	TRUE	LVDS_IG_A_DATA_N<2>	17 74 89
6260	TRUE	LVDS_IG_A_DATA_P<2>	17 74 89
6260	TRUE	LVDS_CONN_A_CLK_F_N	74 85
6260	TRUE	LVDS_CONN_A_CLK_F_P	74 85
6260	TRUE	LED_RETURN_1	74 77
6260	TRUE	LED_RETURN_2	74 77
6260	TRUE	LED_RETURN_3	74 77
6260	TRUE	LED_RETURN_4	74 77
6260	TRUE	LED_RETURN_5	74 77
6260	TRUE	LED_RETURN_6	74 77
(NEED TO ADD 5 GND TP)			

SATA ODD CONN

6654	TRUE	PP5V_SW_ODD (NEED 2 TP)	6 41
6654	TRUE	SMC_ODD_DETECT	41 45
6654	TRUE	SATA_ODD_D2R_C_P	41 85
6654	TRUE	SATA_ODD_D2R_C_N	41 85
6654	TRUE	SATA_ODD_R2D_P	41 89
6654	TRUE	SATA_ODD_R2D_N	41 89
6654	TRUE	SMC_SSD_TEMP_CTL_R	
6654	TRUE	HDD_OOB_TEMP	
(NEED TO ADD 3 GND TP)			

SATA HDD/IR/SIL

6660	TRUE	PP5V_S0_HDD_FLT (NEED 2 TP)	6 41
6660	TRUE	SATA_HDD_R2D_P	41 89
6660	TRUE	SATA_HDD_R2D_N	41 89
6660	TRUE	SATA_HDD_D2R_C_P	41 89
6660	TRUE	SATA_HDD_D2R_C_N	41 89
6660	TRUE	SYS_LED_ANODE_R	41
6660	TRUE	IR_RX_OUT	41 44
6660	TRUE	SMC_SSD_THROTTLE_R	
6660	TRUE	PP5V_S3_IR_R	41
(NEED TO ADD 3 GND TP)			

BATT POWER CONN

6660	TRUE	SMBUS_SMC_5_G3_SCL	6 45 48 84
6660	TRUE	SMBUS_SMC_5_G3_SDA	6 45 48 84
6660	TRUE	SYS_DETECT_L	63
6660	TRUE	PPVBAT_G3H_CONN (NEED 5 TP)	63 64
(NEED TO ADD 5 GND TP)			

BIL CONN

6660	TRUE	PP3V42_G3H	6 7
6660	TRUE	SMBUS_SMC_5_G3_SCL	6 45 48 84
6660	TRUE	SMBUS_SMC_5_G3_SDA	6 45 48 84
6660	TRUE	SMC_BIL_BUTTON_L	45 46 63
6660	TRUE	SMC_LID_R	63
(NEED TO ADD 2 GND TP)			

X19 CONN

6660	TRUE	PP3V3_WLAN (NEED 3 TP)	6 32 46
6660	TRUE	PCIE_AP_D2R_PI_P	32 81
6660	TRUE	PCIE_AP_D2R_PI_N	32 81
6660	TRUE	PCIE_AP_R2D_P	32 81
6660	TRUE	PCIE_AP_R2D_N	32 81
6660	TRUE	PCIE_CLK100M_AP_CONN_P	32 85
6660	TRUE	PCIE_CLK100M_AP_CONN_N	32 85
6660	TRUE	PP3V3_S3RS4_BT_F	32
6660	TRUE	PCIE_WAKE_L	17 24 32
6660	TRUE	USB_BT_CONN_P	32 89
6660	TRUE	USB_BT_CONN_N	32 89
6660	TRUE	AP_CLKREQ_Q_L	32
6660	TRUE	AP_RESET_CONN_L	32
6660	TRUE	AP_TEMP_SMB_SDA_R	32
6660	TRUE	AP_TEMP_SMB_SCL_R	32
6660	TRUE	WIFI_EVENT_L_R	32
(NEED TO ADD 5 GND TP)			

IPD_FLEX_CONN

6660	TRUE	PP3V3_S4	6 7
6660	TRUE	PP18V5_Z2	6 94
6660	TRUE	Z2_CS_L	53 54
6660	TRUE	Z2_DEBUG3	53 54
6660	TRUE	Z2_M0SI	53 54
6660	TRUE	Z2_MISO	53 54
6660	TRUE	Z2_SCL	53 54
6660	TRUE	Z2_BOOTST_EN	54
6660	TRUE	Z2_HOST_INTN	53 54
6660	TRUE	Z2_CLKIN	53 54
6660	TRUE	Z2_KEY_ACT_L	53 54
6660	TRUE	Z2_RESET	53 54
6660	TRUE	PSOC_MISO	53 54
6660	TRUE	PSOC_M0SI	53 54
6660	TRUE	PSOC_SCL	53 54
6660	TRUE	SMBUS_SMC_2_S3_SCL	6 45 48 84
6660	TRUE	SMBUS_SMC_2_S3_SDA	6 45 48 84
6660	TRUE	PSOC_F_CS_L	53 54
6660	TRUE	PICKB_L	53 54
6660	TRUE	PP5V_S5_CUMULUS	54
(NEED TO ADD 2 GND TP)			

KEYBOARD CONN

6660	TRUE	PP3V3_S4	6 7
6660	TRUE	PP3V42_G3H	6 7
6660	TRUE	WS_KBD1	53
6660	TRUE	WS_KBD2	53
6660	TRUE	WS_KBD3	53
6660	TRUE	WS_KBD4	53
6660	TRUE	WS_KBD5	53
6660	TRUE	WS_KBD6	53
6660	TRUE	WS_KBD7	53
6660	TRUE	WS_KBD8	53
6660	TRUE	WS_KBD9	53
6660	TRUE	WS_KBD10	53
6660	TRUE	WS_KBD11	53
6660	TRUE	WS_KBD12	53
6660	TRUE	WS_KBD13	53
6660	TRUE	WS_KBD14	53
6660	TRUE	WS_KBD15_CAP	53
6660	TRUE	WS_KBD16_NUM	53
6660	TRUE	WS_KBD17	53
6660	TRUE	WS_KBD18	53
6660	TRUE	WS_KBD19	53
6660	TRUE	WS_KBD20	53
6660	TRUE	WS_KBD21	53
6660	TRUE	WS_KBD22	53
6660	TRUE	WS_KBD23	53
6660	TRUE	WS_KBD_ONOFF_L	53
6660	TRUE	WS_LEFT_SHIFT_KBD	53
6660	TRUE	WS_LEFT_OPTION_KBD	53
6660	TRUE	WS_CONTROL_KBD	53
(NEED TO ADD 2 GND TP)			

KBD BACKLIGHT CONN

6660	TRUE	KBDLED_ANODE	54
6660	TRUE	SMC_KBDLED_PRESENT_L	54
(NEED TO ADD 1 GND TP)			

CAMERA/ALS CONN

6660	TRUE	PP5V_S3_ALSCAMERA_F	32
6660	TRUE	SMBUS_SMC_2_S3_SCL	6 45 48 84
6660	TRUE	SMBUS_SMC_2_S3_SDA	6 45 48 84
6660	TRUE	USB_CAMERA_CONN_P	32 89
6660	TRUE	USB_CAMERA_CONN_N	32 89
(NEED TO ADD 2 GND TP)			

DEBUG VOLTAGE

6660	TRUE	PPVCORE_S0_CPU	7
6660	TRUE	PPVCORE_S0_AXG	7
6660	TRUE	PP1V2_S3_ENET_INTREG	71
6660	TRUE	PP1V05_S0	7
6660	TRUE	PP1V5_S3RS0	7 85
6660	TRUE	PP1V8_S0	7
6660	TRUE	PP3V3_S0	7 85
6660	TRUE	PP5V_S0	6 7
6660	TRUE	PP3V3_S3	7
6660	TRUE	PP5V_S3	7
6660	TRUE	PPVCCSA_S0_CPU	7
6660	TRUE	PP3V3_S5	7 85
6660	TRUE	PP3V42_G3H	6 7
6660	TRUE	PPBUS_G3H	7
6660	TRUE	PP3V3_ENET	7
6660	TRUE	PP3V3_WLAN	6 32 46
6660	TRUE	PP5V_SW_ODD	6 41
6660	TRUE	PP5V_S0_HDD_FLT	6 41
6660	TRUE	PP18V5_Z2	6 94
6660	TRUE	PP3V3_S0_LCD_F	6 74
6660	TRUE	PP3V3_LCDVDD_SW_F	6 74
6660	TRUE	PP4V5_AUDIO_ANALOG	57 62
6660	TRUE	PP1V5_S3	7
6660	TRUE	SMC_PM_G2_EN	45 73
6660	TRUE	PM_SLP_S4_L	8 17 26 32 45 73
6660	TRUE	PM_SLP_S3_L	8 17 26 45 73
(NEED TO ADD 6 GND TP)			

DC POWER CONN

6660	TRUE	PP18V5_DCIN_FUSE	63
6660	TRUE	ADAPTER_SENSE	63
(NEED TO ADD 4 GND TP)			

LPC+SPI DEBUG CONN

6660	TRUE	LPC_AD<0>	16 45 47 81
6660	TRUE	LPC_AD<1>	16 45 47 81
6660	TRUE	LPC_AD<2>	16 45 47 81
6660	TRUE	LPC_AD<3>	16 45 47 81
6660	TRUE	LPC_CLK33M_LPCPLUS	24 47 81
6660	TRUE	LPC_FRAME_L	16 45 47 81
6660	TRUE	LPC_PWRDOWN_L	17 45 47
6660	TRUE	LPC_SERIRQ	16 45 47
6660	TRUE	LPCPLUS_GPIO	19 47
6660	TRUE	LPCPLUS_RESET_L	24 47
6660	TRUE	PM_CLKRUN_L	17 45 47
6660	TRUE	PP3V42_G3H	6 7
6660	TRUE	PP5V_S0	6 7
6660	TRUE	SMC_RX_L	45 46 47
6660	TRUE	SMC_TCK	45 46 47
6660	TRUE	SMC_TDI	45 46 47
6660	TRUE	SMC_TDO	45 46 47
6660	TRUE	SMC_TMS	45 46 47
6660	TRUE	SMC_TX_L	45 46 47
6660	TRUE	SPI_ALT_CLK	47
6660	TRUE	SPI_ALT_CS_L	47
6660	TRUE	SPI_ALT_MISO	47
6660	TRUE	SPI_ALT_MOSI	47
6660	TRUE	SPIROM_USE_MLB	19 47 56
(NEED TO ADD 2 GND TP)			

NC NO_TESTS

NO_TEST			
17	TP_CRT_IG_BLUE	== TRUE	NC_CRT_IG_BLUE
17	TP_CRT_IG_GREEN	== MAKE_BASE=TRUE	NC_CRT_IG_GREEN
17	TP_CRT_IG_RED	== MAKE_BASE=TRUE	NC_CRT_IG_RED
17	TP_CRT_IG_DDC_CLK	== TRUE	NC_CRT_IG_DDC_CLK
17	TP_CRT_IG_DDC_DATA	== MAKE_BASE=TRUE	NC_CRT_IG_DDC_DATA
17	TP_CRT_IG_HSYNC	== TRUE	NC_CRT_IG_HSYNC
17	TP_CRT_IG_VSYNC	== MAKE_BASE=TRUE	NC_CRT_IG_VSYNC
17	TP_LVDS_IG_CTRL_CLK	== TRUE	NC_LVDS_IG_CTRL_CLK
17	TP_LVDS_IG_CTRL_DATA	== MAKE_BASE=TRUE	NC_LVDS_IG_CTRL_DATA
17	TP_PCH_LVDS_VBG	== MAKE_BASE=TRUE	NC_PCH_LVDS_VBG
16	TP_HDA_SDIN1	== TRUE	NC_HDA_SDIN1
16	TP_HDA_SDIN2	== MAKE_BASE=TRUE	NC_HDA_SDIN2
16	TP_HDA_SDIN3	== MAKE_BASE=TRUE	NC_HDA_SDIN3
18	TP_PCI_PME_L	== TRUE	NC_PCI_PME_L
18	TP_PCI_CLK33M_OUT3	== MAKE_BASE=TRUE	NC_PCI_CLK33M_OUT3

16	TP_CLINK_CLK	== TRUE	NC_CLINK_CLK
16	TP_CLINK_DATA	== MAKE_BASE=TRUE	NC_CLINK_DATA
16	TP_CLINK_RESET_L	== MAKE_BASE=TRUE	NC_CLINK_RESET_L
16	TP_PCIE_CLK100M_PEBN	== TRUE	NC_PCIE_CLK100M_PEBN
16	TP_PCIE_CLK100M_PEBP	== MAKE_BASE=TRUE	NC_PCIE_CLK100M_PEBP

38	TP_FW643_SDA	== TRUE	NC_FW643_SDA
38	TP_FW643_SM	== MAKE_BASE=TRUE	NC_FW643_SM
38	TP_FW643_TCK	== MAKE_BASE=TRUE	NC_FW643_TCK
38	TP_FW643_TMS	== TRUE	NC_FW643_TMS
38	TP_FW643_FW620_L	== MAKE_BASE=TRUE	NC_FW643_FW620_L
38	TP_FW643_VBUE	== MAKE_BASE=TRUE	NC_FW643_VBUE
38	TP_FW643_OCR10_CTL	== TRUE	NC_FW643_OCR10_CTL
38	TP_FW643_AVREG	== TRUE	NC_FW643_AVREG
38	TP_FW643_TDI	== MAKE_BASE=TRUE	NC_FW643_TDI

23	TP_XDP_PCH_OBSFN_A<0..1>	== TRUE	NC_TP_XDP_PCH_OBSFN_A<0..1>
23	TP_XDP_PCH_OBSFN_B<0..1>	== MAKE_BASE=TRUE	NC_TP_XDP_PCH_OBSFN_B<0..1>
23	TP_XDPPCH_H00K2	== MAKE_BASE=TRUE	NC_TP_XDPPCH_H00K2
23	TP_XDPPCH_H00K3	== MAKE_BASE=TRUE	NC_TP_XDPPCH_H00K3
23	TP_XDP_PCH_OBSFN_D<0..1>	== MAKE_BASE=TRUE	NC_TP_XDP_PCH_OBSFN_D<0..1>
23	TP_XDP_PCH_H00K4	== TRUE	NC_TP_XDP_PCH_H00K4
23	TP_XDP_PCH_H00K5	== MAKE_BASE=TRUE	NC_TP_XDP_PCH_H00K5

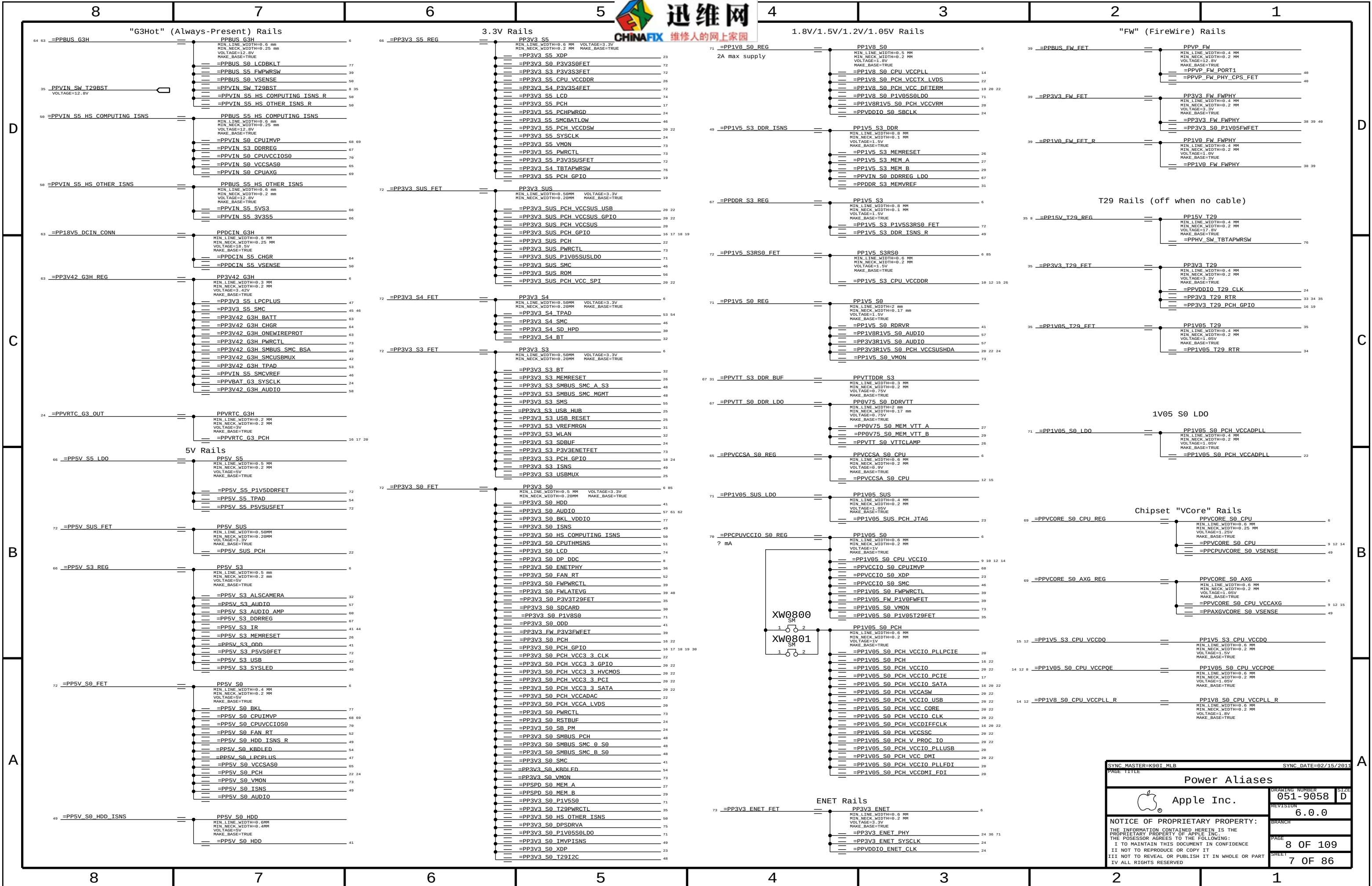
16	TP_PCH_GPI064_CLKOUTFLEX0	== TRUE	NC_PCH_GPI064_CLKOUTFLEX0
16	TP_PCH_GPI065_CLKOUTFLEX1	== MAKE_BASE=TRUE	NC_PCH_GPI065_CLKOUTFLEX1
16	TP_PCH_GPI066_CLKOUTFLEX2	== MAKE_BASE=TRUE	NC_PCH_GPI066_CLKOUTFLEX2
16	TP_PCH_GPI067_CLKOUTFLEX3	== MAKE_BASE=TRUE	NC_PCH_GPI067_CLKOUTFLEX3

NC NO_TESTS

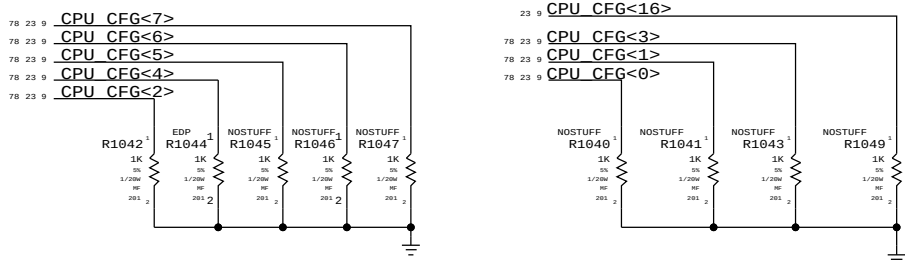
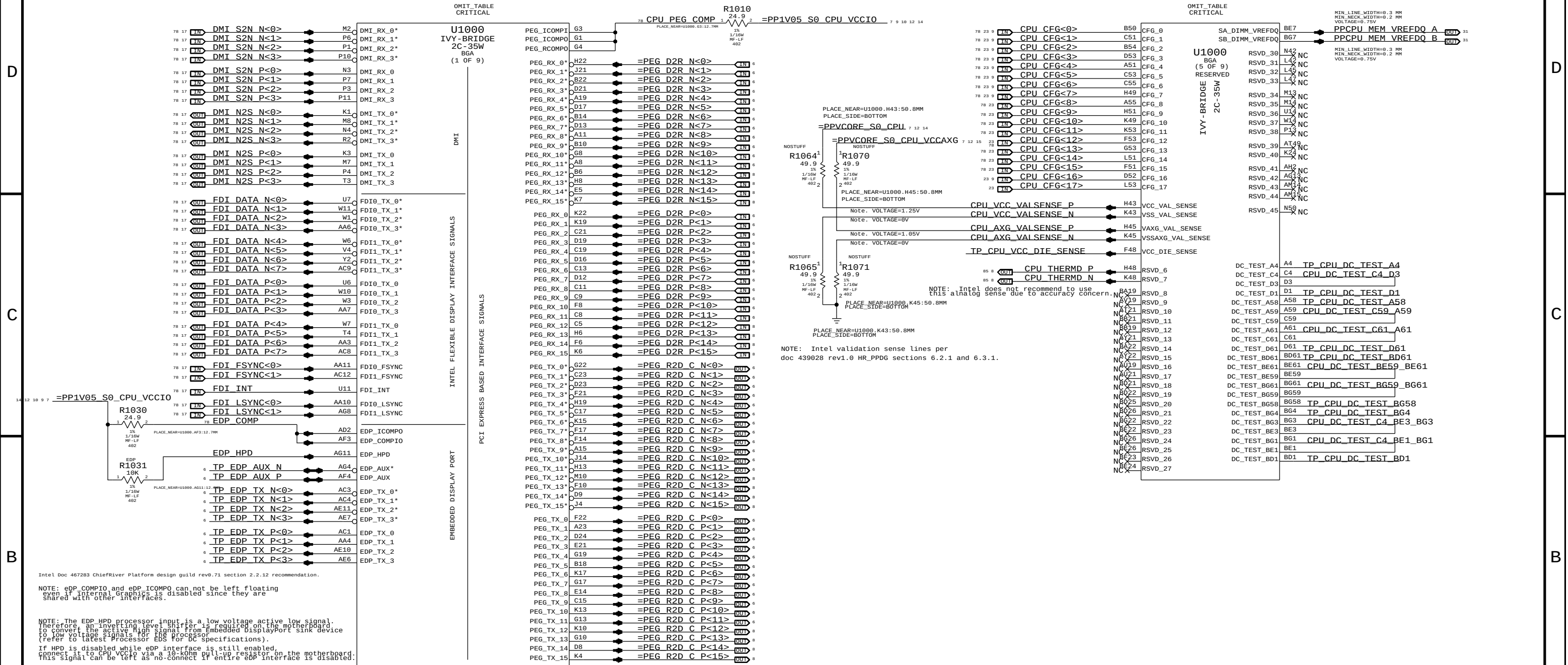
6660	TRUE	NC_FW2_TBPB	48
6660	TRUE	NC_FW2_TPBH	48
6660	TRUE	NC_FW2_TPBAS	48
6660	TRUE	NC_FW2_TPAP	48
6660	TRUE	NC_FW2_TPAN	48
6660	TRUE	NC_FW0_TBPB	48
6660	TRUE	NC_FW0_TPBH	48
6660	TRUE	NC_FW0_TPAP	48

6660	TRUE	XDP_PCH_AP_PWR_EN	
6660	TRUE	XDP_PCH_USB_HUB_SOFT_RST_L	
6660	TRUE	XDP_PCH_SDCONN_STATE_RST_L	
6660	TRUE	XDP_PCH_ENET_PWR_EN	
6660	TRUE	XDP_PCH_SDCONN_DET_L	
6660	TRUE	XDP_PCH_S5_PWRGD	23
6660	TRUE	XDP_PCH_PWRBTN_L	23
6660	TRUE	XDP_PCH_ISOLATE_CPU_MEM_L	
6660	TRUE	XDP_FW_CLKREQ_L	
6660	TRUE	XDP_AP_CLKREQ_L	
6660	TRUE	XDP_PCH_AUD_IPHS_SWITCH_EN	

17	TP_SDVO_TVCLKINN	== TRUE	NC_SDVO_TVCLKINN
17	TP_SDVO_TVCLKINP	== MAKE_BASE=TRUE	NC_SDVO_TVCLKINP
17	TP_SDVO_STALLN	== TRUE	NC_SDVO_STALLN
17	TP_SDVO_STALLP	== MAKE_BASE=TRUE	NC_SDVO_STALLP
17	TP_SDVO_INTN	== TRUE	NC_SDVO_INTN
17	TP_SDVO_INTP	== MAKE_BASE=TRUE	NC_SDVO_INTP



NOTE: Intel provides an internal pull-up OF 5-15k to VCCI0 on all CFG signals.



These can be Placed close to J2500 and Only for debug access

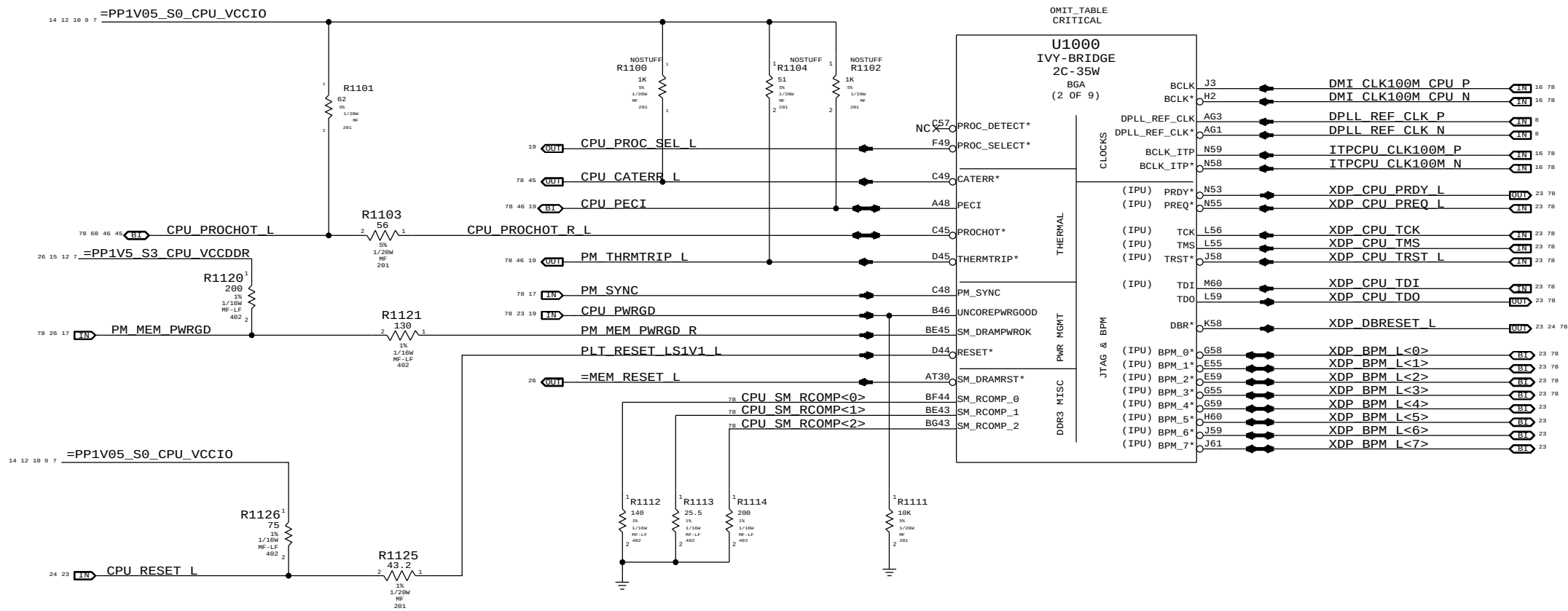
FOR IVYBRIDGE PROCESSOR

```

CFG [7]:PEG DEFER TRAINING                1 = (DEFAULT) IMMEDIATELY AFTER xxBSETB    0 = WAIT FOR BIOS
CFG [6:5]:PCIe BIFURCATION                11 = 1 X16 (DEFAULT)    10 = 2 X8    01 = RSVD    00 = X8, X4, X4
CFG [4]:eDP ENABLE/DISABLE                1 = DISABLED    0 = ENABLED
CFG [3]:PCIe x4 LANE REVERSAL              1 = NORMAL OPERATION    0 = LANES REVERSED
CFG [2]:PCIe x16 LANE REVERSAL            1 = NORMAL OPERATION    0 = LANES REVERSED

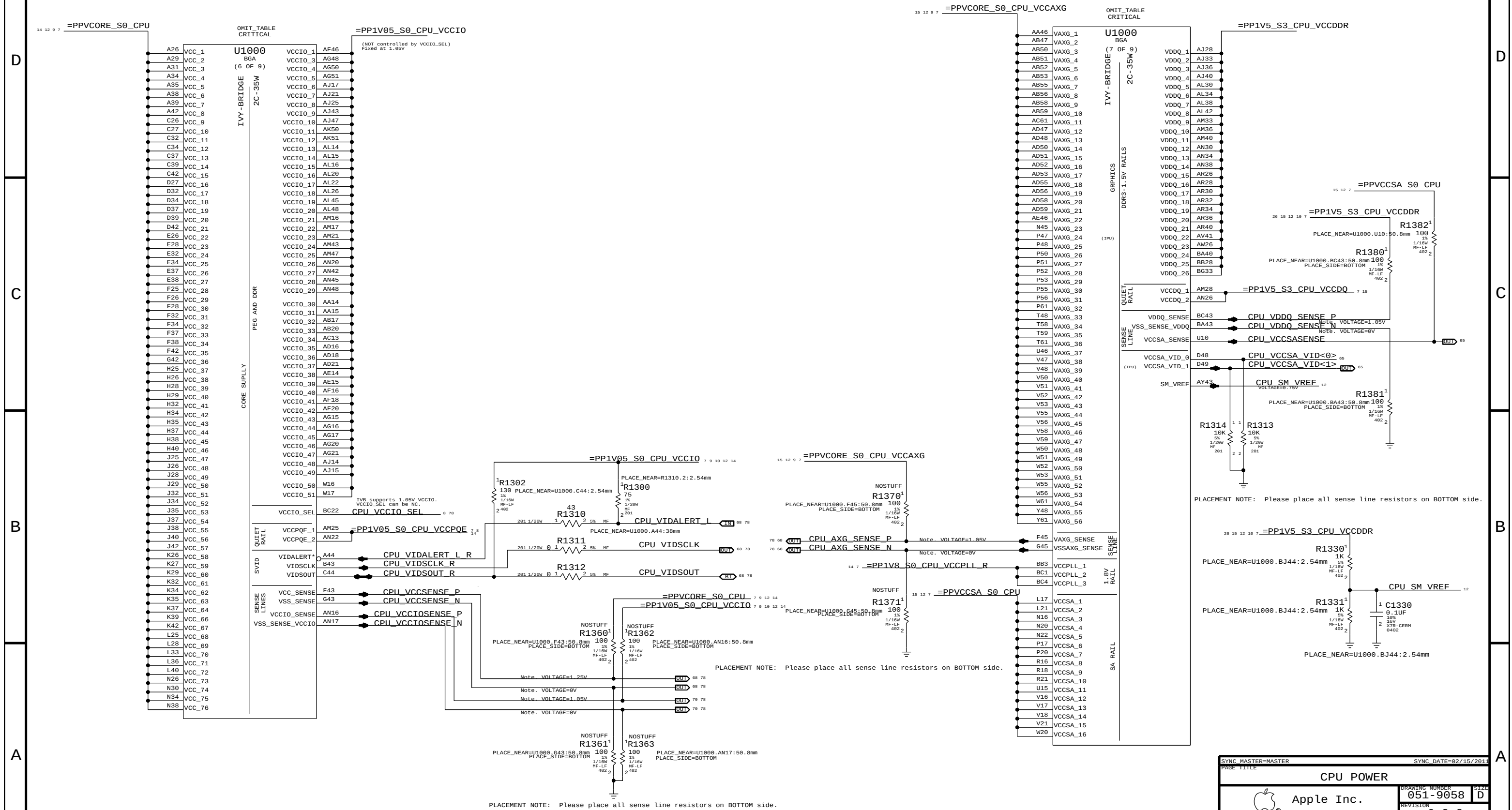
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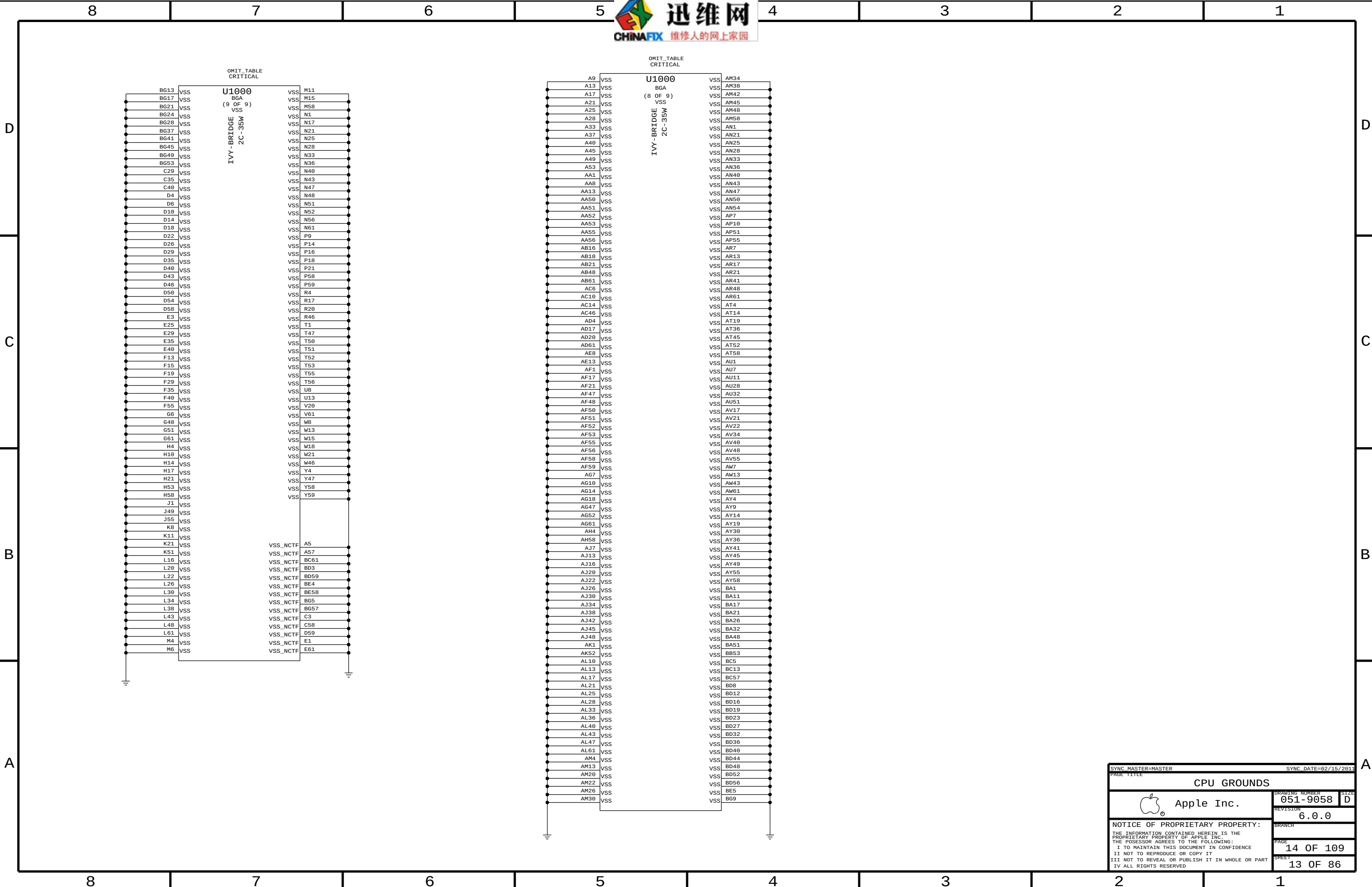
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		REVISION	
		6.0.0	
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IV ALL RIGHTS RESERVED			



OMIT_TABLE CRITICAL			
79 28	81	MEM A D0<0>	AG6 SA_D0_0
79 28	81	MEM A D0<1>	AJ6 SA_D0_1
79 28	81	MEM A D0<2>	AP11 SA_D0_2
79 28	81	MEM A D0<3>	AL6 SA_D0_3
79 28	81	MEM A D0<4>	AJ10 SA_D0_4
79 28	81	MEM A D0<5>	AJ8 SA_D0_5
79 28	81	MEM A D0<6>	AL8 SA_D0_6
79 28	81	MEM A D0<7>	AL7 SA_D0_7
79 28	81	MEM A D0<8>	AR11 SA_D0_8
79 28	81	MEM A D0<9>	AP6 SA_D0_9
79 28	81	MEM A D0<10>	AU6 SA_D0_10
79 28	81	MEM A D0<11>	AV9 SA_D0_11
79 28	81	MEM A D0<12>	AR6 SA_D0_12
79 28	81	MEM A D0<13>	AP8 SA_D0_13
79 28	81	MEM A D0<14>	AT13 SA_D0_14
79 28	81	MEM A D0<15>	AU13 SA_D0_15
79 28	81	MEM A D0<16>	BC7 SA_D0_16
79 28	81	MEM A D0<17>	BB7 SA_D0_17
79 28	81	MEM A D0<18>	BA13 SA_D0_18
79 28	81	MEM A D0<19>	BB11 SA_D0_19
79 28	81	MEM A D0<20>	BA7 SA_D0_20
79 28	81	MEM A D0<21>	BA9 SA_D0_21
79 28	81	MEM A D0<22>	BB9 SA_D0_22
79 28	81	MEM A D0<23>	AY13 SA_D0_23
79 28	81	MEM A D0<24>	AV14 SA_D0_24
79 28	81	MEM A D0<25>	AR14 SA_D0_25
79 28	81	MEM A D0<26>	AY17 SA_D0_26
79 28	81	MEM A D0<27>	AR19 SA_D0_27
79 28	81	MEM A D0<28>	BA14 SA_D0_28
79 28	81	MEM A D0<29>	AU14 SA_D0_29
79 28	81	MEM A D0<30>	BB14 SA_D0_30
79 28	81	MEM A D0<31>	BB17 SA_D0_31
79 28	81	MEM A D0<32>	BA45 SA_D0_32
79 28	81	MEM A D0<33>	AR43 SA_D0_33
79 28	81	MEM A D0<34>	AW48 SA_D0_34
79 28	81	MEM A D0<35>	BC48 SA_D0_35
79 28	81	MEM A D0<36>	BC45 SA_D0_36
79 28	81	MEM A D0<37>	AR45 SA_D0_37
79 28	81	MEM A D0<38>	AT48 SA_D0_38
79 28	81	MEM A D0<39>	AY48 SA_D0_39
79 28	81	MEM A D0<40>	BA49 SA_D0_40
79 28	81	MEM A D0<41>	AV49 SA_D0_41
79 28	81	MEM A D0<42>	BB51 SA_D0_42
79 28	81	MEM A D0<43>	AY53 SA_D0_43
79 28	81	MEM A D0<44>	BB49 SA_D0_44
79 28	81	MEM A D0<45>	AU49 SA_D0_45
79 28	81	MEM A D0<46>	BA53 SA_D0_46
79 28	81	MEM A D0<47>	BB55 SA_D0_47
79 28	81	MEM A D0<48>	BA55 SA_D0_48
79 28	81	MEM A D0<49>	AV56 SA_D0_49
79 28	81	MEM A D0<50>	AP59 SA_D0_50
79 28	81	MEM A D0<51>	AP53 SA_D0_51
79 28	81	MEM A D0<52>	AV54 SA_D0_52
79 28	81	MEM A D0<53>	AT54 SA_D0_53
79 28	81	MEM A D0<54>	AP56 SA_D0_54
79 28	81	MEM A D0<55>	AP52 SA_D0_55
79 28	81	MEM A D0<56>	AN57 SA_D0_56
79 28	81	MEM A D0<57>	AN53 SA_D0_57
79 28	81	MEM A D0<58>	AG56 SA_D0_58
79 28	81	MEM A D0<59>	AG53 SA_D0_59
79 28	81	MEM A D0<60>	AN55 SA_D0_60
79 28	81	MEM A D0<61>	AN52 SA_D0_61
79 28	81	MEM A D0<62>	AG55 SA_D0_62
79 28	81	MEM A D0<63>	AK56 SA_D0_63
79 27	OUT	MEM A BA<0>	BD37 SA_BS_0
79 27	OUT	MEM A BA<1>	BF36 SA_BS_1
79 27	OUT	MEM A BA<2>	BA28 SA_BS_2
79 27	OUT	MEM A CAS_L	BE39 SA_CAS*
79 27	OUT	MEM A RAS_L	BD39 SA_RAS*
79 27	OUT	MEM A WE_L	AT41 SA_WE*

U1000 BGA (3 OF 9)			
SA_CK_0	AU36	MEM A CLK P<0>	79 27
SA_CK_0*	AV36	MEM A CLK N<0>	79 27
SA_CKE_0	AY26	MEM A CKE<0>	79 27
SA_CK_1	AT40	MEM A CLK P<1>	79 27
SA_CK_1*	AU40	MEM A CLK N<1>	79 27
SA_CKE_1	BB26	MEM A CKE<1>	79 27
SA_CS_0*	BB40	MEM A CS L<0>	27 79
SA_CS_1*	BC41	MEM A CS L<1>	27 79
SA_ODT_0	AY40	MEM A ODT<0>	79 27
SA_ODT_1	BA41	MEM A ODT<1>	79 27
SA_DQS_0*	AL11	MEM A DQS N<0>	27 79
SA_DQS_1*	AR8	MEM A DQS N<1>	27 79
SA_DQS_2*	AV11	MEM A DQS N<2>	27 79
SA_DQS_3*	AT17	MEM A DQS N<3>	27 79
SA_DQS_4*	AV45	MEM A DQS N<4>	27 79
SA_DQS_5*	AY51	MEM A DQS N<5>	27 79
SA_DQS_6*	AT55	MEM A DQS N<6>	27 79
SA_DQS_7*	AK55	MEM A DQS N<7>	27 79
SA_DQS_0	AJ11	MEM A DQS P<0>	27 79
SA_DQS_1	AR10	MEM A DQS P<1>	27 79
SA_DQS_2	AY11	MEM A DQS P<2>	27 79
SA_DQS_3	AU17	MEM A DQS P<3>	27 79
SA_DQS_4	AW45	MEM A DQS P<4>	27 79
SA_DQS_5	AV51	MEM A DQS P<5>	27 79
SA_DQS_6	AT56	MEM A DQS P<6>	27 79
SA_DQS_7	AK54	MEM A DQS P<7>	27 79
SA_MA_0	BG35	MEM A A<0>	27 79
SA_MA_1	BB34	MEM A A<1>	27 79
SA_MA_2	BE35	MEM A A<2>	27 79
SA_MA_3	BD35	MEM A A<3>	27 79
SA_MA_4	AT34	MEM A A<4>	27 79
SA_MA_5	AU34	MEM A A<5>	27 79
SA_MA_6	BB32	MEM A A<6>	27 79
SA_MA_7	AT32	MEM A A<7>	27 79
SA_MA_8	AY32	MEM A A<8>	27 79
SA_MA_9	AV32	MEM A A<9>	27 79
SA_MA_10	BE37	MEM A A<10>	27 79
SA_MA_11	BA30	MEM A A<11>	27 79
SA_MA_12	BC30	MEM A A<12>	27 79
SA_MA_13	AW41	MEM A A<13>	27 79
SA_MA_14	AY28	MEM A A<14>	27 79
SA_MA_15	AU26	MEM A A<15>	27 79





Intel recommendation (Section 6.2): 35x 2.2uF, 25x 22uF, 4x 470uF



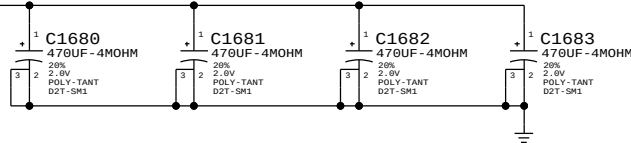
Place close to U1000 on top side.



Place close to U1000 on bottom side.



Place near inductors on bottom side.



Intel recommendation (Section 6.5): 26x 1uF, 10x 10uF, 2x 330uF

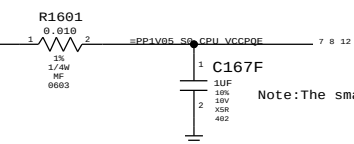
PLACEMENT_NOTE (C1684-C167F):



Place near U1000 on bottom side



Intel recommendation: 1x 10mOhm resistor, 1x 1uF 0402

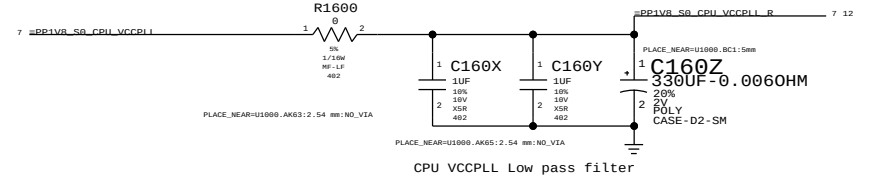


Note: The smallest 10m0hm available in the library are 0805s

Intel recommendation (section 6.4): 2x 1uF, 1x 330uF

PLACEMENT_NOTE (C1646-C1671):

Place near U1000 on top side

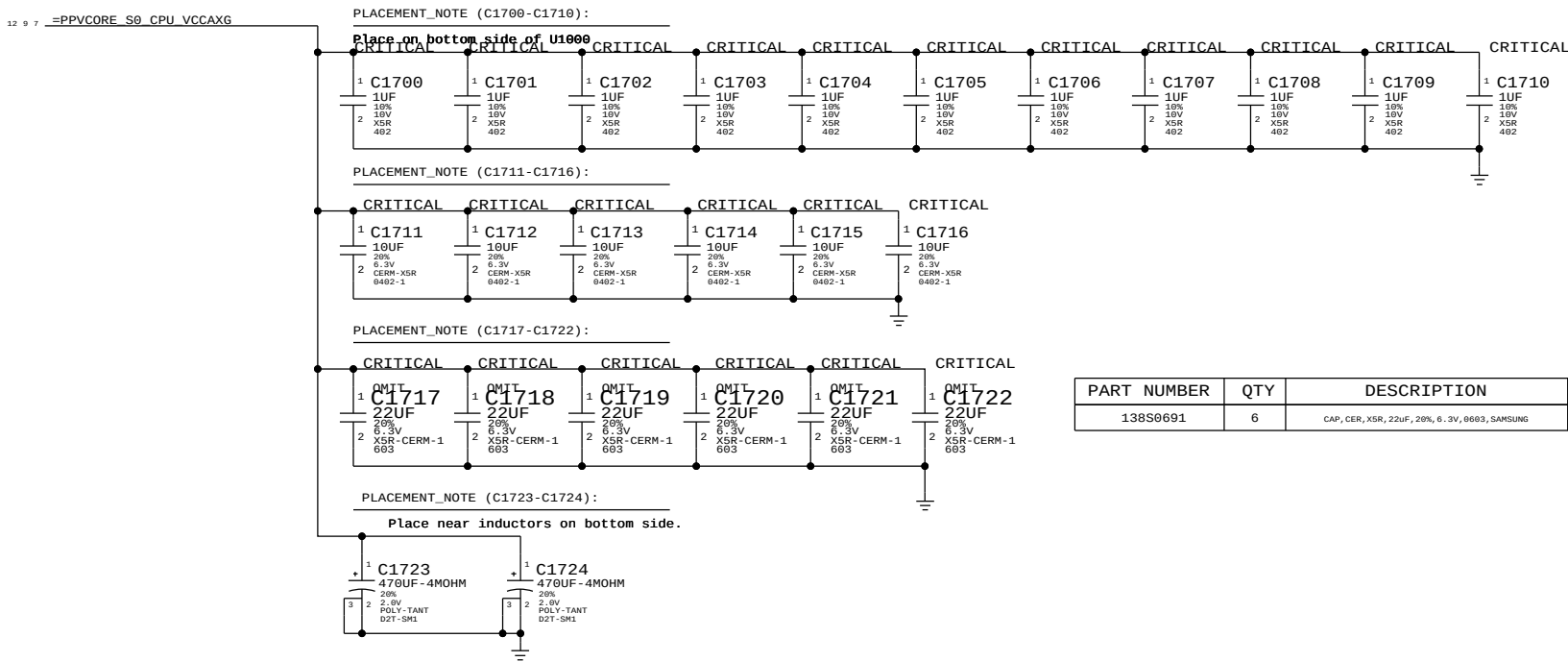


C1655, C1660, C1661, C1664, C1666, C1667, C1670, C1677, C1678, C1679, C1657, C1672, C1658, C1669, C1668, C1656

SYNC MASTER=JACK J30		SYNC DATE=09/27/2011	
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CPU DECOUPLING-I			
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		REVISION	
		6.0.0	
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		14	OF 86

VAXG DECOUPLING

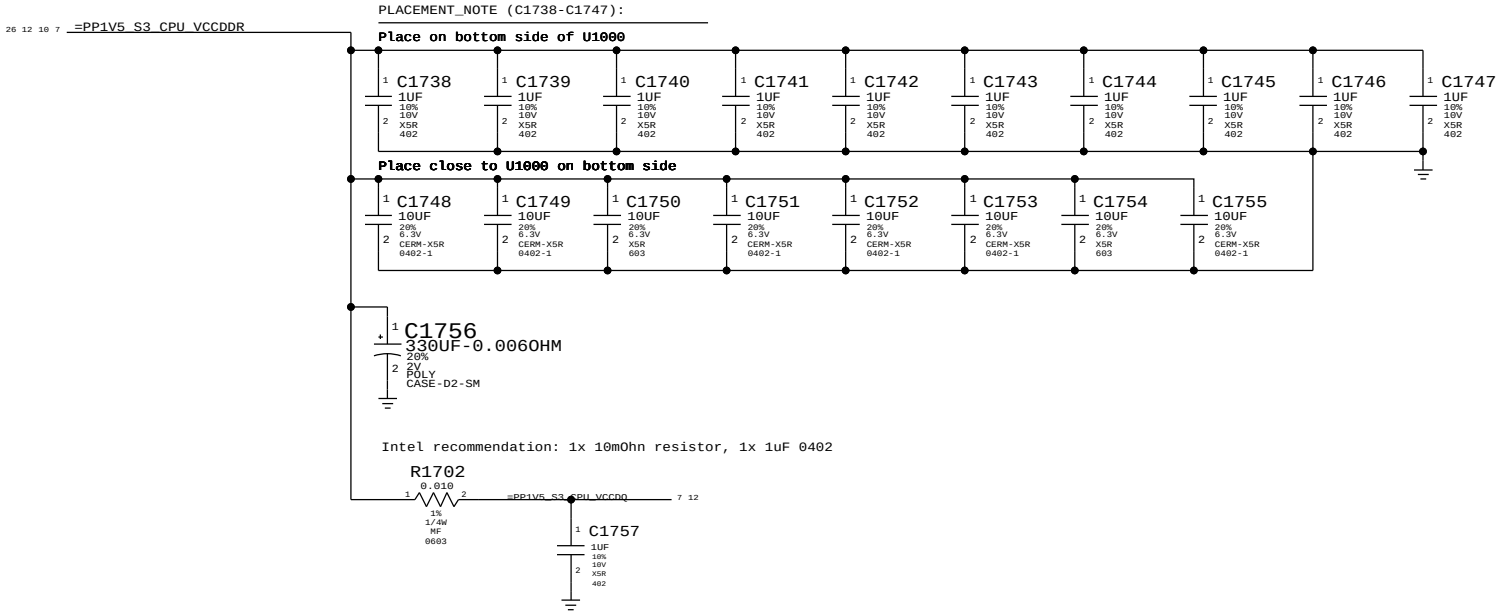
Intel recommendation (section 6.3): 21x 1uF, 6x 10uF, 6x 22uF, 2x 470uF



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0691	6	CAP, CER, XSR, 22uF, 20%, 6.3V, 0603, SAMSUNG	C1717, C1718, C1719, C1720, C1721, C1722	CRITICAL	

CPU VDDQ/VCCDQ DECOUPLING

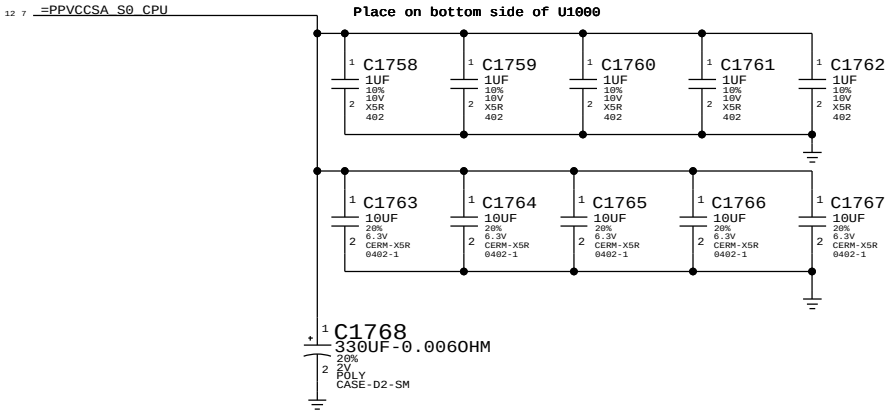
Intel recommendation (Section 6.5): 10x 1uF, 8x 10uF, 1x 330uF

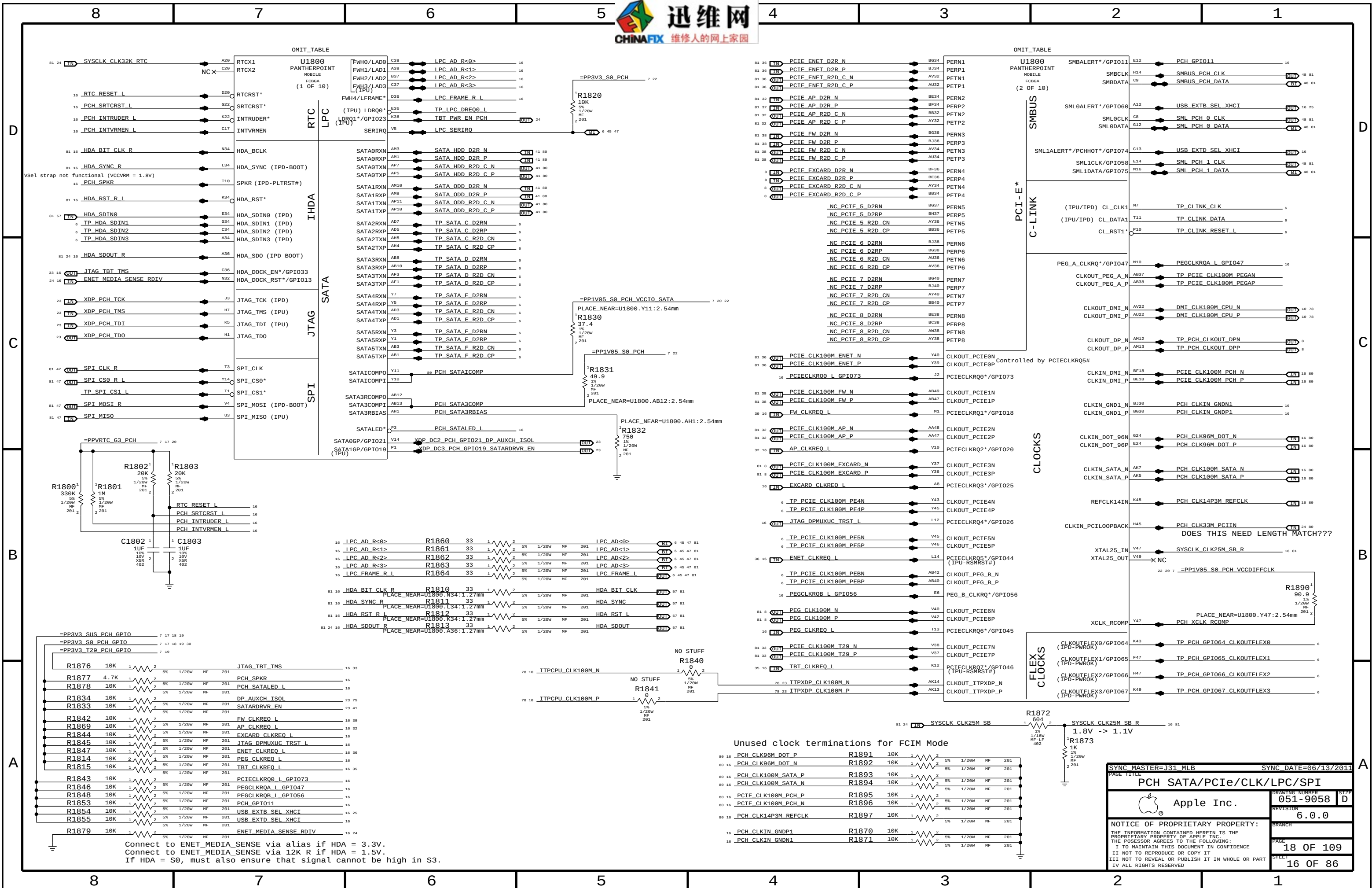


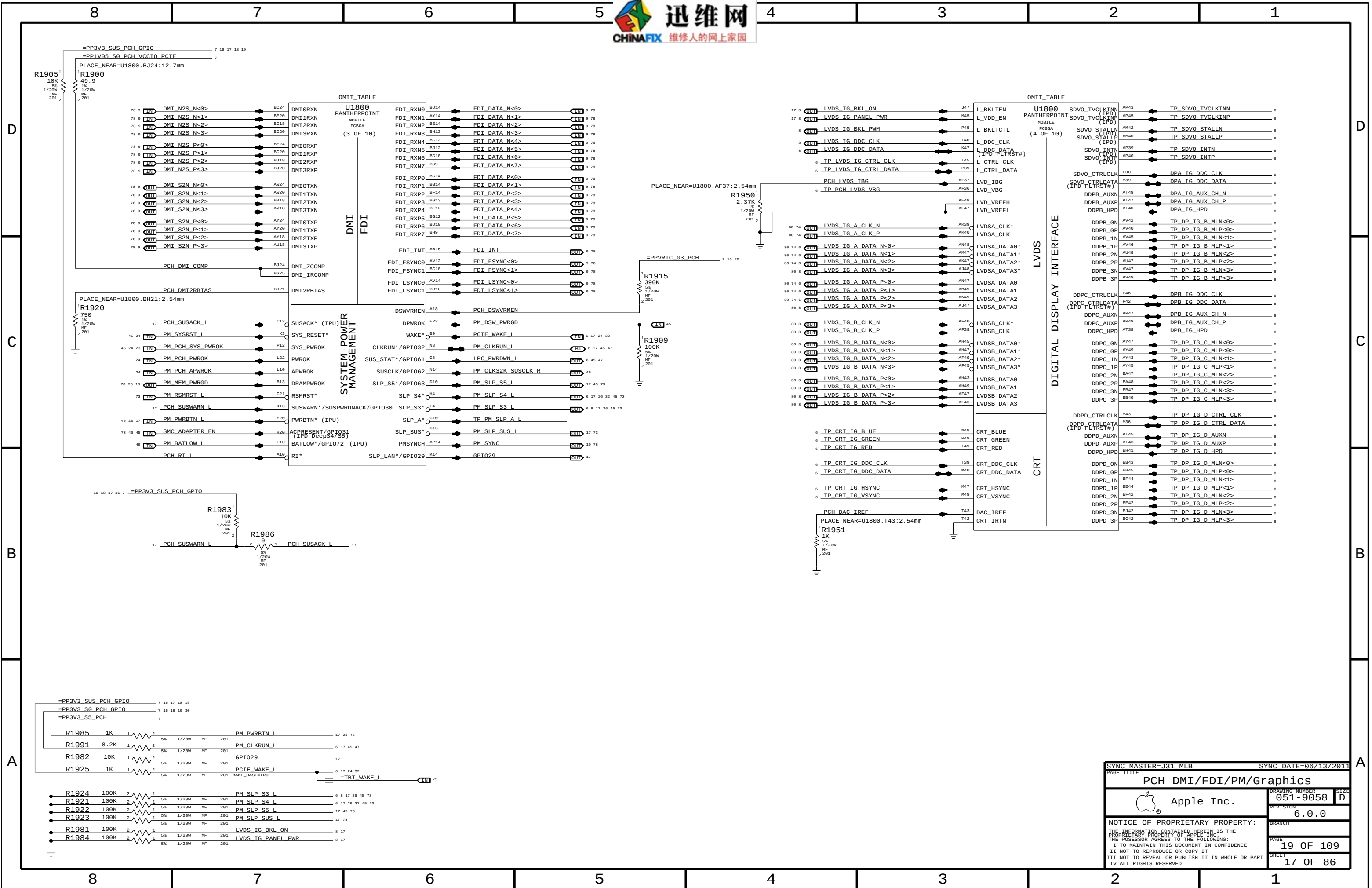
CPU VCCSA DECOUPLING

Intel recommendation (Section 6.6): 6x 1uf, 5x 10uf, 1x 330uf

PLACEMENT_NOTE (C1758-C1762):







U1800
PANTHERPOINT
MOBILE
FCBGA
(5 OF 10)

NC X BG26 TP1
NC X BJ26 TP2
NC X BH25 TP3
NC X BJ16 TP4
NC X BG16 TP5
NC X AH38 TP6
NC X AH37 TP7
NC X AK43 TP8
NC X AK45 TP9
NC X C18 TP10
NC X N30 TP11
NC X H3 TP12
NC X AH12 TP13
NC X AM4 TP14
NC X AM5 TP15
NC X Y13 TP16
NC X K24 TP17
NC X L24 TP18
NC X AB46 TP19
NC X AB45 TP20
NC X B21 TP21
NC X M20 TP22
NC X AY16 TP23
NC X BG46 TP24

RSVD1 AY7 X NC
RSVD2 AV7 X NC
RSVD3 AU3 X NC
RSVD4 BG4 X NC
RSVD5 AT10 X NC
RSVD6 BC8 X NC
RSVD7 AU2 X NC
RSVD8 AT4 X NC
RSVD9 AT3 X NC
RSVD10 AT1 X NC
RSVD11 AY3 X NC
RSVD12 AT5 X NC
RSVD13 AV3 X NC
RSVD14 AV1 X NC
RSVD15 BB1 X NC
RSVD16 BA3 X NC
RSVD17 BB5 X NC
RSVD18 BB3 X NC
RSVD19 BE7 X NC
RSVD20 BE8 X NC
RSVD21 BD4 X NC
RSVD22 BF6 X NC
RSVD23 AV5 X NC
RSVD24 AV10 X NC
RSVD25 AT8 X NC
RSVD26 AY5 X NC
RSVD27 BA2 X NC
RSVD28 AT12 X NC
RSVD29 BF3 X NC

80 42 IN USB3_EXTB_RX_N BE28
80 41 IN USB3_EXTB_RX_N BC30
80 40 IN USB3_EXTB_RX_N BE32
80 39 IN USB3_EXTD_RX_N BJ32
80 42 IN USB3_EXTB_RX_P BC28
80 41 IN USB3_EXTB_RX_P BE30
80 40 IN USB3_EXTC_RX_P BF32
80 39 IN USB3_EXTD_RX_P BG32
80 42 OUT USB3_EXTB_TX_N AV26
80 41 OUT USB3_EXTB_TX_N BB26
80 40 OUT USB3_EXTC_TX_N AU28
80 39 OUT USB3_EXTD_TX_N AY30
80 42 OUT USB3_EXTB_TX_P AU26
80 41 OUT USB3_EXTB_TX_P AY26
80 40 OUT USB3_EXTC_TX_P AV28
80 39 OUT USB3_EXTD_TX_P AH38

USB3RN1
USB3RN2
USB3RN3
USB3RN4
USB3RP1
USB3RP2
USB3RP3
USB3RP4
USB3TN1
USB3TN2
USB3TN3
USB3TN4
USB3TP1
USB3TP2
USB3TP3
USB3TP4

USB

USBP0N C24 USB_EXTB_N 80 42 80
USBP0P A24 USB_EXTB_P 80 42 80
USBP1N C25 USB_EXTB_XHCI_N 80 25 80
USBP1P B25 USB_EXTB_XHCI_P 80 25 80
USBP2N C26 USB_EXTB_N 80 8 80
USBP2P A26 USB_EXTB_P 80 8 80
USBP3N K28 USB_EXTD_XHCI_N 80 8 80
USBP3P H28 USB_EXTD_XHCI_P 80 8 80
USBP4N E28 TP_USB_4N
USBP4P D28 TP_USB_4P
USBP5N C28 TP_USB_SDN
USBP5P A28 TP_USB_SDP
USBP6N C29 TP_USB_WLANN
USBP6P B29 TP_USB_WLANP
USBP7N N28 USB_HUB_UP_N 80 25 80
USBP7P M28 USB_HUB_UP_P 80 25 80
USBP8N L30 USB_CAMERA_N 80 32 80
USBP8P K30 USB_CAMERA_P 80 32 80
USBP9N G30 USB_EXTB_EHCI_N 80 25 80
USBP9P E30 USB_EXTB_EHCI_P 80 25 80
USBP10N C30 USB_EXTD_EHCI_N 80 8
USBP10P A30 USB_EXTD_EHCI_P 80 8
USBP11N L32 TP_USB_BT_HSN
USBP11P K32 TP_USB_BT_HSP
USBP12N G32 TP_USB_12N
USBP12P E32 TP_USB_12P
USBP13N C32 TP_USB_13N
USBP13P A32 TP_USB_13P
USBP13P (IPD)
USBRBIAS* C33 PCH_USB_RBIAS
USBRBIAS B33

Ext A (XHCI/EHCI)
Ext B (XHCI)
Ext C (XHCI/EHCI)
Ext D (XHCI) (Mobiles: Trackpad?)
Unused
RSVD: SD
RSVD: WiFi
USB Hub (All LS/FS Devices)
Camera
Ext B (EHCI)
Ext D (EHCI)
RSVD: BT (HS)
Unused
Unused

30 19 18 17 16 7 =PP3V3_S0_PCH_GPIO
R2010 10K 1 2 5% 1/20W MF 201
R2011 10K 1 2 5% 1/20W MF 201
R2012 10K 1 2 5% 1/20W MF 201
R2013 10K 1 2 5% 1/20W MF 201

PCI INTA L K40
PCI INTB L K38
PCI INTC L H30
PCI INTD L G38
PIRQA*
PIRQB*
PIRQC*
PIRQD*

18 OUT JTAG_GMUX_TMS C46
18 OUT BLC_I2C_MUX_SEL C44
18 OUT USE_HDD_00B_L E40
TP_PCH_STRP_BBS1 D47
TP_PCH_STRP_ESI_L E42
PCH_STRP_TOPBLK_SWP_L F46

18 IN BLC_GPIO G42
62 18 IN AUD_IP_PERIPHERAL_DET G40
75 18 IN TBT_PWR_REQ_L C42
62 18 IN AUD_I2C_INT_L D44
6 TP_PCI_PME_L K10
26 24 OUT PLT_RESET_L C6
81 24 OUT LPC_CLK33M_SMC_R H49
24 OUT LPC_CLK33M_LPCPLUS_R H43
6 TP_PCI_CLK33M_OUT2 J48
6 TP_PCI_CLK33M_OUT3 K42
24 OUT PCH_CLK33M_PCIOUT H40

PCI

REQ1*/GPIO50
REQ2*/GPIO52
REQ3*/GPIO54
GNT1*/GPIO51
GNT2*/GPIO53
GNT3*/GPIO55
(IPU-PCIERST#)
PIRQE*/GPIO2
PIRQF*/GPIO3
PIRQG*/GPIO4
PIRQH*/GPIO5
PME* (IPU)
PLTRST*
CLKOUT_PCI0
CLKOUT_PCI1
CLKOUT_PCI2
CLKOUT_PCI3
CLKOUT_PCI4 (IPD)

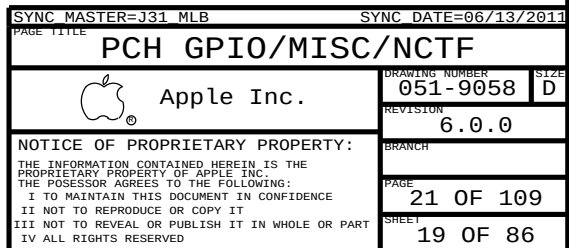
R2054 10K NO STUFF 2 1 5% 1/20W MF 201

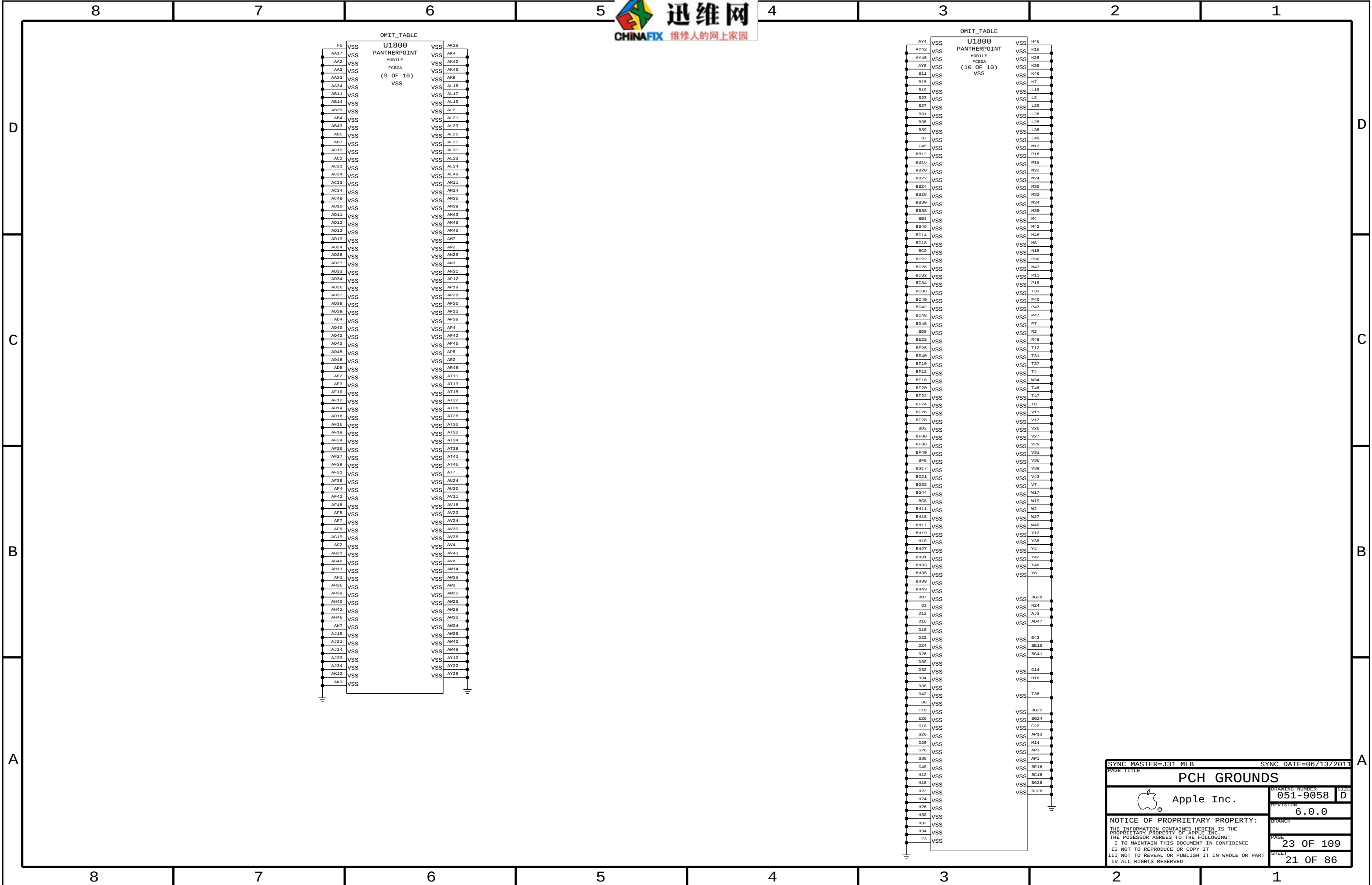
=PP3V3_SUS_PCH_GPIO 7 16 17 19
=PP3V3_S3_PCH_GPIO 7 24
=PP3V3_S0_PCH_GPIO 7 16 17 18 19 30
R2016 10K 1 2 5% 1/20W MF 201
R2017 10K 1 2 5% 1/20W MF 201
R2018 10K 1 2 5% 1/20W MF 201
R2030 10K 1 2 5% 1/20W MF 201
R2014 10K NO STUFF 1 2 5% 1/20W MF 201
R2031 10K 1 2 5% 1/20W MF 201
R2033 10K NO STUFF 1 2 5% 1/20W MF 201
R2069 10K 1 2 5% 1/20W MF 201
R2060 10K 1 2 5% 1/20W MF 201
R2061 10K 1 2 5% 1/20W MF 201
R2062 10K 1 2 5% 1/20W MF 201
R2068 10K 1 2 5% 1/20W MF 201
R2067 10K 2 1 5% 1/20W MF 201

18 OUT JTAG_GMUX_TMS
18 OUT BLC_I2C_MUX_SEL
18 OUT USE_HDD_00B_L
18 OUT BLC_GPIO
18 62 AUD_IP_PERIPHERAL_DET
18 75 TBT_PWR_REQ_L
18 62 AUD_I2C_INT_L
18 6 TP_PCI_PME_L
18 26 24 PLT_RESET_L
18 81 24 LPC_CLK33M_SMC_R
18 24 LPC_CLK33M_LPCPLUS_R
18 6 TP_PCI_CLK33M_OUT2
18 6 TP_PCI_CLK33M_OUT3
18 24 PCH_CLK33M_PCIOUT

XDP_DA0_PCH_GPIO59_USB_EXTB_OC_L 18 23
XDP_DA1_PCH_GPIO40_USB_EXTB_OC_L 18 23
XDP_DA2_PCH_GPIO41_USB_EXTB_OC_L 18 23
XDP_DA3_PCH_GPIO42_USB_EXTD_OC_L 18 23
XDP_DB0_PCH_GPIO43_USB_EXTB_OC_EHCI_L 18 23
XDP_DB1_PCH_GPIO09_USB_EXTD_OC_EHCI_L 23
XDP_DB2_PCH_GPIO10_AP_PWR_EN 23
XDP_DB3_PCH_GPIO14_SDCONN_STATE_CHANGE 18 23

PLACE_NEAR=U1800.B33:2.54mm
R2070 22.6 1% 1/20W MF 201





SYNC MASTER=J31 MLB

SYNC DATE=06/13/2011

PAGE TITLE

PCH GROUNDS

 Apple Inc.

DRAWING NUMBER
051-9058

REVISION
6.0.0

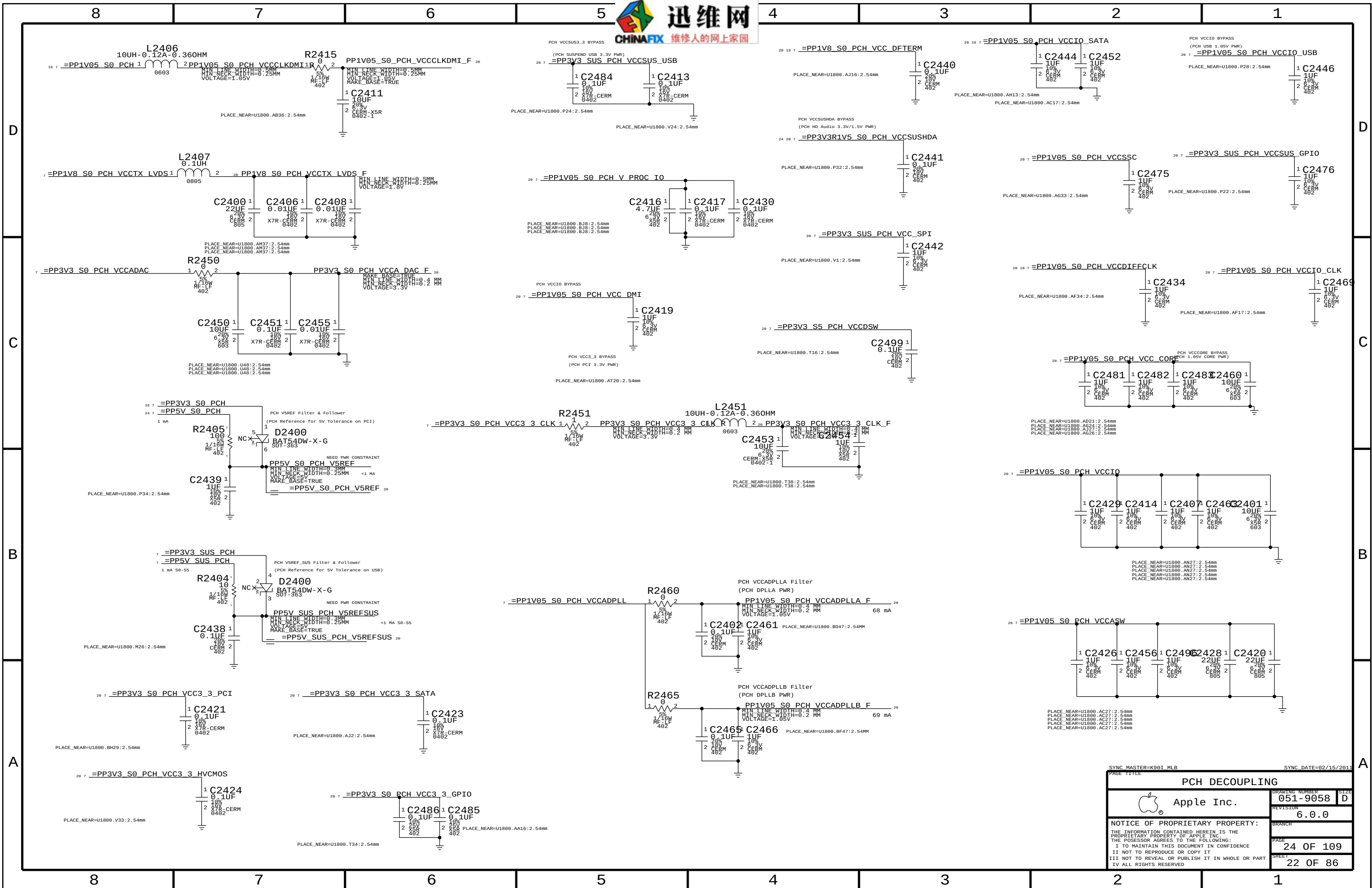
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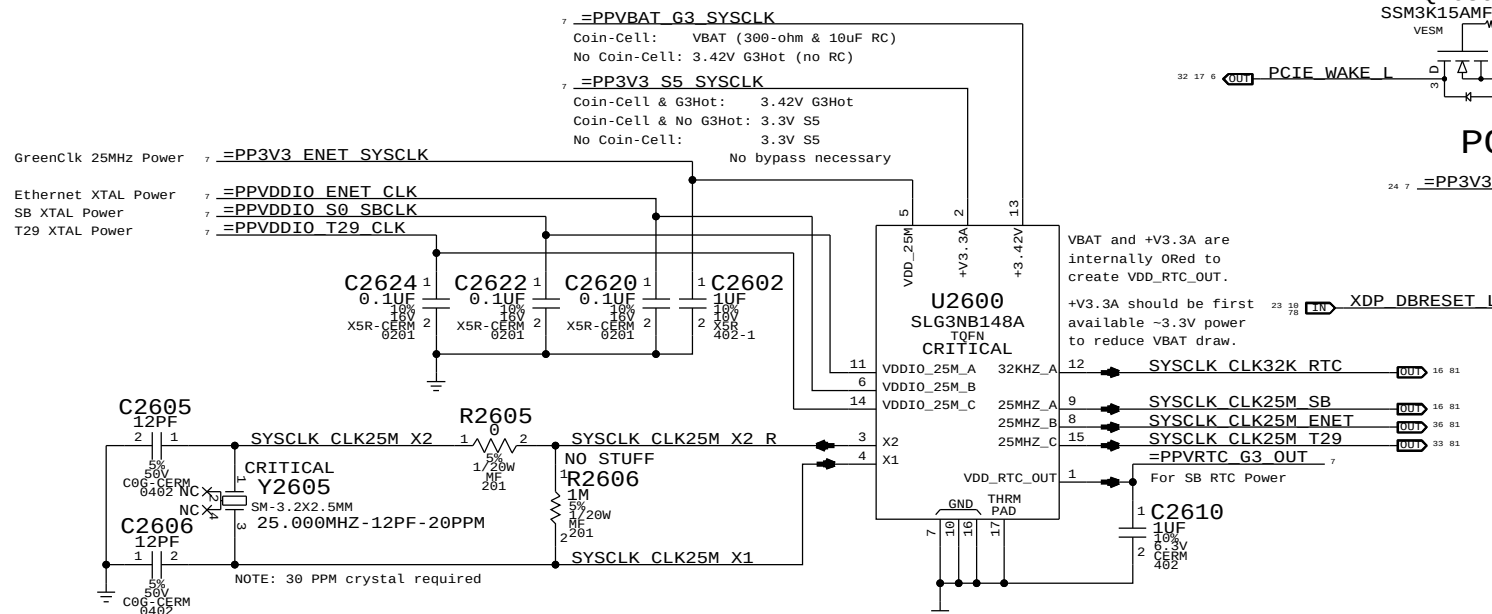
PAGE
23 OF 109

SHEET

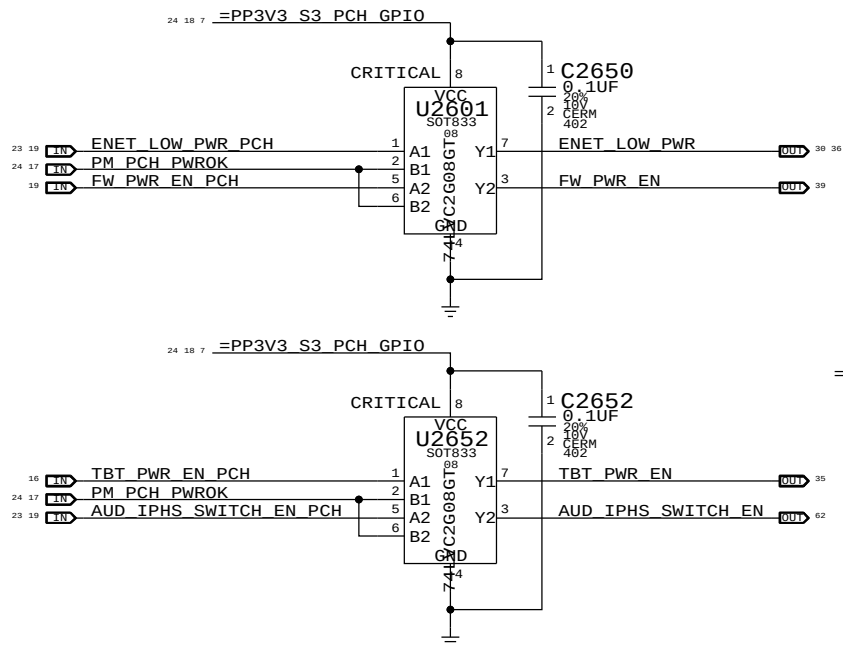
21 OF 86



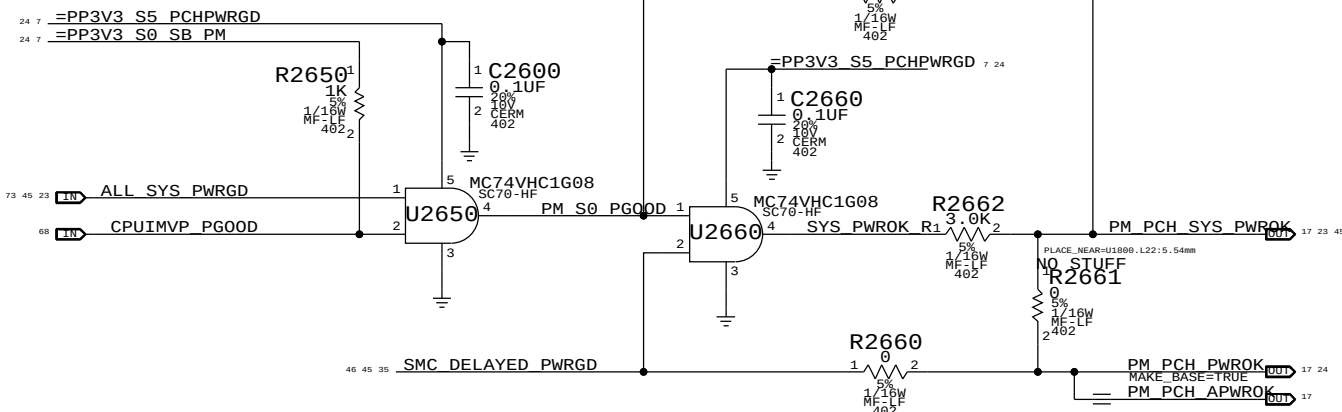
System RTC Power Source & 32kHz / 25MHz Clock Generator



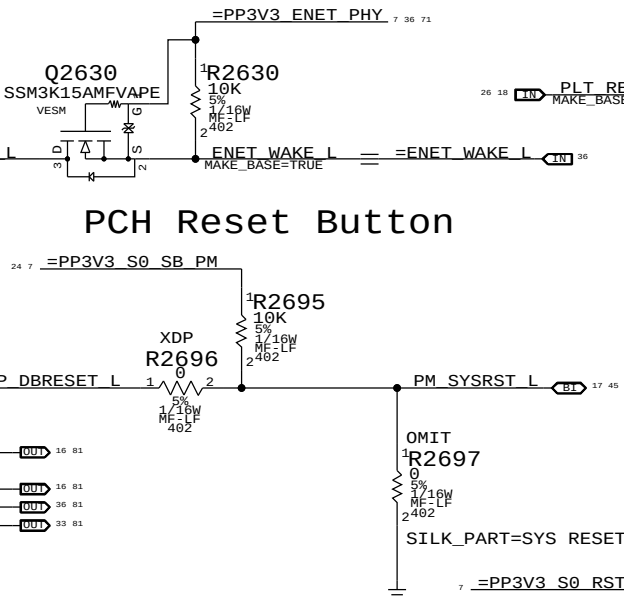
GPIO Glitch Prevention



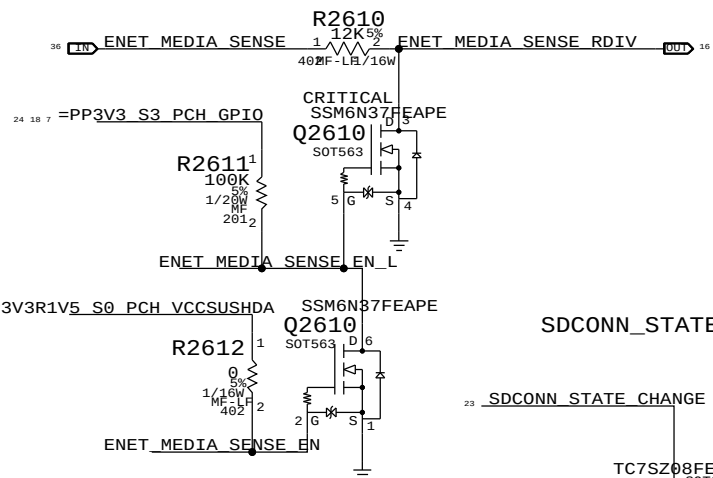
PCH S0 PWRGD



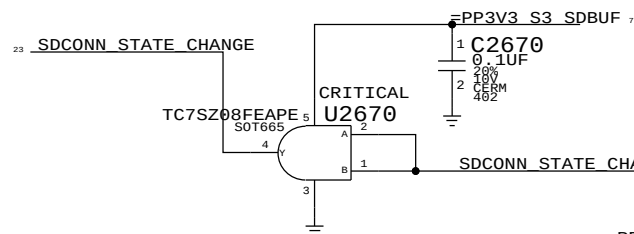
PCH Reset Button



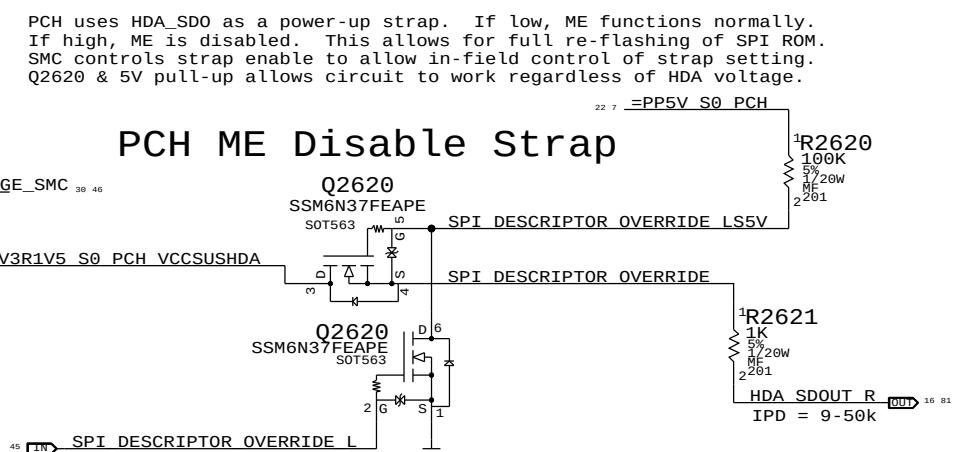
ENET_MEDIA_SENSE ISOLATION CIRCUIT



SDCONN_STATE_CHANGE ISOLATION



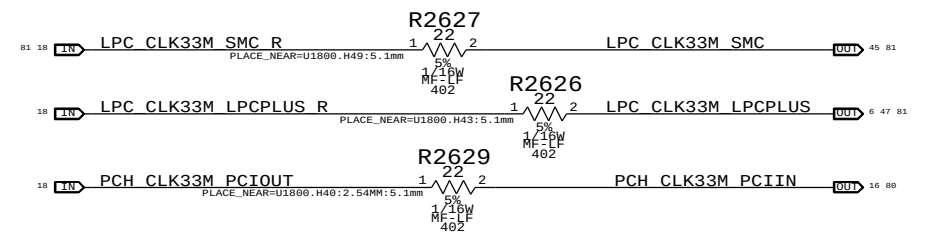
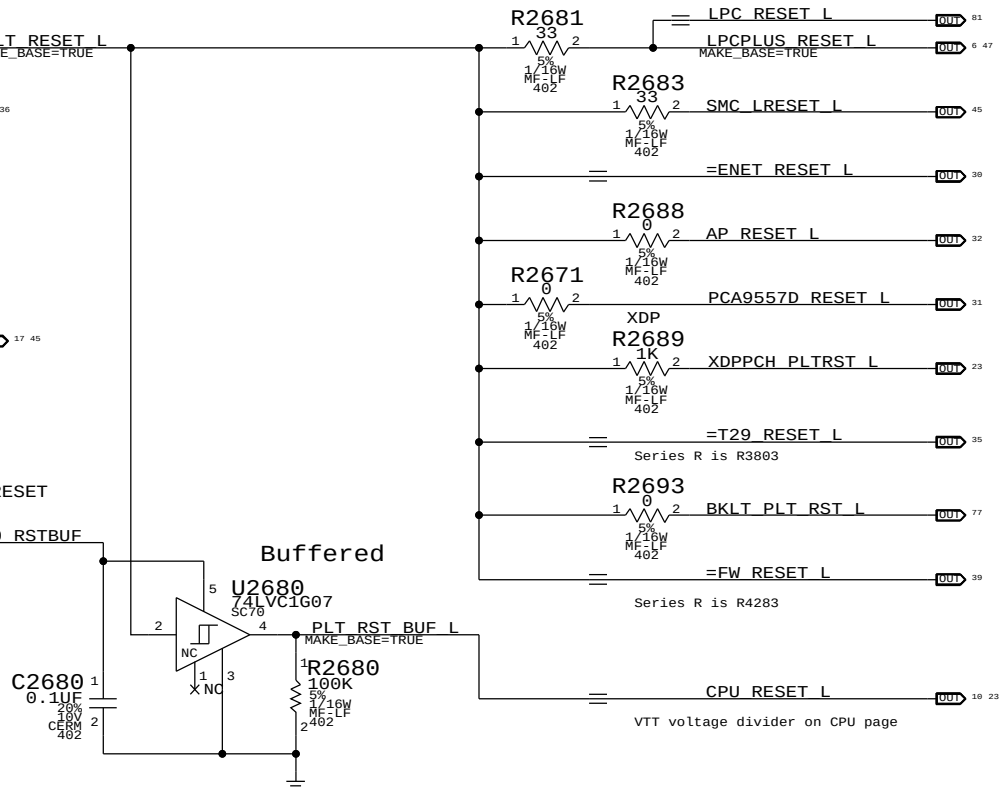
PCH ME Disable Strap



PCH uses HDA_SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. Q2620 & 5V pull-up allows circuit to work regardless of HDA voltage.

Platform Reset Connections

Unbuffered



SYNC MASTER=K901 MLB		SYNC DATE=02/15/2011	
PAGE TITLE		Chipset Support	
Apple Inc.		DRAWING NUMBER	051-9058
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USB MUX FOR LS/FS INTERNAL DEVICES

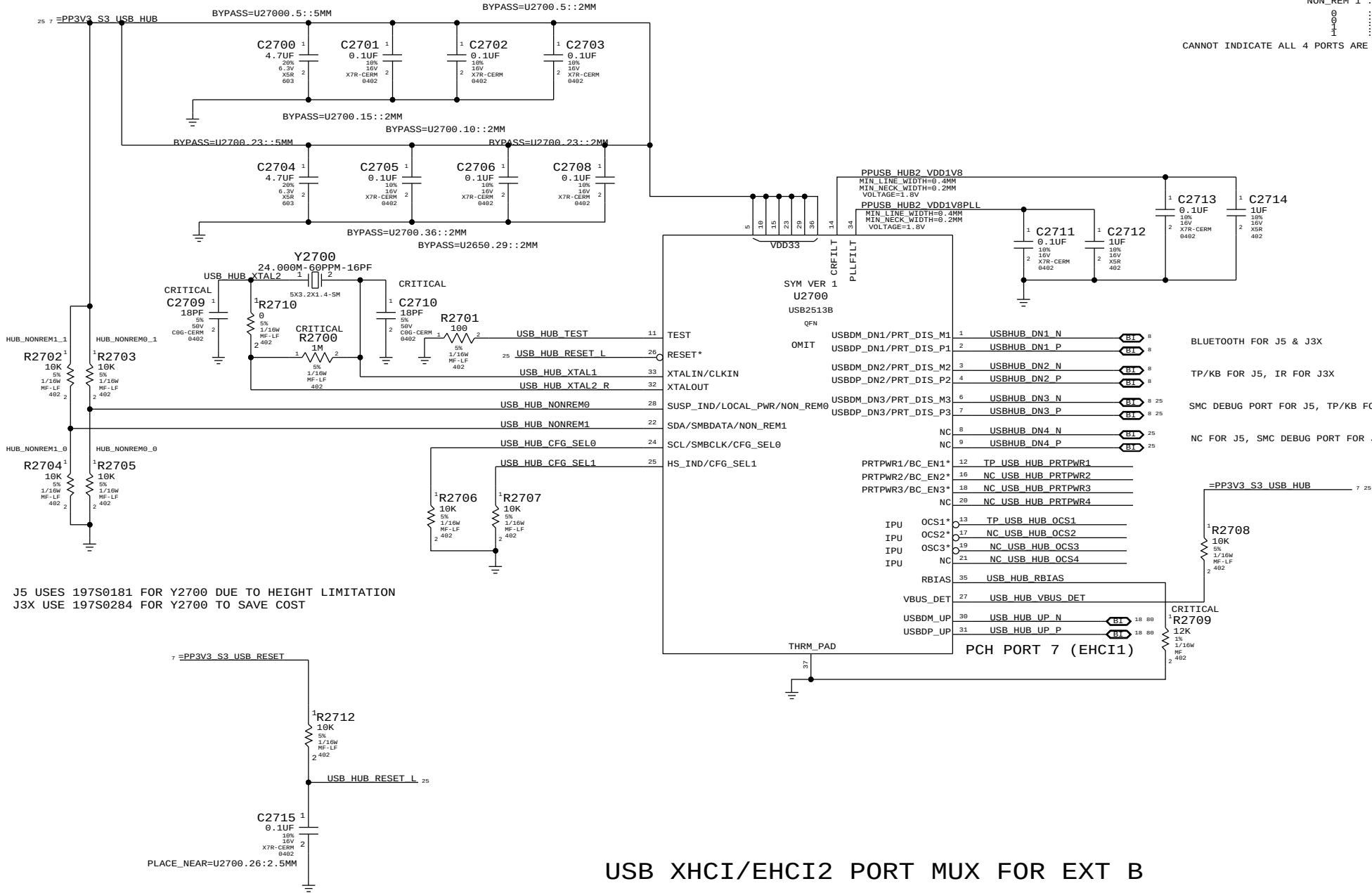
BOM GROUP	BOM OPTIONS
HUB_ALLREM	HUB_NONREM1_0, HUB_NONREM0_0
HUB_1NONREM	HUB_NONREM1_0, HUB_NONREM0_1
HUB_2NONREM	HUB_NONREM1_1, HUB_NONREM0_0
HUB_3NONREM	HUB_NONREM1_1, HUB_NONREM0_1

NON_REM 1 : NON_REM 0
0 0
1 1
STRAP PIN CFG
ALL PORTS ARE REMOVABLE
PORT 1&2 ARE NON REMOVABLE
PORT 1&2&3 ARE NON REMOVABLE
CANNOT INDICATE ALL 4 PORTS ARE NON REMOVABLE ON USB2514B VIA STARPPING, PROGRAM NON_REMOVABLE DEVICE REGISTER 09H

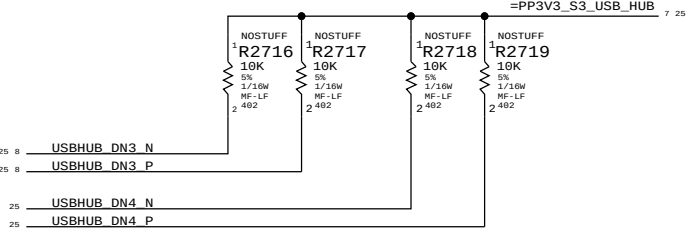
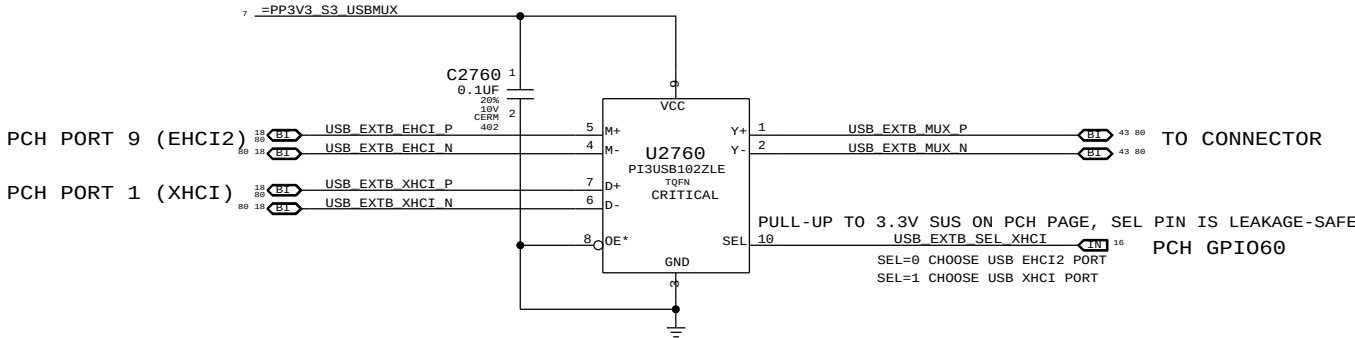
BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S9824	1	USB HUB 2514B	U2700	CRITICAL	USBHUB2514B
338S9923	1	USB HUB 2513B	U2700	CRITICAL	USBHUB2513B
338S9983	1	USB HUB 2512B	U2700	CRITICAL	USBHUB2512B

J5 ENGINEERING: USE USB2513B PRODUCTION: USE USB2512B
J3X ENGINEERING: USE USB2514B PRODUCTION: USE USB2513B



USB XHCI/EHCI2 PORT MUX FOR EXT B



SYNC MASTER=LINDA J30		SYNC DATE=09/19/2011	
PAGE TITLE		USB HUB & MUX	
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The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.

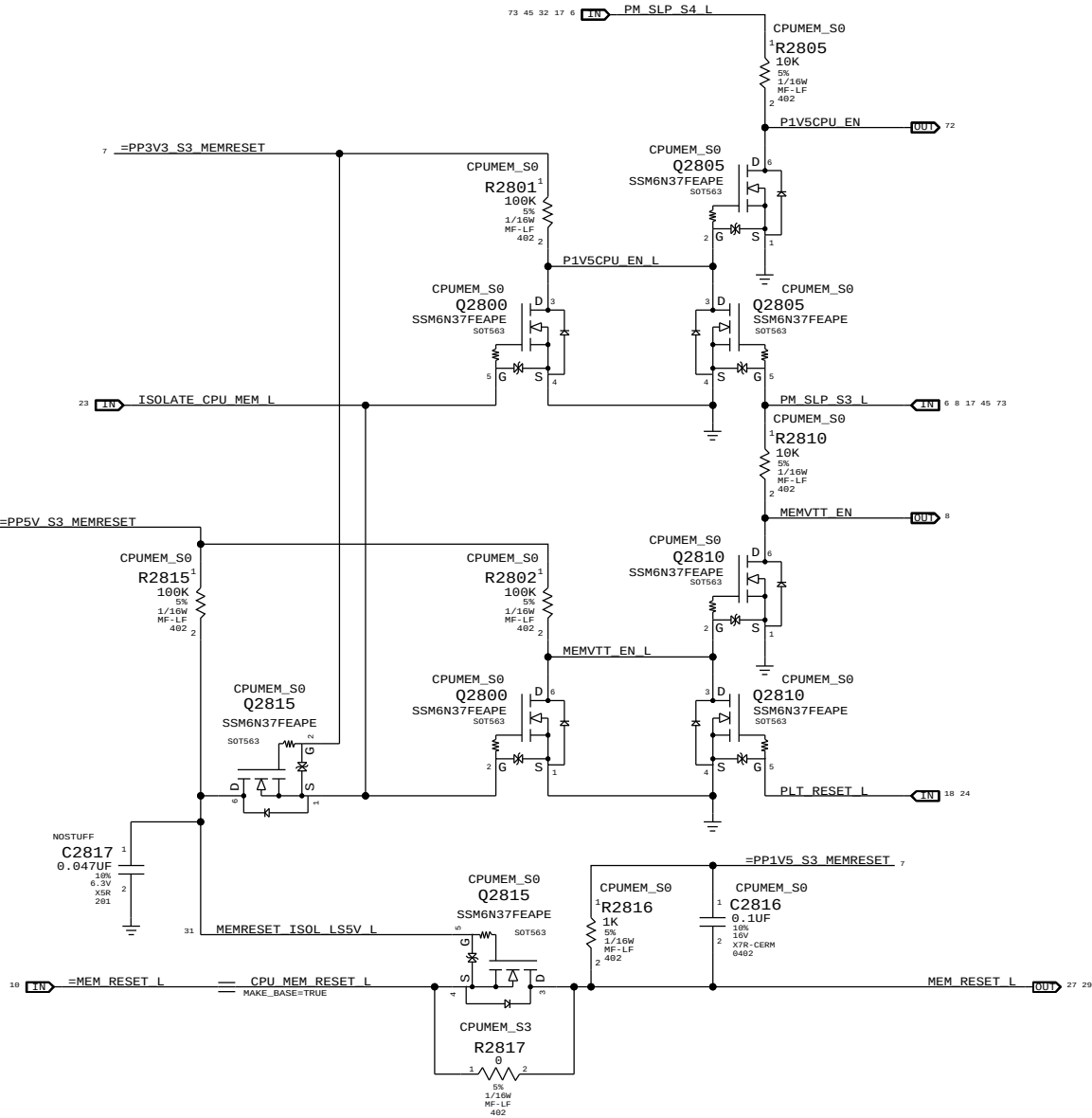
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

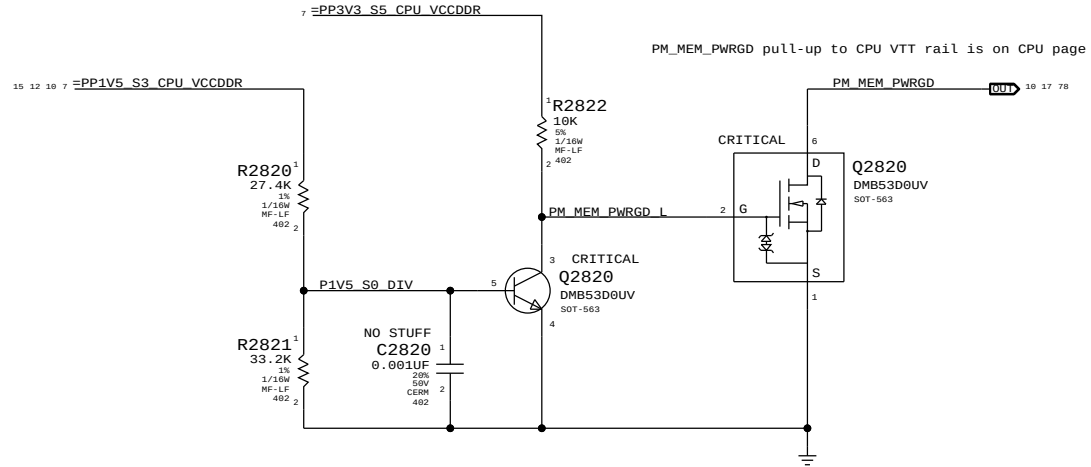
$P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L$

$MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L$

$MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L$

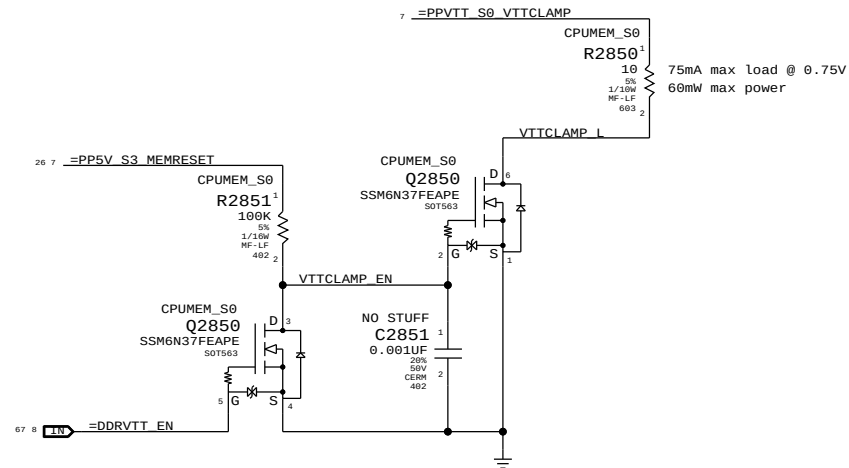


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
1	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=K90T MLB		SYNC DATE=02/15/2011	
PAGE TITLE		CPU Memory S3 Support	
Apple Inc.		DRAWING NUMBER	051-9058
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ases required by this page:

Power aliases required by this page:

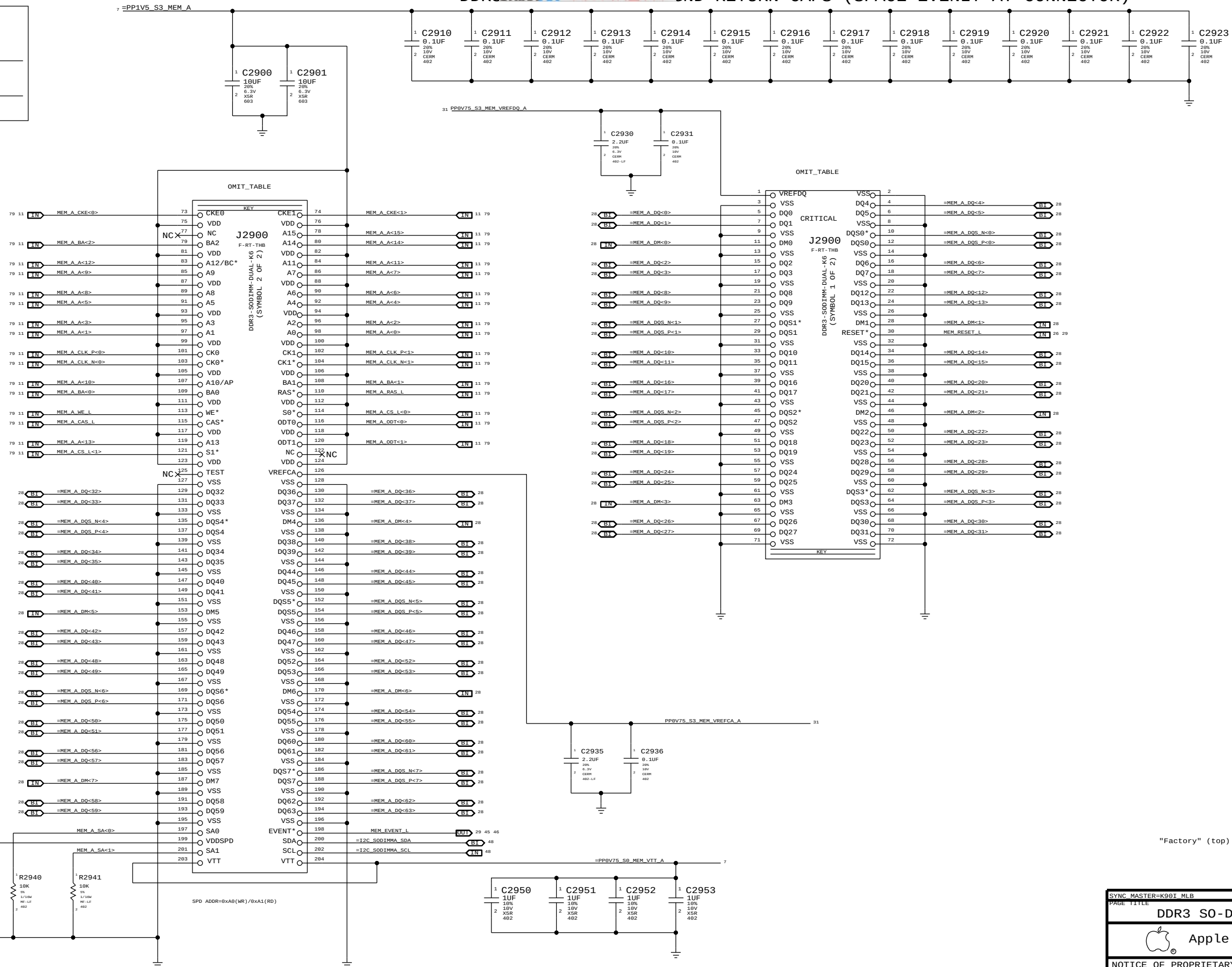
- #PPIV5_S0_MEM_A
- #PPIV5_S3_MEM_A
- #PGV75_S0_MEM_VTT_A
- #PPSP0_S0_MEM_A (2.5 - 3.3v)

Signal aliases required by this page

- #I2C_S00IMMA_SCL
- #I2C_S00IMMA_SDA

BCM functions provided by this page:

(NONE)



"Factory" (top) slot

SYMC MASTER-K901 MLB		SYMC DATE=02/15/2011	
PAGE TITLE			
DDR3 SO-DIMM Connector A			
 Apple Inc.		DRAWING NUMBER 051-9058	
		SIZE D	
		REVISION 6.0.0	
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		SHEET 27 OF 86	

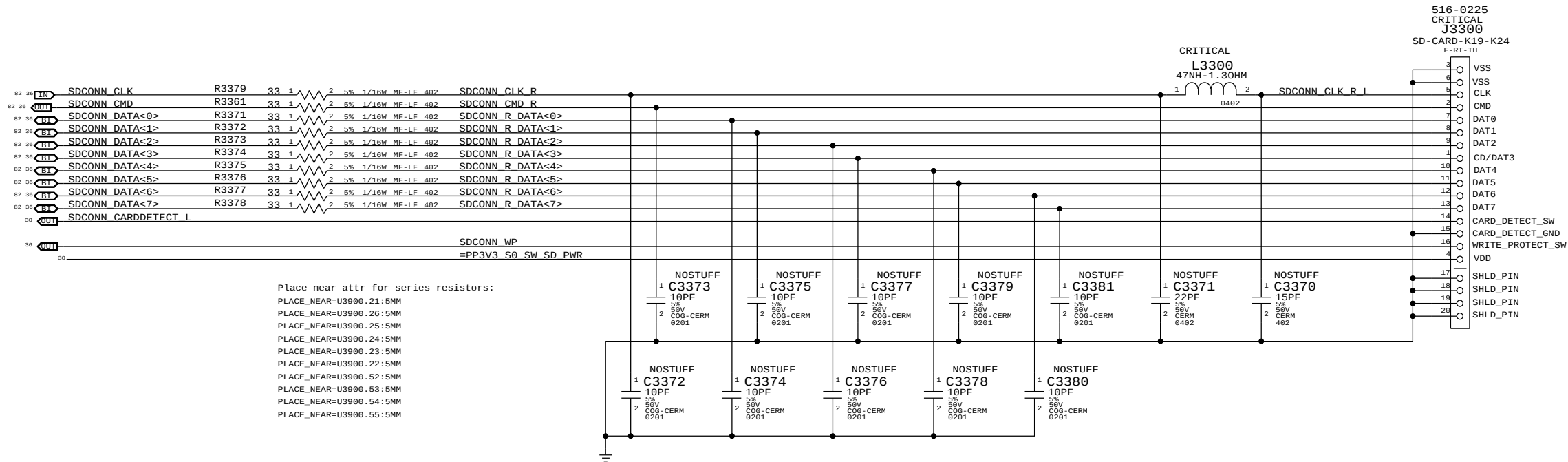
8	7	6	5
CPU CHANNEL A DQS 0 -> DIMM A DQS 0			
79 11 MEM_A_DQS_N<0>	==MEM_A_DQS_N<0>	27	
79 11 MEM_A_DQS_P<0>	==MEM_A_DQS_P<0>	27	
CPU CHANNEL A DQS 1 -> DIMM A DQS 1			
79 11 MEM_A_DQS_N<1>	==MEM_A_DQS_N<1>	27	
79 11 MEM_A_DQS_P<1>	==MEM_A_DQS_P<1>	27	
CPU CHANNEL A DQS 2 -> DIMM A DQS 2			
79 11 MEM_A_DQS_N<2>	==MEM_A_DQS_N<2>	27	
79 11 MEM_A_DQS_P<2>	==MEM_A_DQS_P<2>	27	
CPU CHANNEL A DQS 3 -> DIMM A DQS 3			
79 11 MEM_A_DQS_N<3>	==MEM_A_DQS_N<3>	27	
79 11 MEM_A_DQS_P<3>	==MEM_A_DQS_P<3>	27	
CPU CHANNEL A DQS 4 -> DIMM A DQS 4			
79 11 MEM_A_DQS_N<4>	==MEM_A_DQS_N<4>	27	
79 11 MEM_A_DQS_P<4>	==MEM_A_DQS_P<4>	27	
CPU CHANNEL A DQS 5 -> DIMM A DQS 5			
79 11 MEM_A_DQS_N<5>	==MEM_A_DQS_N<5>	27	
79 11 MEM_A_DQS_P<5>	==MEM_A_DQS_P<5>	27	
CPU CHANNEL A DQS 6 -> DIMM A DQS 6			
79 11 MEM_A_DQS_N<6>	==MEM_A_DQS_N<6>	27	
79 11 MEM_A_DQS_P<6>	==MEM_A_DQS_P<6>	27	
CPU CHANNEL A DQS 7 -> DIMM A DQS 7			
79 11 MEM_A_DQS_N<7>	==MEM_A_DQS_N<7>	27	
79 11 MEM_A_DQS_P<7>	==MEM_A_DQS_P<7>	27	

6	5	4	3	2	1
CPU CHANNEL B DQS 0 -> DIMM B DQS 0					
79 11 MEM_B_DQS_N<0>	==MEM_B_DQS_N<0>	29			
79 11 MEM_B_DQS_P<0>	==MEM_B_DQS_P<0>	29			
CPU CHANNEL B DQS 1 -> DIMM B DQS 1					
79 11 MEM_B_DQS_N<1>	==MEM_B_DQS_N<1>	29			
79 11 MEM_B_DQS_P<1>	==MEM_B_DQS_P<1>	29			
CPU CHANNEL B DQS 2 -> DIMM B DQS 2					
79 11 MEM_B_DQS_N<2>	==MEM_B_DQS_N<2>	29			
79 11 MEM_B_DQS_P<2>	==MEM_B_DQS_P<2>	29			
CPU CHANNEL B DQS 3 -> DIMM B DQS 3					
79 11 MEM_B_DQS_N<3>	==MEM_B_DQS_N<3>	29			
79 11 MEM_B_DQS_P<3>	==MEM_B_DQS_P<3>	29			
CPU CHANNEL B DQS 4 -> DIMM B DQS 4					
79 11 MEM_B_DQS_N<4>	==MEM_B_DQS_N<4>	29			
79 11 MEM_B_DQS_P<4>	==MEM_B_DQS_P<4>	29			
CPU CHANNEL B DQS 5 -> DIMM B DQS 5					
79 11 MEM_B_DQS_N<5>	==MEM_B_DQS_N<5>	29			
79 11 MEM_B_DQS_P<5>	==MEM_B_DQS_P<5>	29			
CPU CHANNEL B DQS 6 -> DIMM B DQS 6					
79 11 MEM_B_DQS_N<6>	==MEM_B_DQS_N<6>	29			
79 11 MEM_B_DQS_P<6>	==MEM_B_DQS_P<6>	29			
CPU CHANNEL B DQS 7 -> DIMM B DQS 7					
79 11 MEM_B_DQS_N<7>	==MEM_B_DQS_N<7>	29			
79 11 MEM_B_DQS_P<7>	==MEM_B_DQS_P<7>	29			

NOTE: Ivybridge does not use DM signals per doc 460452 CR SFF DG Section 2.6.14

SYNC MASTER=K90T.MLB		SYNC DATE=02/15/2011	
PAGE TITLE		SIZE	
DDR3 Byte/Bit Swaps		DRAWING NUMBER	
Apple Inc.		051-9058	
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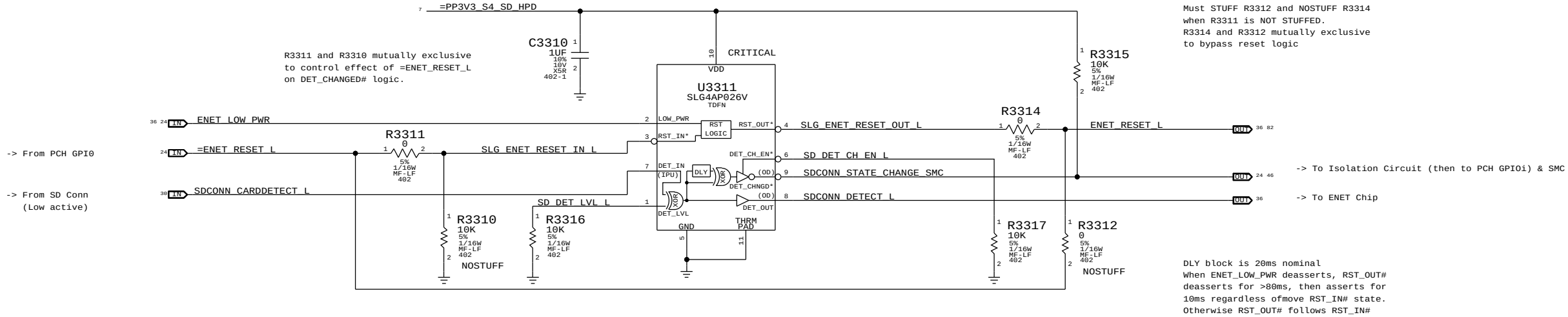
SD Card Connector



SD Not Inserted, CARD_DETECT is OPEN.
CAESAR-IV Card Detect is programmable,
but a Silicon bug makes the active
high case unusable.

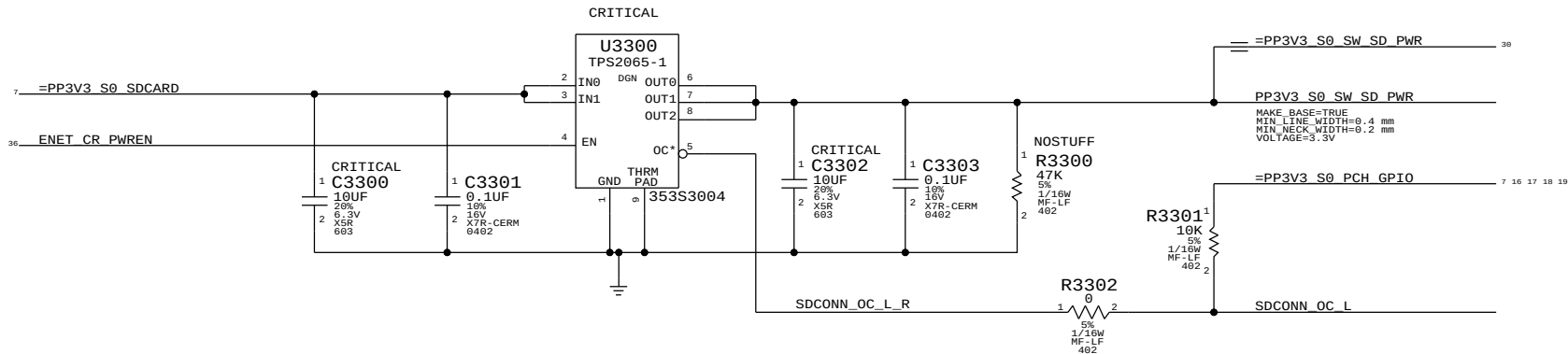
SD Detect & Reset Logic

SDCONN_DETECT Debounce, Inversion, Detect-Changed PCH GPIO Latch Circuit
Converts SDCONN from active-low level signal to active-high pulses.



SD Card 3.3V Overcurrent Protection

TPS2065-1 (1.0A limit) has active load discharge so R4810 is NOSTUFF.



SYNC MASTER=YONAS J30		SYNC DATE=11/03/2011	
PAGE TITLE		SD Card Connector	
Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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		PAGE	33 OF 109
		SHEET	30 OF 86

NOTE: Must not enable more than two S0-DIMM margining buffers at once or VRef source may be overloaded.

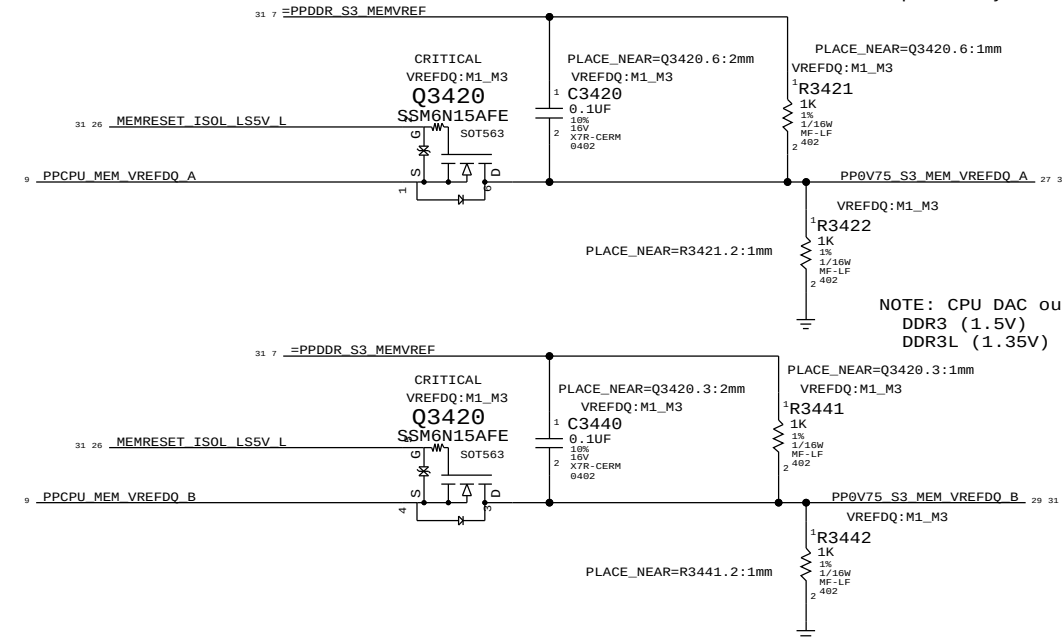
Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
DDRREF_DAC - Stuffs Apple margining circuit.
VREFDQ:LDO - LDO outputs sent to DQ inputs.
VREFDQ:LDO_DAC - Margined LDO outputs sent to DQ inputs.
VREFDQ:M1_M3 - CPU margined DDR voltage divider sent to DQ inputs.
VREFDQ:M1_DAC - DAC margined DDR voltage divider sent to DQ inputs.
VREFCA:LDO - LDO outputs sent to CA inputs.
VREFCA:LDO_DAC - DAC margined LDO outputs sent to CA inputs.

RST* on 'platform reset' so that system watchdog will disable margining.
NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

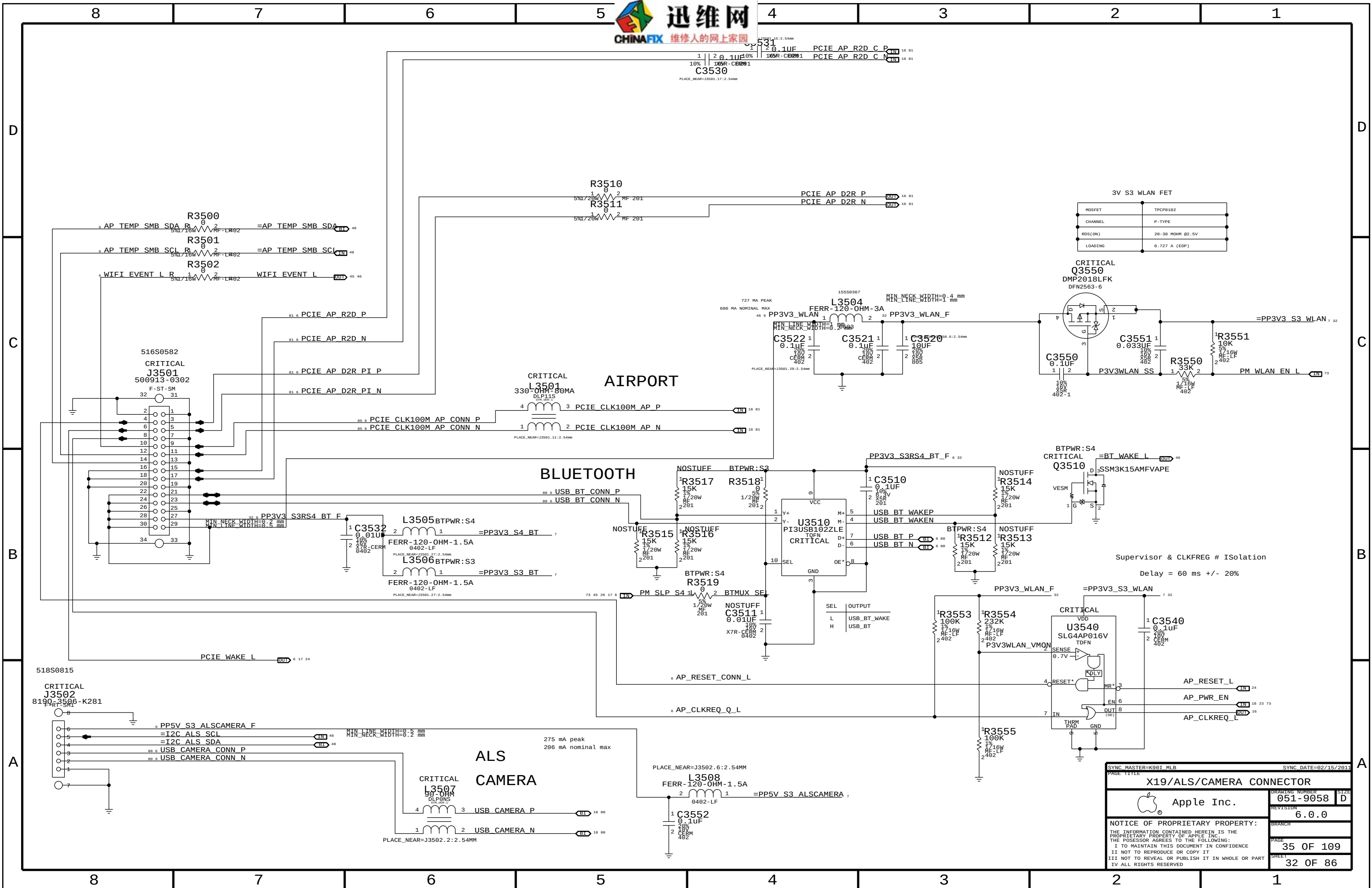


	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value	0.75V (DAC: 0x3A)					1.267V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)					1.056V - 1.442V (+/- 180mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)					0.000V - 3.300V (0x00 - 0xFF)
VRef current:	+3.4mA - -3.4mA (- = sourced)					+6.0mA - -5.0mA (- = sourced)
DAC step size:	7.69mV / step @ output					1.51mV / step @ output

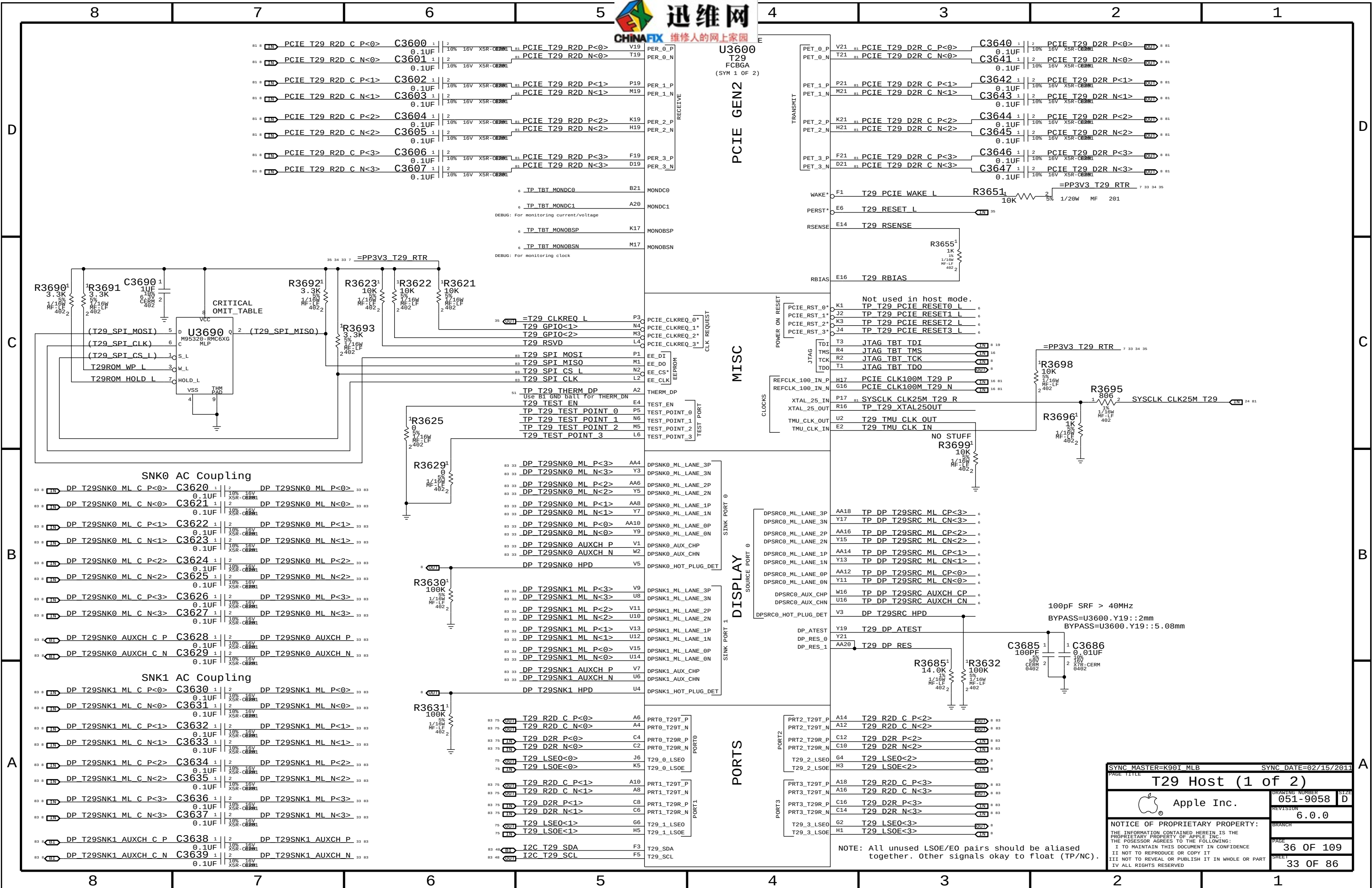
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3403, R3405		VREFDQ:LDO
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3409, R3411		VREFCA:LDO

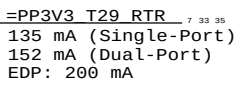
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0218	4	RES, MTL FILM, 1K, 1%, 0402, SM, LF	R3421, R3422, R3441, R3442		VREFDQ:M1_DAC
114S0171	2	RES, MTL FILM, 332, 1%, 0402, SM, LF	R3404, R3406		VREFDQ:M1_DAC

PAGE TITLE		SYNC DATE=06/13/2011	
DDR3/FRAMEBUF VREF MARGINING			
Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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SYNC MASTER=K901 MLB		SYNC DATE=02/15/2011	
PAGE TITLE		X19/ALS/CAMERA CONNECTOR	
Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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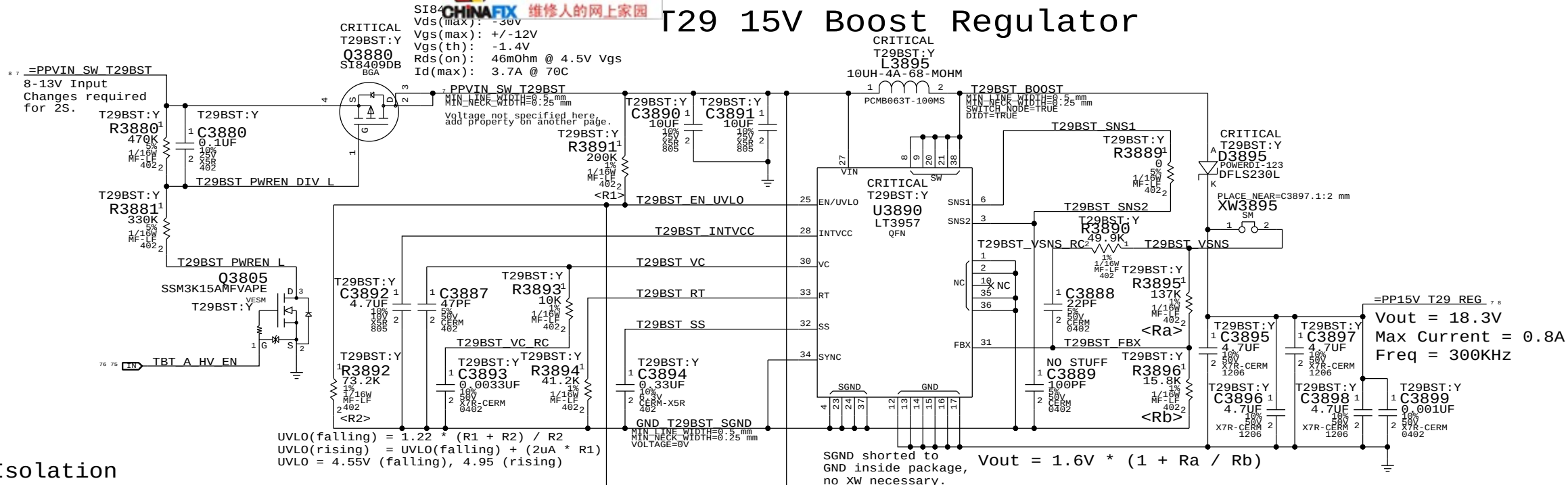
Page Notes

Power aliases required by this page:
 -PPVIN_SW_T29BST (8-13V Boost Input)
 -PP18V_T29_REG (18V Boost Output)
 -PP3V3_T29_P3V3T29FET (3.3V FET Input)
 -PP3V3_T29_FET (3.3V FET Output)
 -PP3V3_S0_T29PWRCTL (1.05V FET Input)
 -PP1V05_T29_P1V05T29FET (1.05V FET Input)
 -PP1V05_T29_FET (1.05V FET Output)

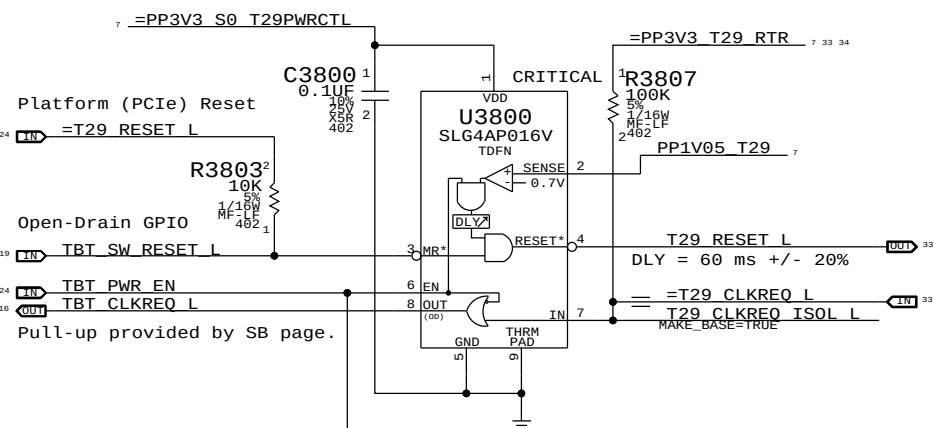
Signal aliases required by this page:
 -T29_CLKREQ_L
 -T29_RESET_L

BOM options provided by this page:
 T29BST:Y - Stuffs 18V boost circuitry.

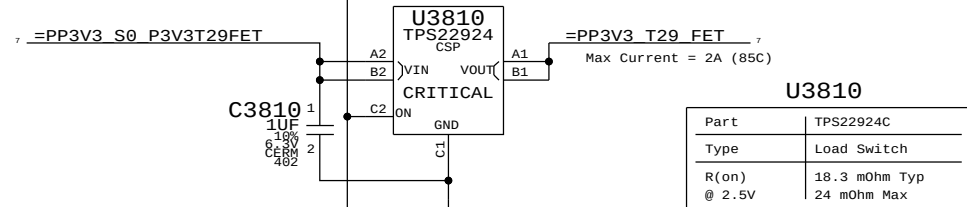
T29 15V Boost Regulator



Supervisor & CLKREQ# Isolation

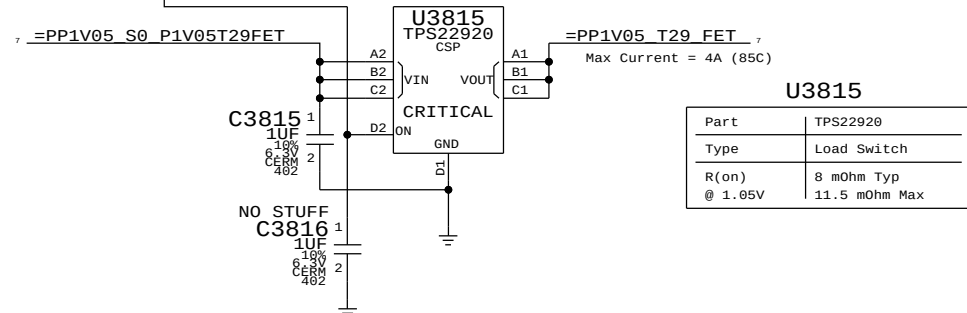


3.3V T29 Switch

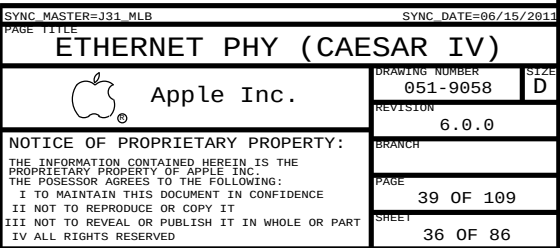


Part	TPS22924C
Type	Load Switch
R(on)	18.3 mOhm Typ 24 mOhm Max
@ 2.5V	

1.05V T29 Switch



Part	TPS22920
Type	Load Switch
R(on)	8 mOhm Typ 11.5 mOhm Max
@ 1.05V	

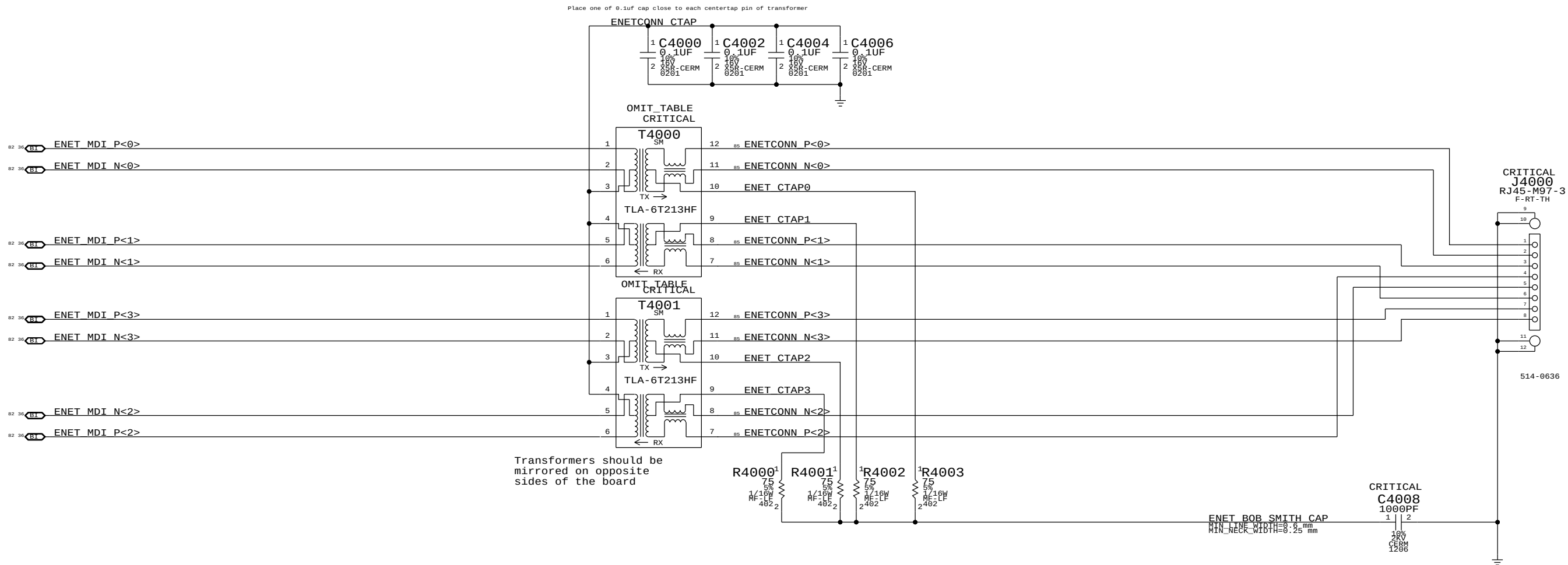


Page Notes

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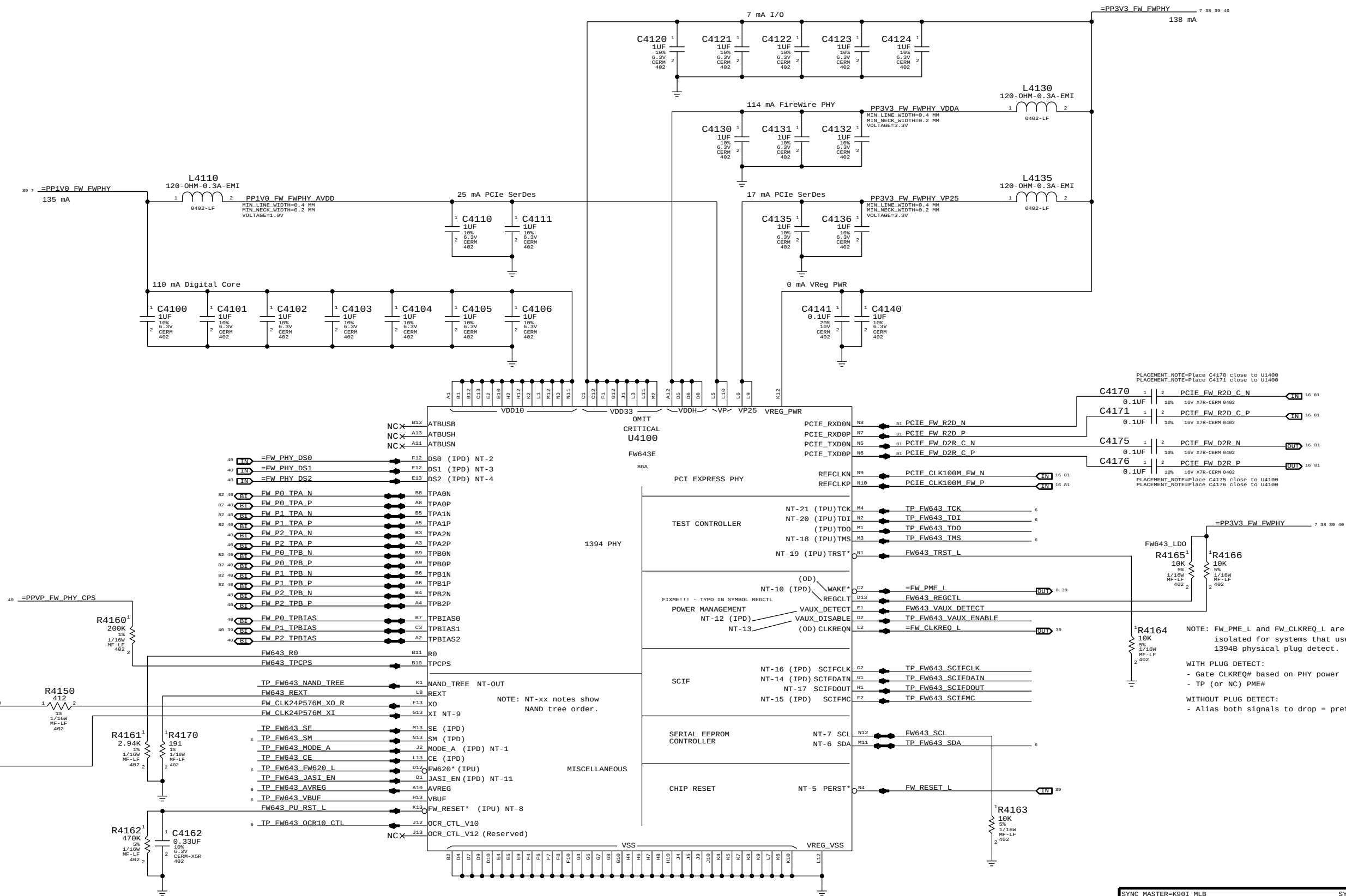
Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
157S0084	2	XFMR, ISO, HALF-PORT, 1000T, 12P, SMD, HF	T4000, T4001	CRITICAL	

SYNC MASTER=K901 MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
Ethernet Connector			
 Apple Inc.		DRAWING NUMBER	051-9058
		SIZE	D
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		BRANCH	
		PAGE	40 OF 109
		SHEET	37 OF 86



Page Notes

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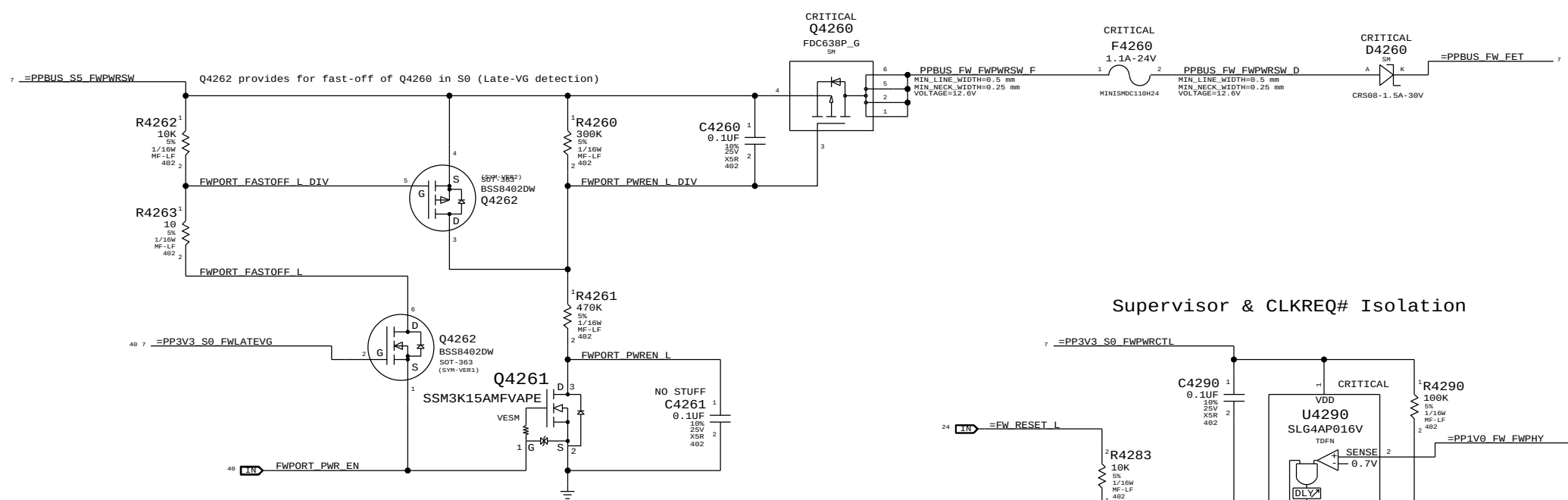
Power aliases required by this page:
- =PPBUS_S0_F5_FWPNRWSW (FW VP FET Input)
- =PPBUS_FW_FET (FW VP FET Output)
- =PP3V3_FW_PP3V3FWFET (3.3V FET Input)
- =PP3V3_FW_FET (3.3V FET Output)
- =PP3V3_FW_FWPHY (PHY 3.3V Power)
- =PP3V3_S0_FWLATEVG
- =PP3V3_S0_FWPNRCTL
- =PP1V05_S0_FW1V05FWFET (5KPD Bias Rail)
- =PP1V0_FW_FET_R (1.0V FET Input)
- =PP1V0_FW_FET_OUT (1.0V FET Output)
- =PP1V0_FW_FWPHY (PHY 1.0V)

```

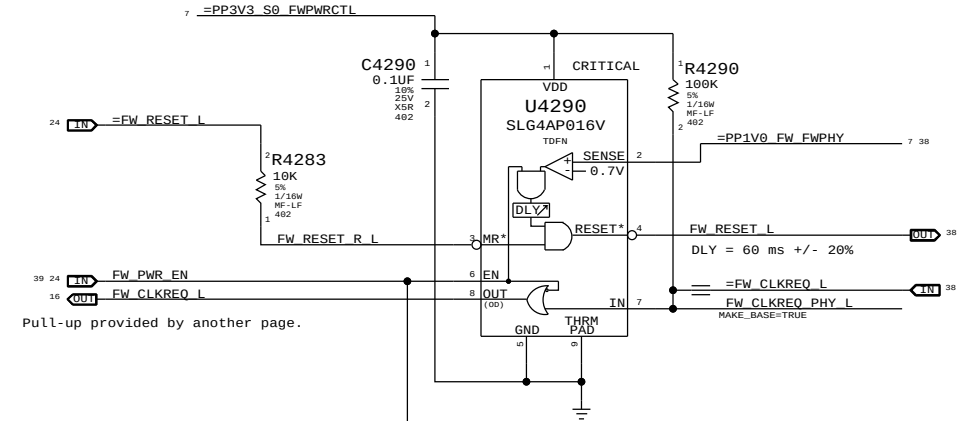
Signal aliases required by this page:

- =FW_CLKREQ_L
- =FW_PME_L

BOM options provided by this page:
(NONE)

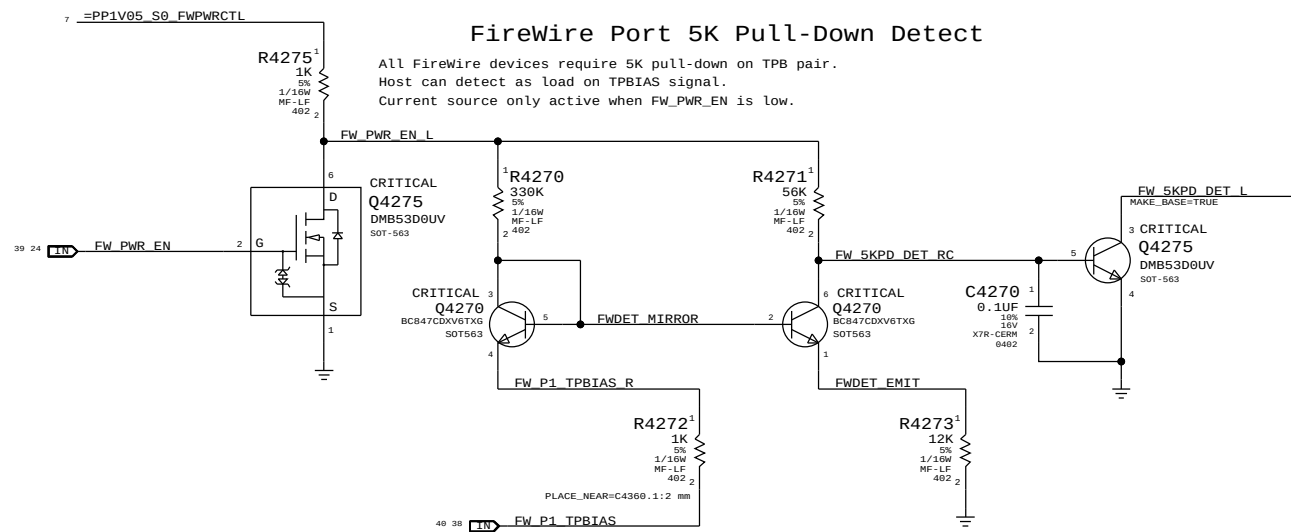


Supervisor & CLKREQ# Isolation



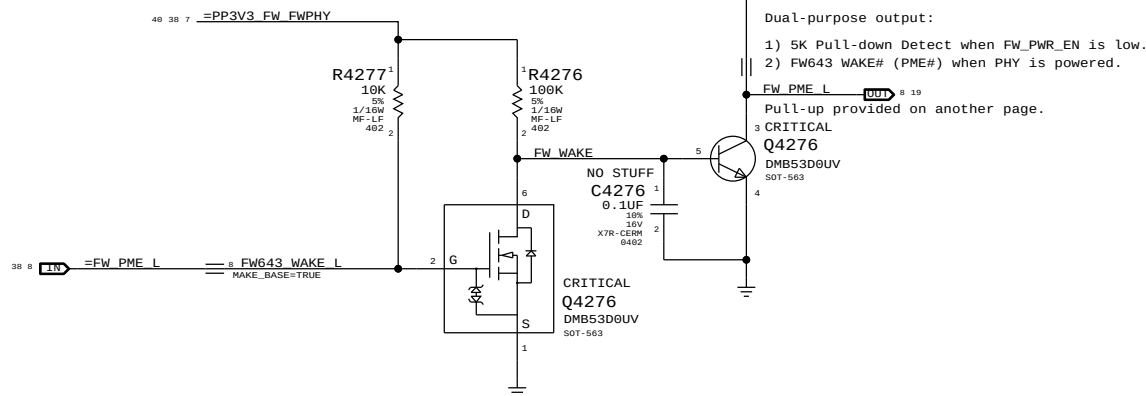
FireWire Port 5K Pull-Down Detect

All FireWire devices require 5K pull-down on TPB pair.
Host can detect as load on TPBIAS signal.
Current source only active when FW_PWR_EN is low.



FireWire PHY WAKE# Support

When PHY is powered, FW_5KPD_DET_L acts as legacy PME# signal.



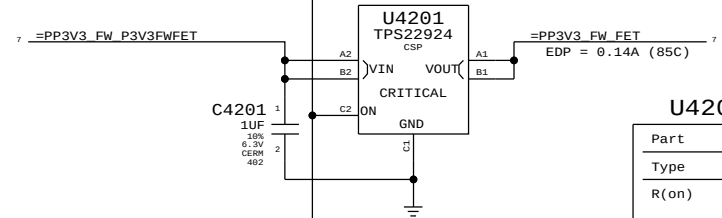
Dual-purpose output:

- 1) 5K Pull-down Detect when FW_PWR_EN is low.
- 2) FW643 WAKE# (PME#) when PHY is powered.

FW PME_L  8 19

Pull-up provided on another page.

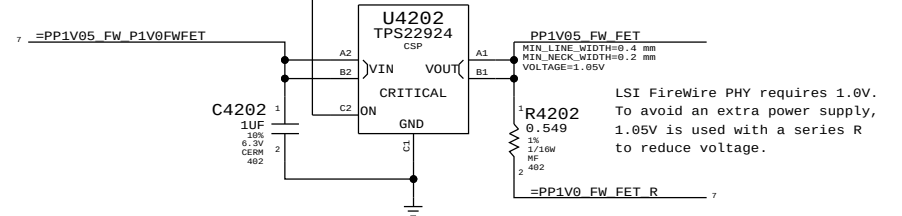
3.3V FW Switch



U4201 & U4202

Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

1.0V FW Switch



LSI FireWire PHY requires 1.0V.
To avoid an extra power supply,
1.05V is used with a series R
to reduce voltage.

Page Notes

Power aliases required by this page:

- PPVP_FW_PORT1
- PPVP_FW_PHY_CPS_FET (From Port)
- PPVP_FW_PHY_CPS (To PHY)
- PP3V3_FW_FWPHY
- PP3V3_S0_FWLATEVG

Signal aliases required by this page:

- FW_PHY_DS0
- FW_PHY_DS1
- FW_PHY_DS2

NOTE: This page is expected to contain the necessary aliases to map the FireWire TPA/TPB pairs to their appropriate connectors and/or to properly terminate unused signals.

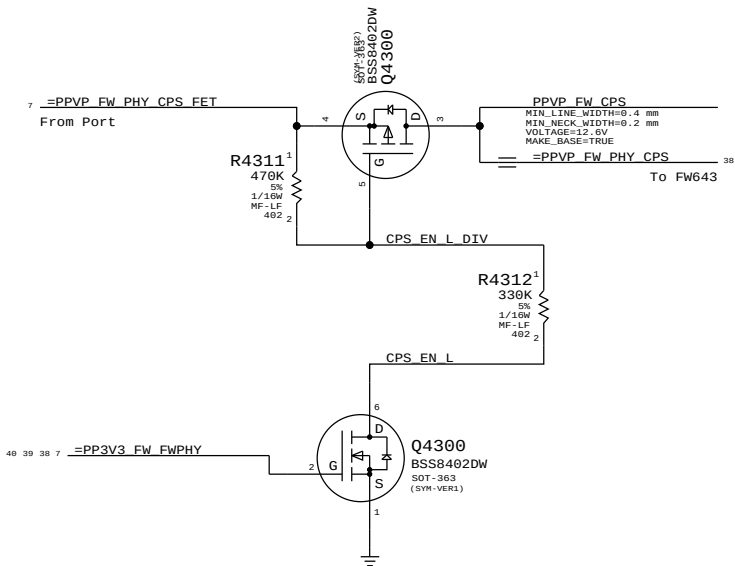
BOM options provided by this page:

(NONE)

1394b implementation based on Apple FireWire Design Guide (FWDG 0.6, 5/14/03)

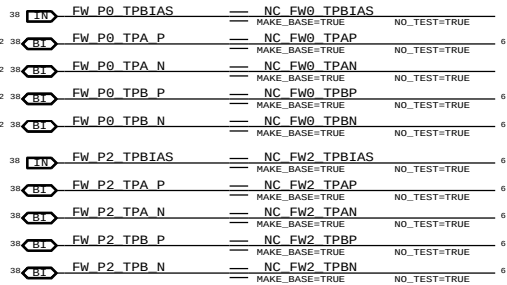
FW643 TPCPS Leakage Protection

FW643 has internal leakage path from TPCPS pin to VDD33.
FET blocks current to TPCPS until VDD33 is powered.



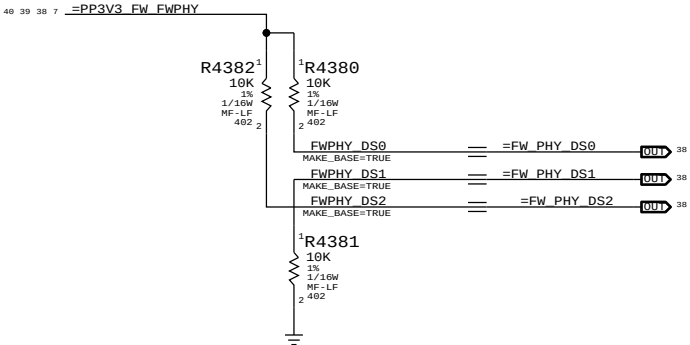
Disabled FireWire Ports

Disabled per LSI instructions
(All unused port signals TP/NC)



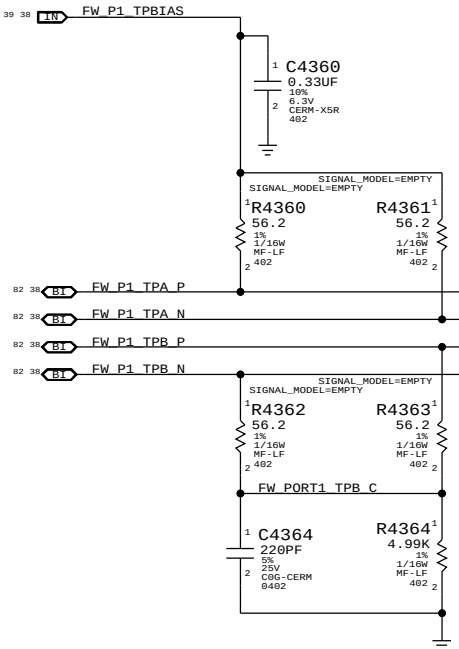
FireWire PHY Config Straps

Configures PHY for:
- Port "1" Bilingual (1394B)



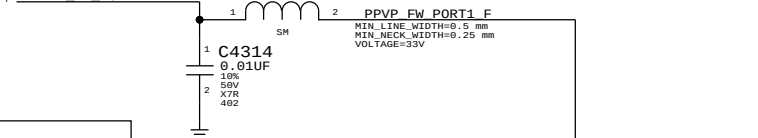
Termination

Place close to FireWire PHY

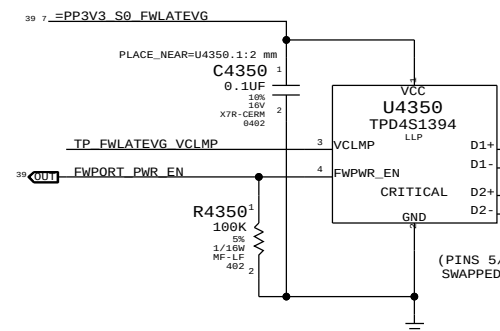


Cable Power

Note: Trace PPVP_FW_PORT1 must handle up to 5A



"Snapback" & "Late VG" Protection



PORT 1

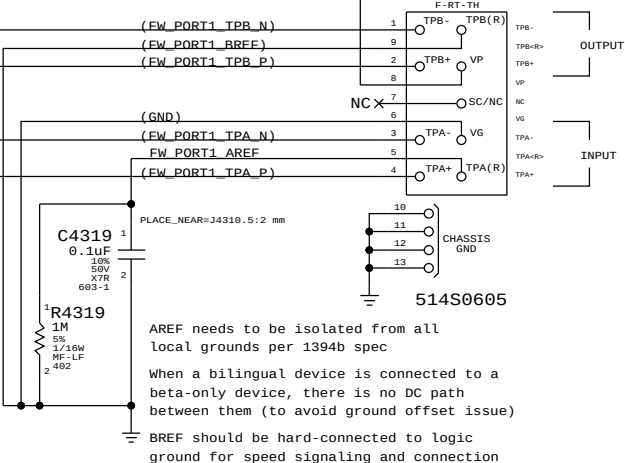
BILINGUAL

CRITICAL

J4310

1394B-M97

F-RT-TH



AREF needs to be isolated from all local grounds per 1394b spec
When a bilingual device is connected to a beta-only device, there is no DC path between them (to avoid ground offset issue)
BREF should be hard-connected to logic ground for speed signaling and connection

CANNOT SYNC THIS PAGE FROM T27, TPA AND TPB FOR U4350 IS SWAPPED

FireWire Connector

Apple Inc.

DRAWING NUMBER

051-9058

REVISION

6.0.0

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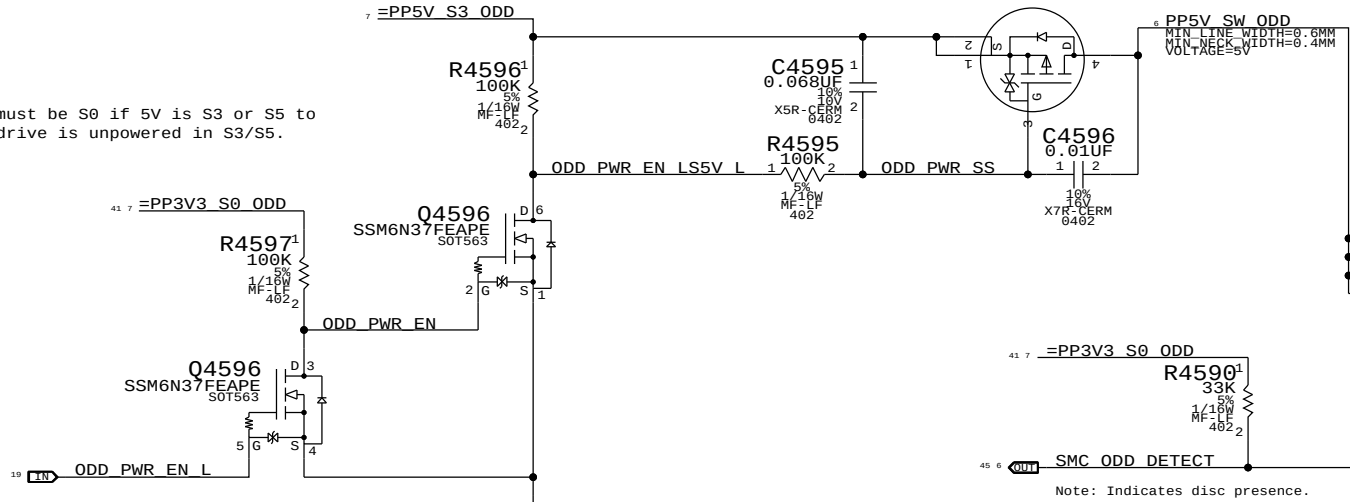
43 OF 109

SHEET

40 OF 86

ODD Power Control

Note: 3.3V must be S0 if 5V is S3 or S5 to ensure the drive is unpowered in S3/S5.



SATA ODD Connector

SATA OOB Comparator

Notes:
00BD2R was OOB_TEMP, from SSD, to SMC
00BR2D was TEMP_CTL, from SMC, to SSD

SATA HDD Connector (Gen3)

SATA Redriver

Internally PD -150K
Write:0xB6 Read:0xB7

ADDR1	ADD0	Address (R/W)
L	L	0x96/0x97
L	H	0x98/0x99
H	L	0xB6/0xB7
H	H	0xB8/0xB9

D

D

C

C

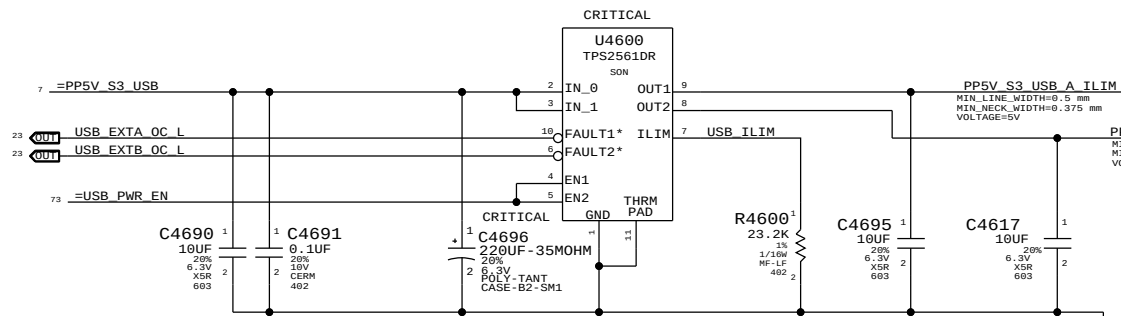
B

B

A

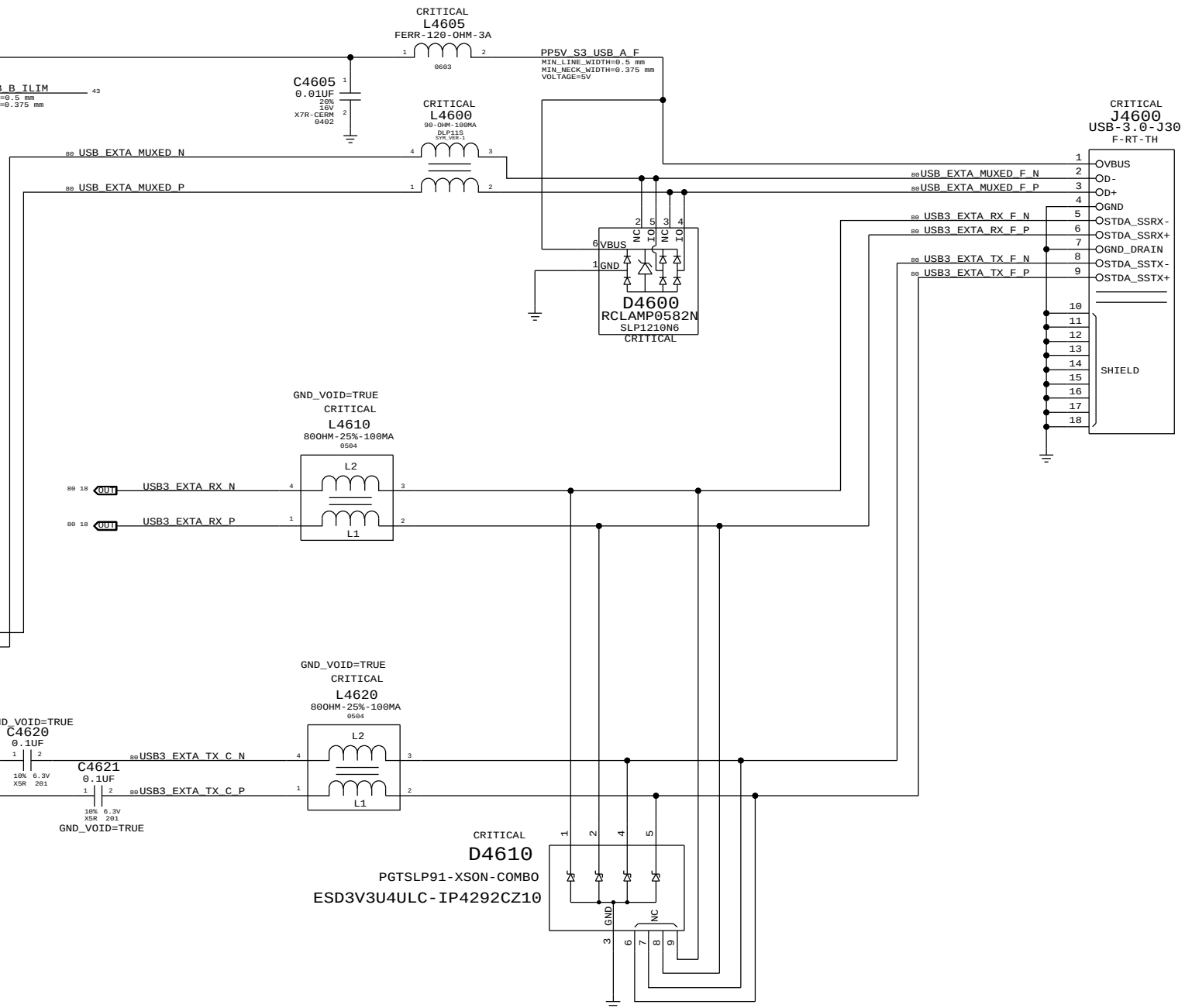
A

USB Port Power Switch

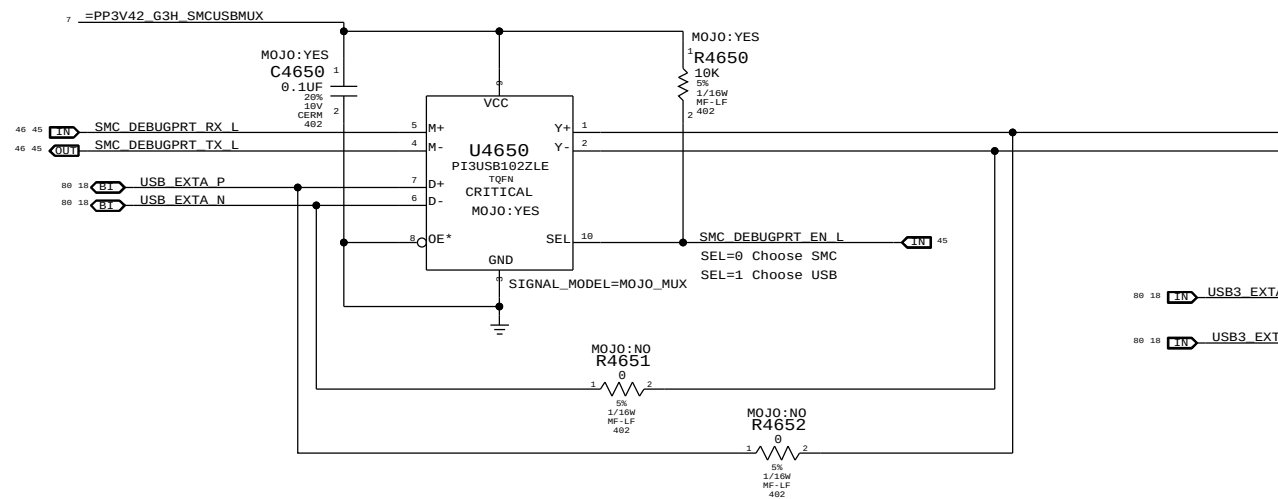


Current limit per port (R4600): 2.18A min / 2.63A max

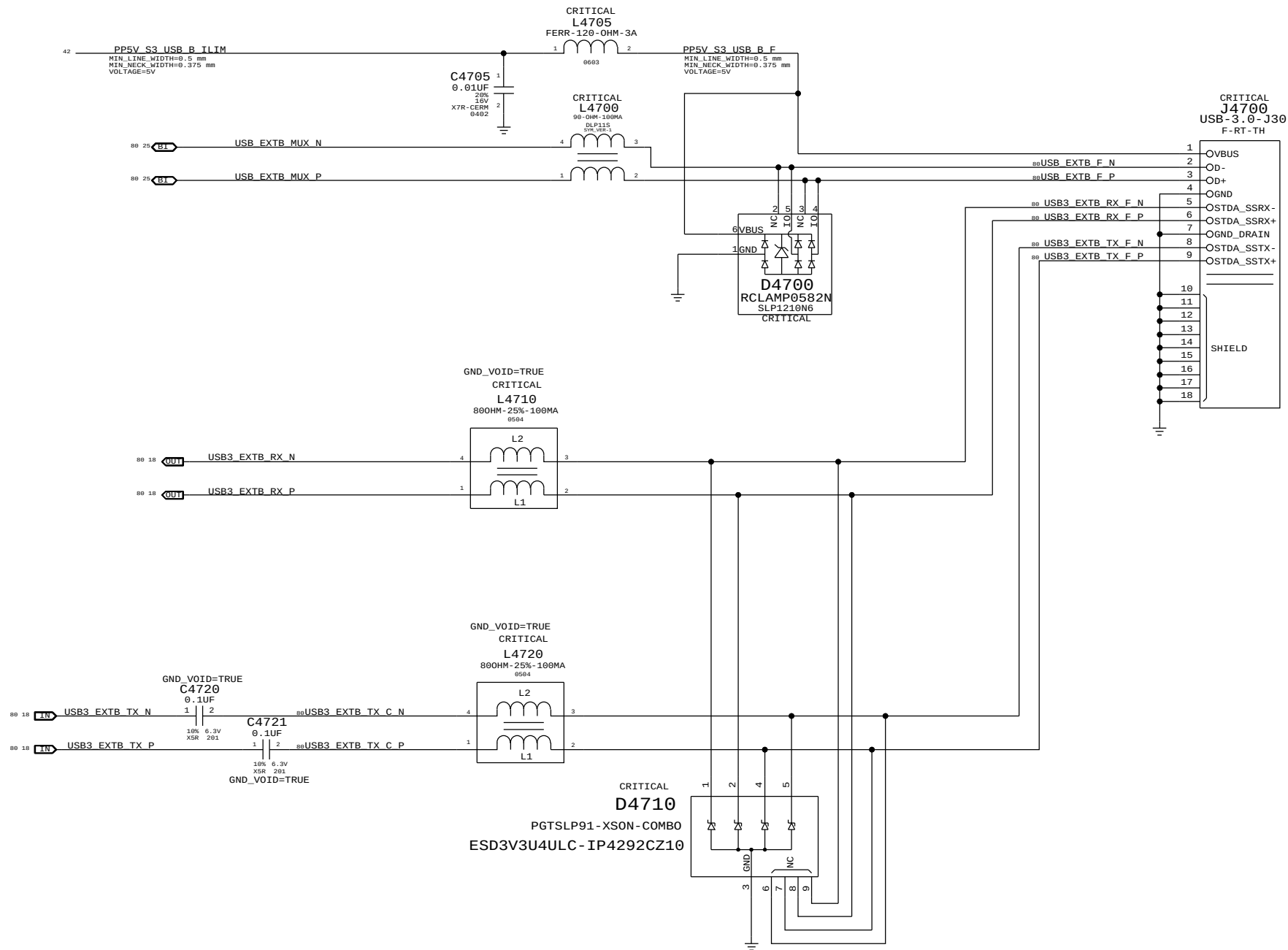
USB Port A (Front Port)



Mojo SMC Debug Mux



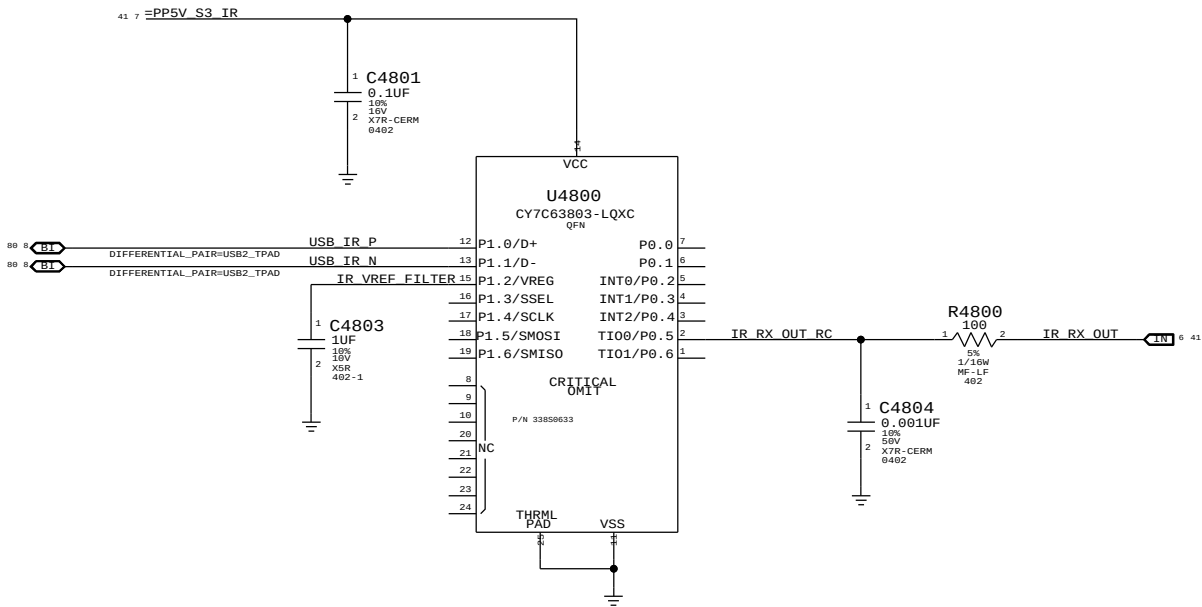
USB Port B (Back Port)

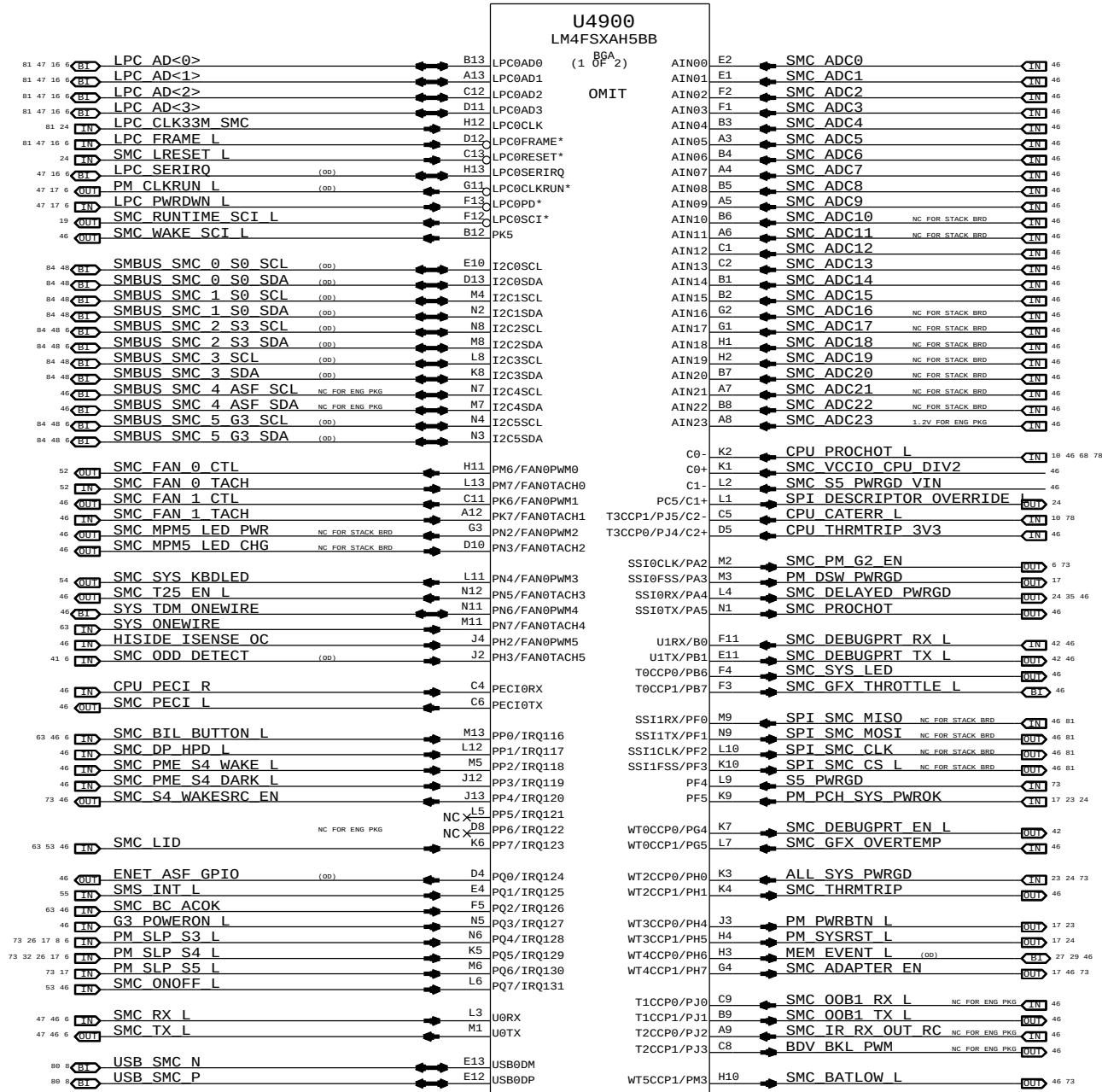


NOTE: Swapped pin4 and 5, pin6 and 7 for layout.

SYNC MASTER=J31 MLB		SYNC DATE=07/08/2011	
PAGE TITLE			
External B USB3 Connector			
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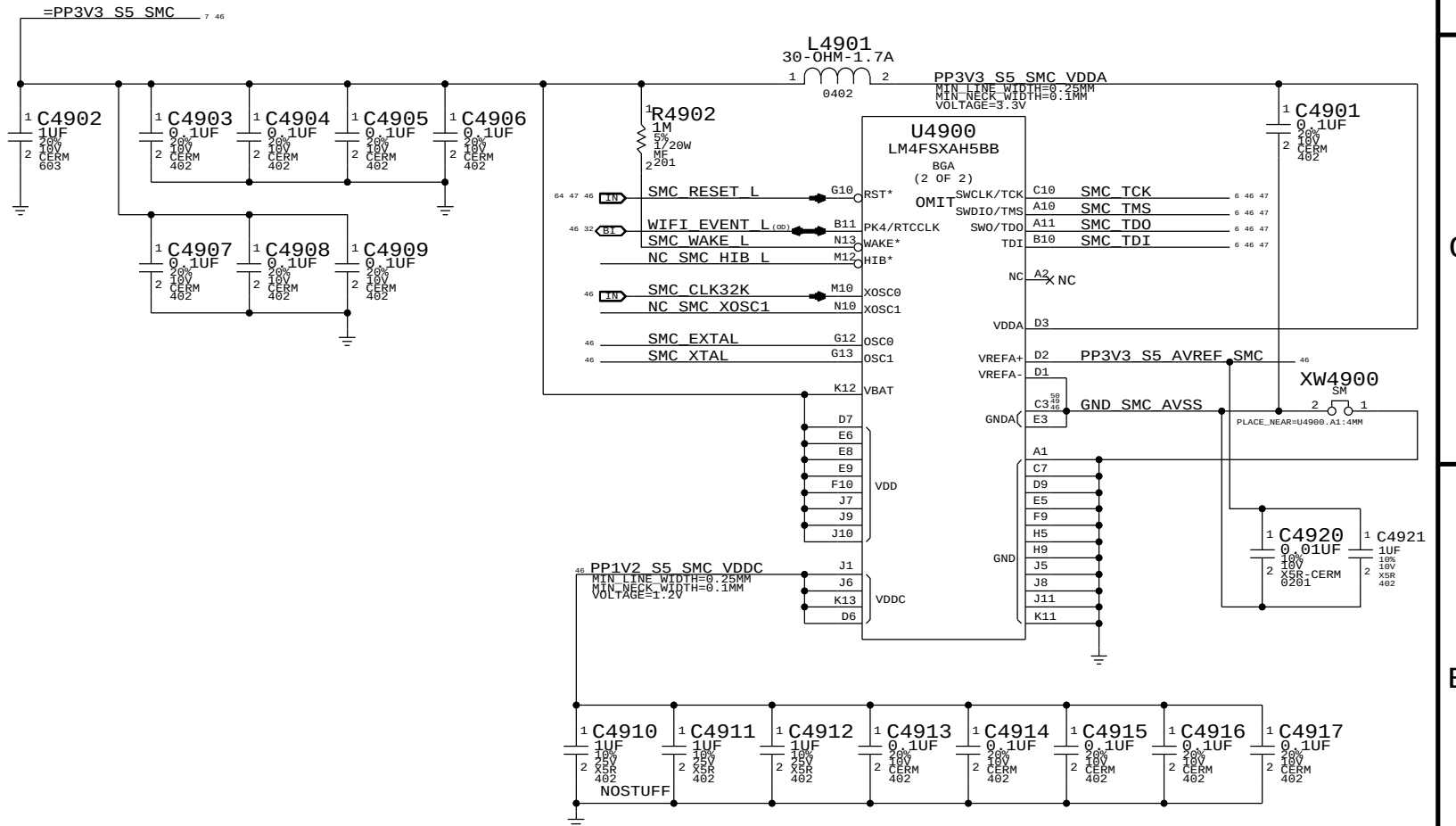
IR SUPPORT



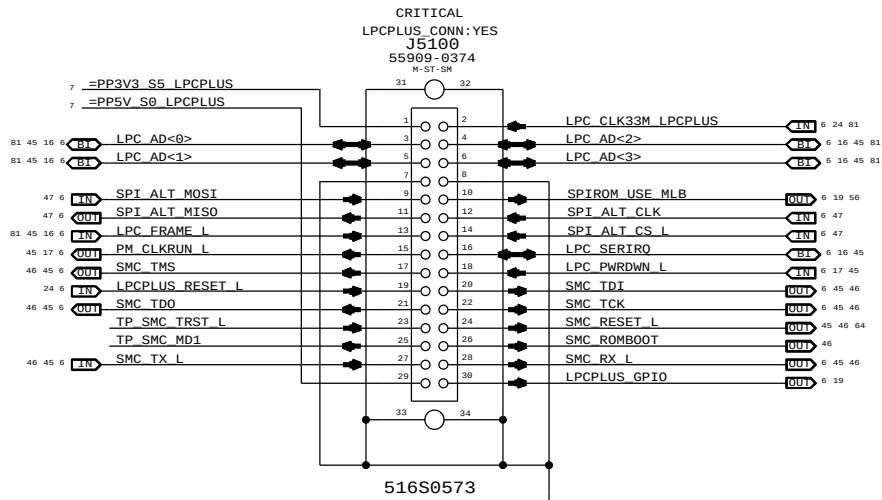


NOTE:
SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

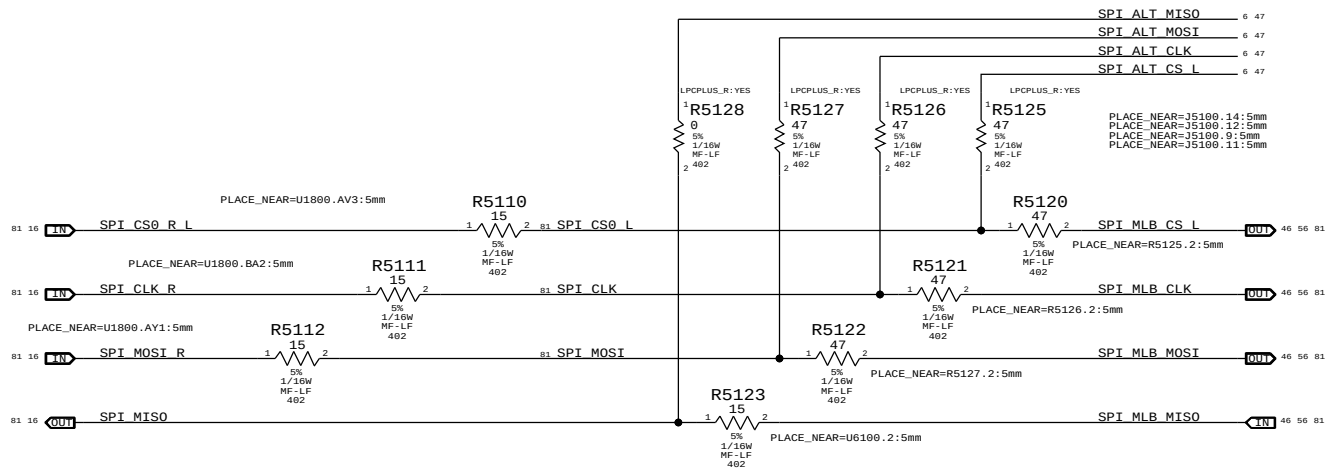
NOTE:
Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

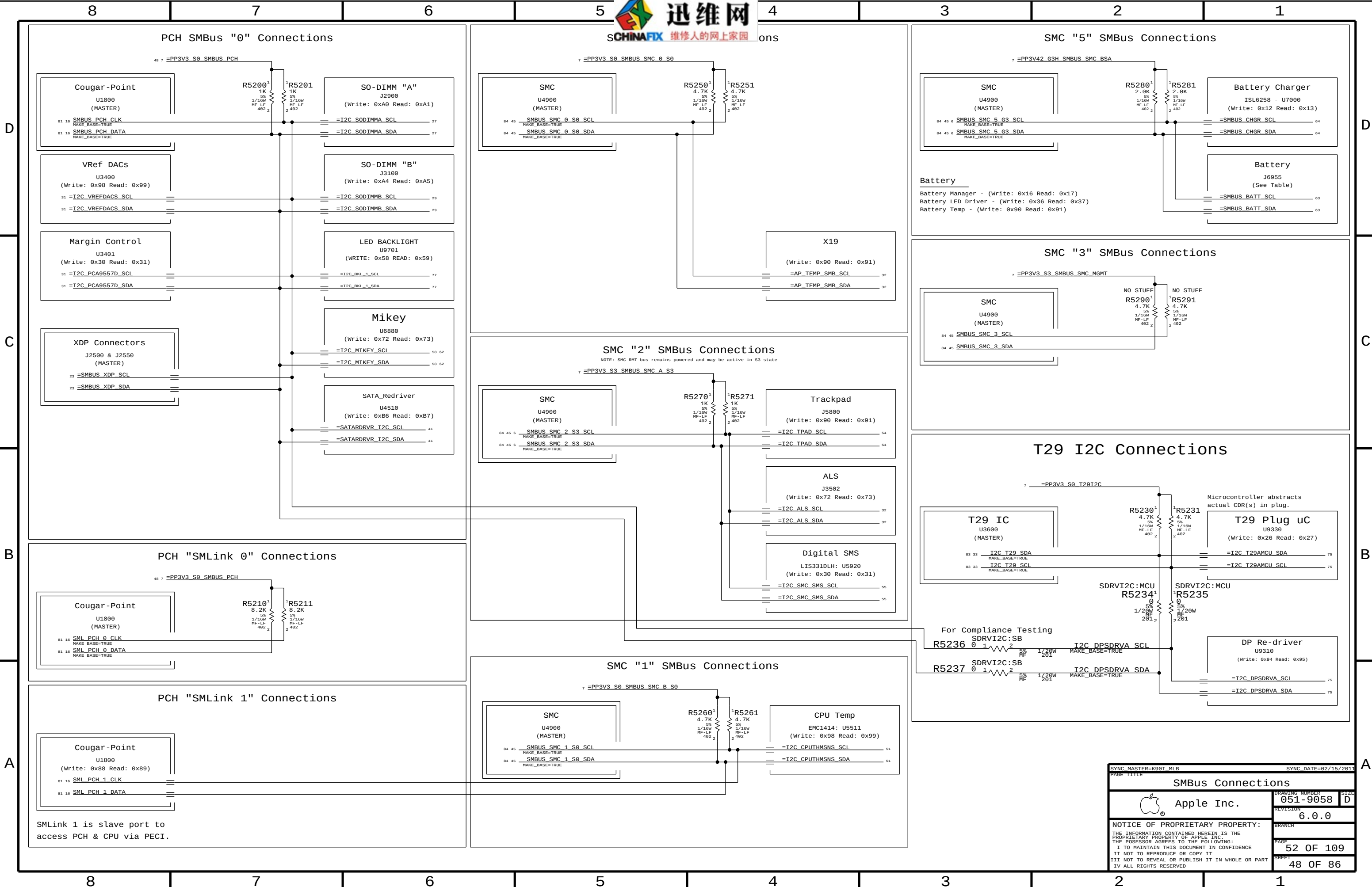


LPC+SPI Connector



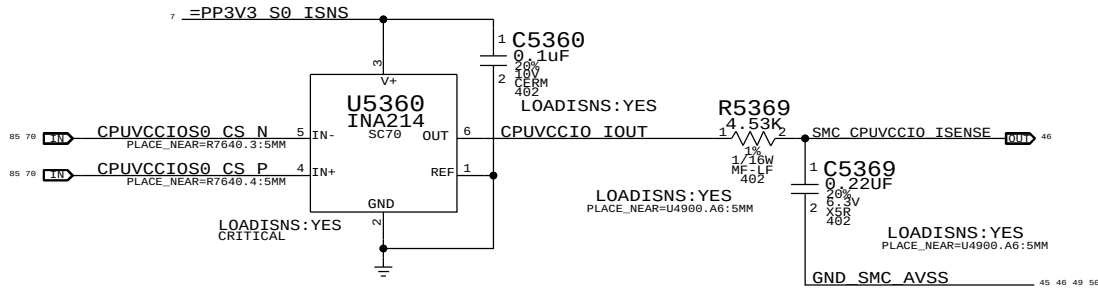
SPI Bus Series Termination





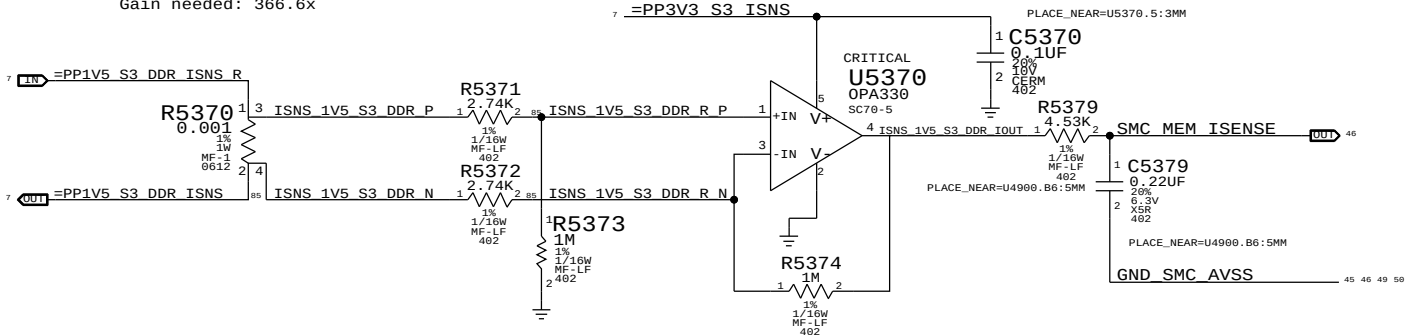
CPU VCCIO 1.05V Load Side Current Sense (IC1C)

Gain: 100x, EDP: 20.1 A
Rsense: 0.001 (R7640)
V across Rsense: 20.1 mV
Gain needed: 164.2x



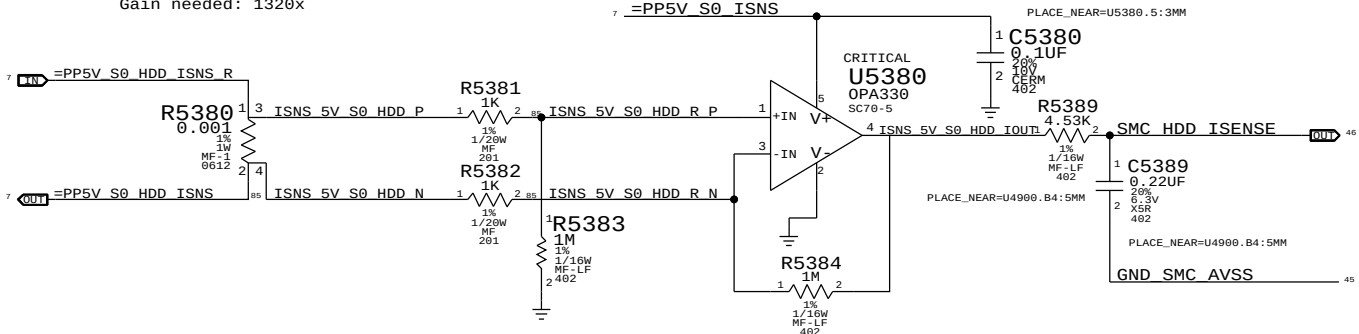
DDR 1.5V S3 (Memory) Current Sense (IM0C)

Gain: 364.9x, EDP: 9 A
Rsense: 0.001 (R5370)
V across Rsense: 9 mV
Gain needed: 366.6x

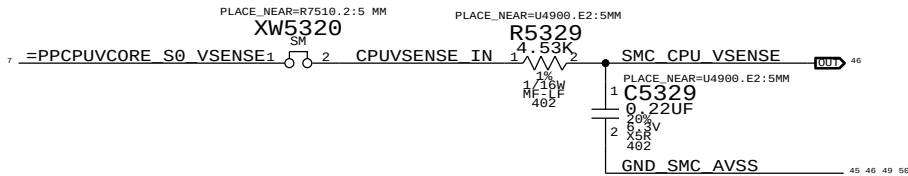


HDD Current Sense (IHDC)

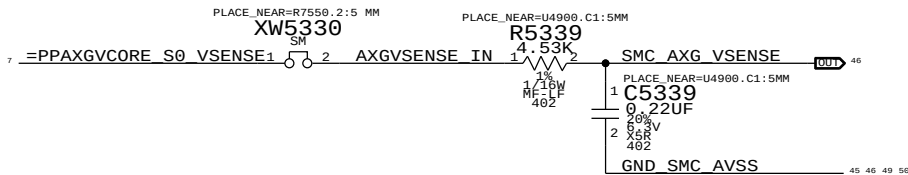
Gain: 1000x, EDP: 2.5 A (12.5 W)
Rsense: 0.001 (R5380)
V across Rsense: 2.5 mV
Gain needed: 1320x



CPU Core Voltage Sense (VC0C)

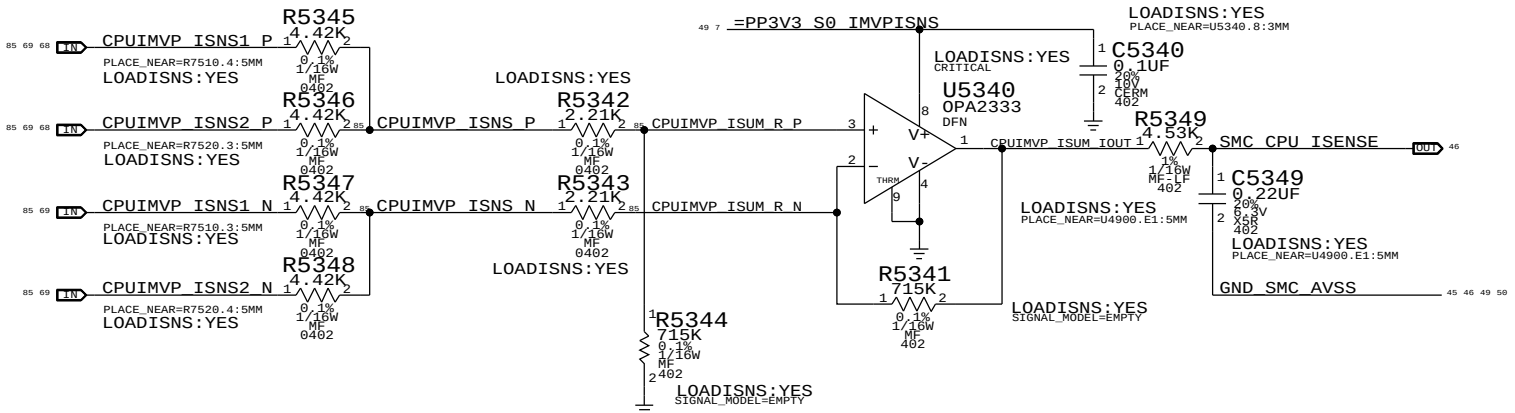


AXG Core Voltage Sense (VN0C)



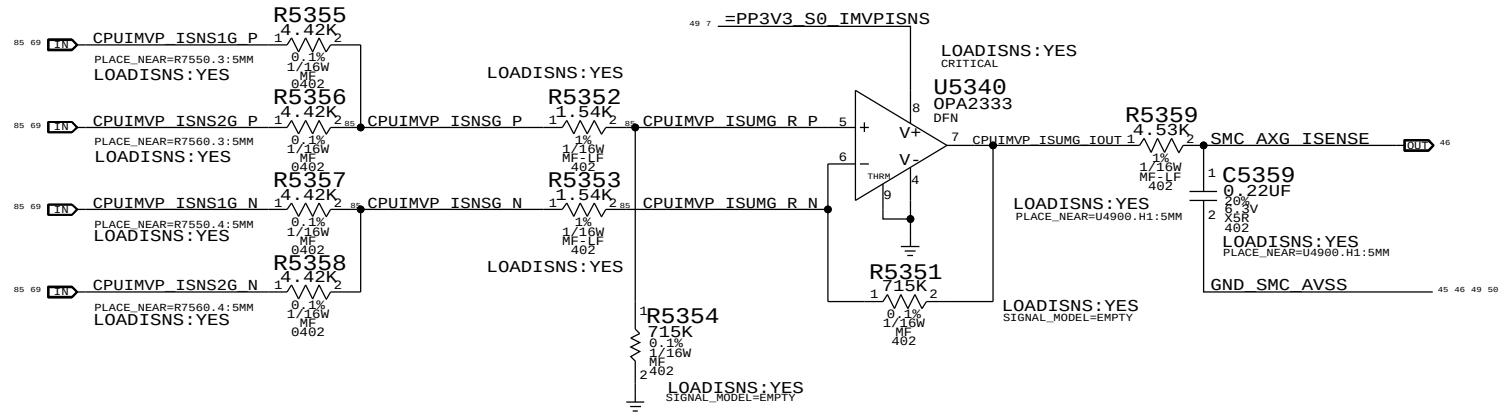
CPU Core Load Side Current Sense (IC0C)

Gain: 161.5x, EDP: 53 A
Rsense: 2x of 0.00075 (R7510, R7520), Rsum: 0.000375
V across Rsense: 19.8 mV
Gain needed: 166.1x



AXG Core Load Side Current Sense (IN0C)

Gain: 190.6x, EDP: 46 A
Rsense: 2x of 0.00075 (R7550, R7560), Rsum: 0.000375
V across Rsense: 17.25 mV
Gain needed: 191.3x

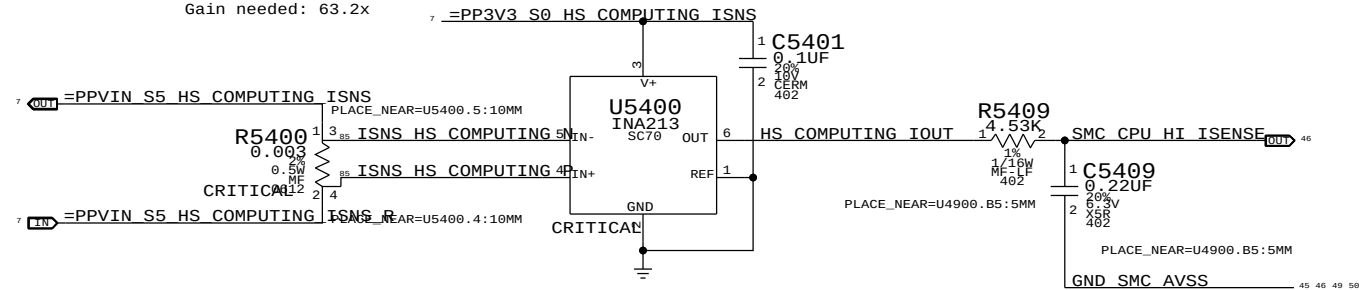


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0114	3	RES, MTL, FLIM, 100K, 1/16W, 0402, SMD, LF	C5349, C5359, C5369		LOADISNS:NO

SYNC MASTER=LINDA J36		SYNC DATE=09/28/2011	
PAGE TITLE			
Power Sensors: Load Side			
 Apple Inc.		DRAWING NUMBER	051-9058
		SIZE	D
		REVISION	6.0.0
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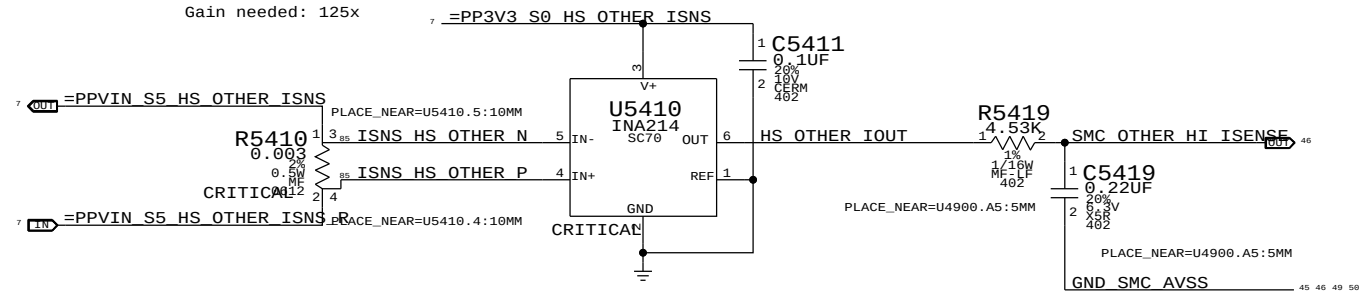
CPU High Side Current Sense (IC0R)

Gain: 50x, EDP: 17.4 A
Rsense: 0.003 (R5400)
V across Rsense: 52.2 mV
Gain needed: 63.2x



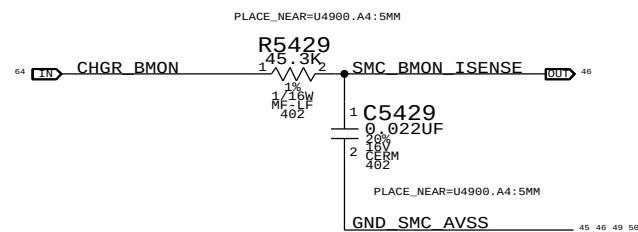
OTHER High Side Current Sense (IO0R)

Gain: 100x, EDP: 8.8 A
Rsense: 0.003 (R5410)
V across Rsense: 26.4 mV
Gain needed: 125x



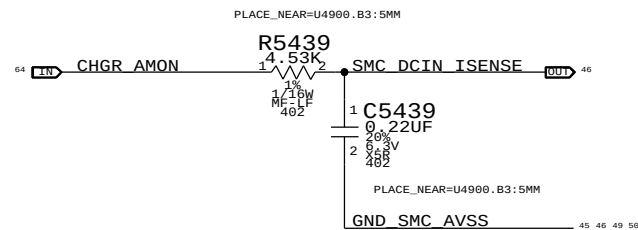
Charger (BMON Production) Current Sense (IPBR)

Charger Gain: 36x
Rsense: 0.010 (R7050)
Max Current Measured: 9.2 A

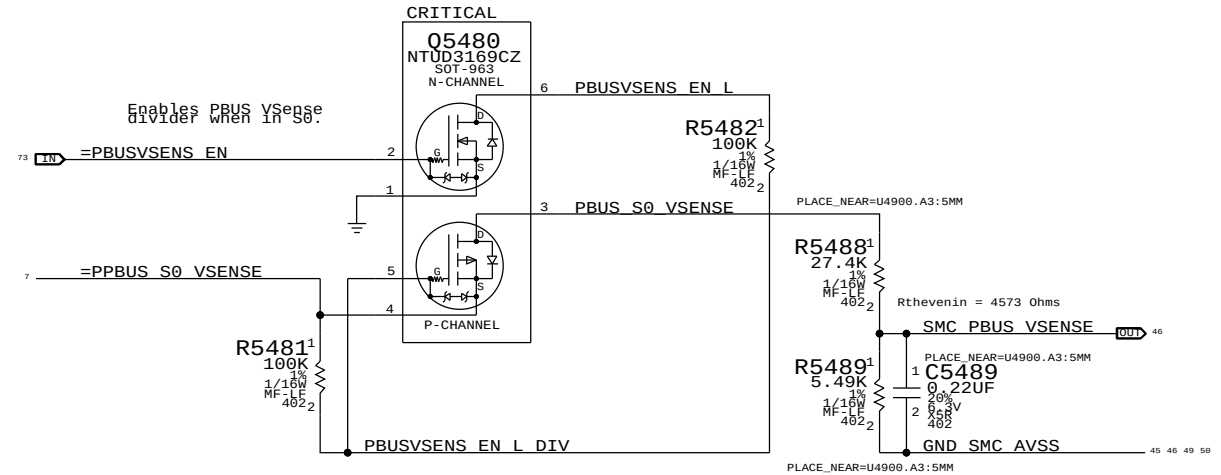


DC-In (AMON) Current Sense (ID0R)

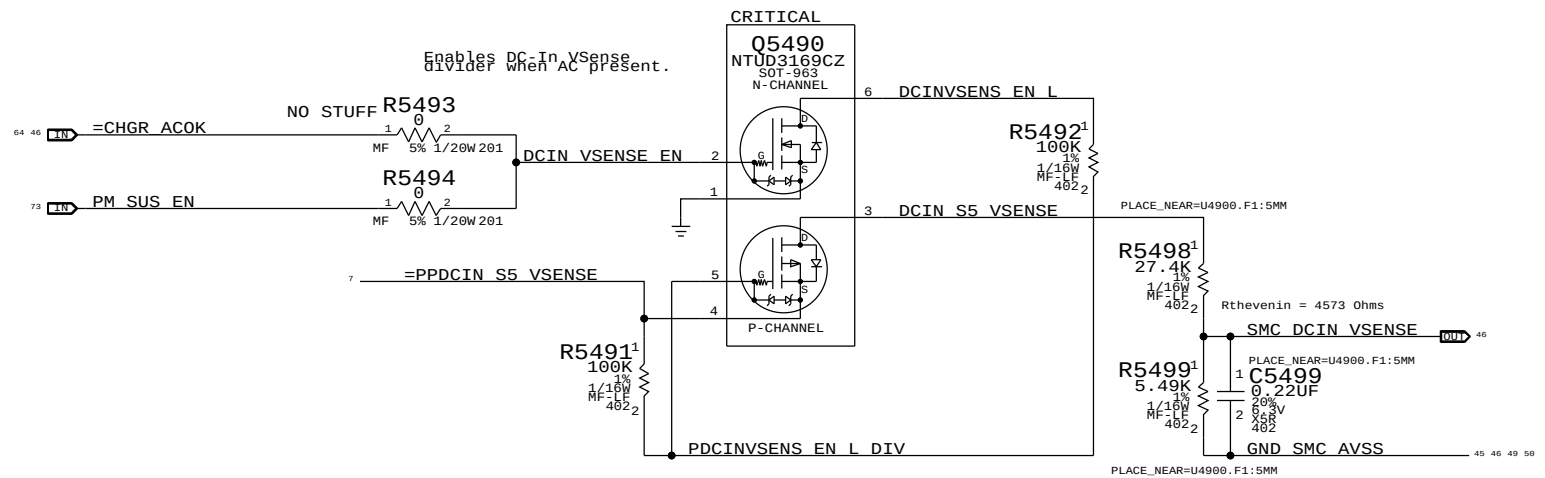
Charger Gain: 20x
Rsense: 0.020 (R7020)
Max Current Measured: 8.3 A



PBUS Voltage Sense & Enable (VP0R)

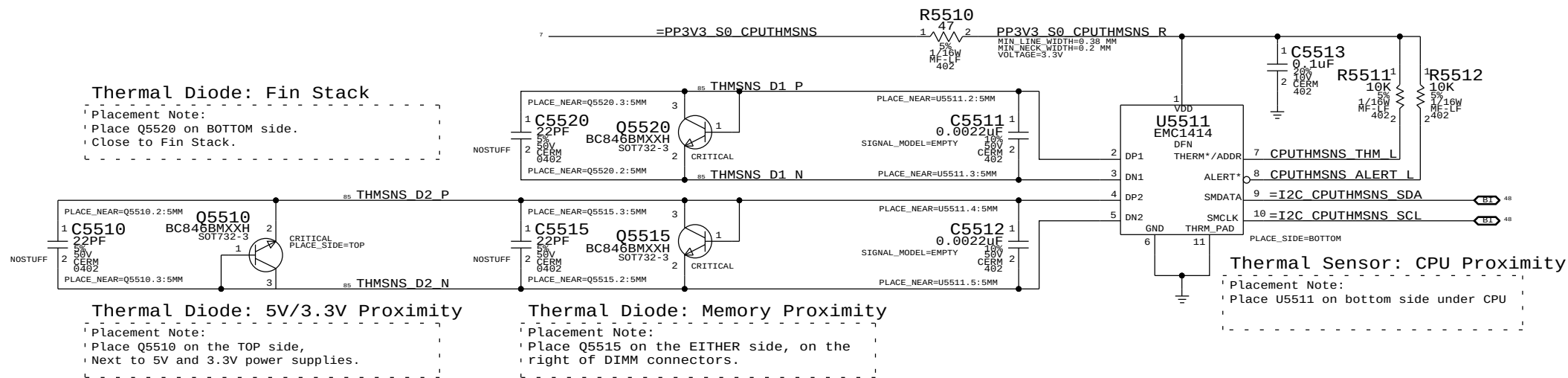


DC In Voltage Sense & Enable (VD0R)

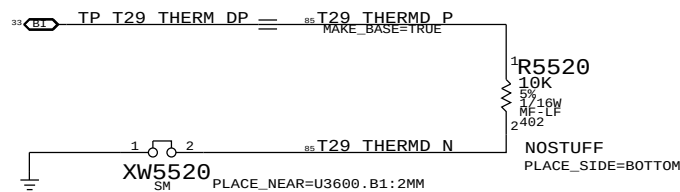


Thermal Sensor: CPU Proximity, Fin Stack, Memory Proximity, 5V/3.3V Proximity

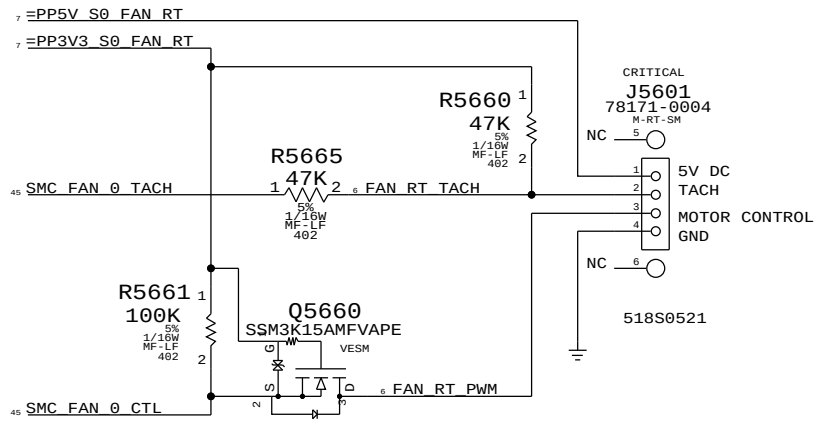
I2C Write: 0x98, I2C Read: 0x99



Thermal Sensor: T29 Die

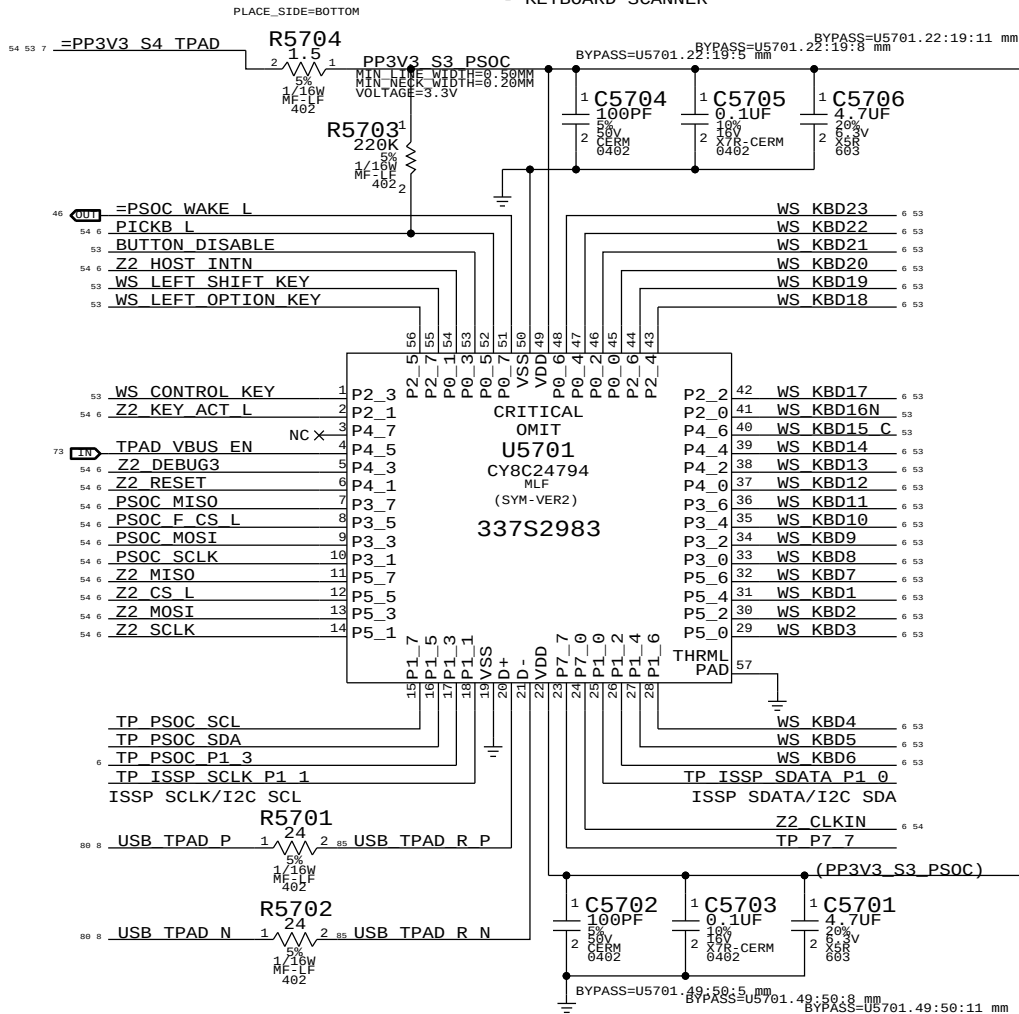


Note: Use GND pin B1 on U3600 for N leg.



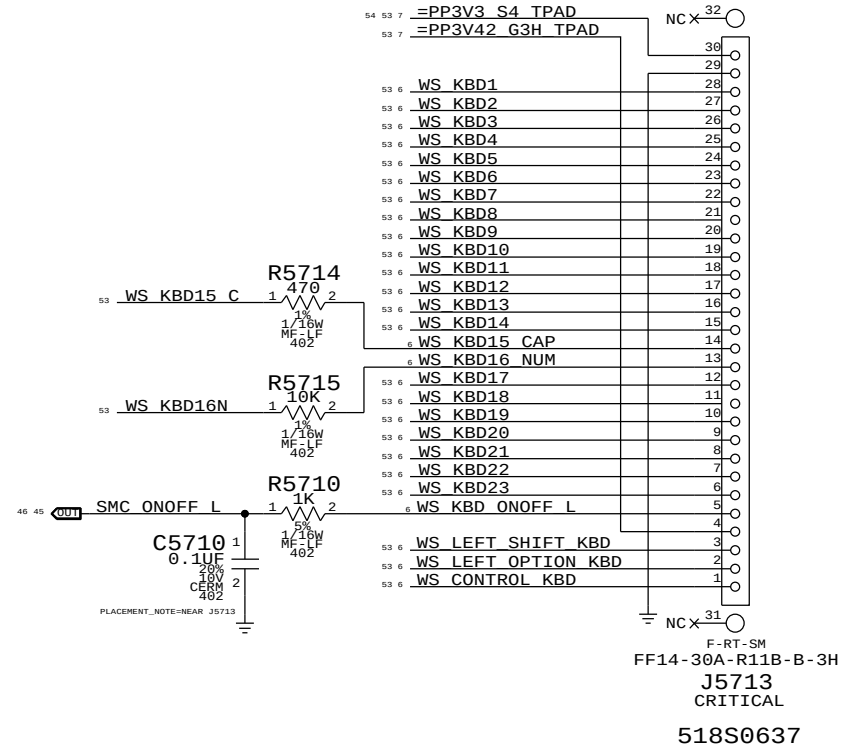
PSOC USB CONTROLLER

- USB INTERFACES TO MLB
- SPI HOST TO Z2
- TRACKPAD PICK BUTTONS
- KEYBOARD SCANNER



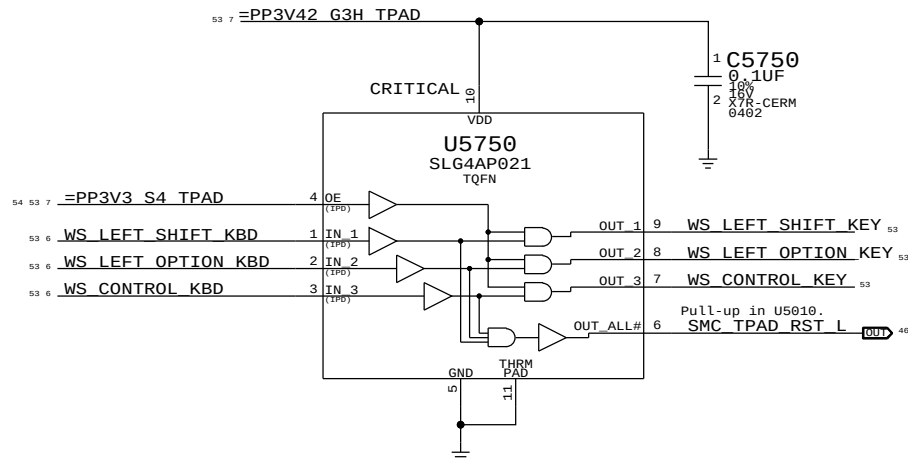
IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA 80UA	2.55 KOHM	0.0255 V 0.204 V	0.255E-6 W 16.32E-6 W
3V3 LDO	VDD VOUT	60MA (MAX) 60MA (MAX)	10 OHM 0.2 OHM	0.6 V 0.012 V	36E-3 W 0.72E-3 W
PSOC	VDD	8MA (TYP) 14MA (MAX)	1.5 OHM	0.012 V 0.021 V	96E-6 W 294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

Keyboard Connector



SMC Manual Reset & Isolation

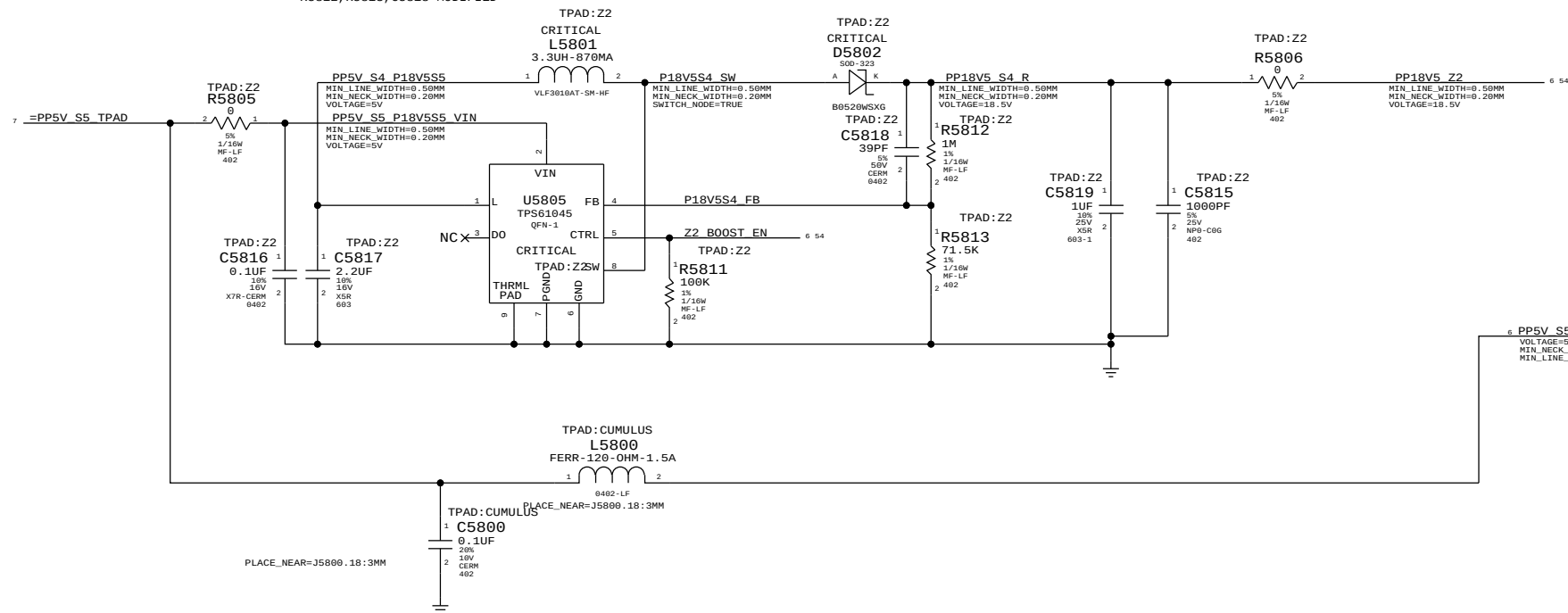
Left shift, option & control keys combined with power button cause SMC RESET# assertion.
Keys ANDed with MSP power to isolate when MSP is not powered. No IPD on OE input pin PP3V3_S4 (symbol error).



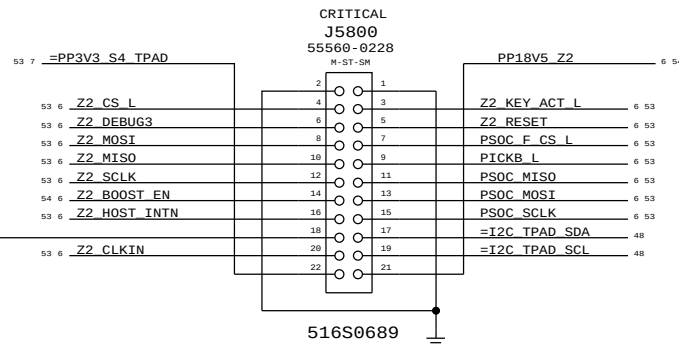
BOOSTER +18.5VDC FOR SENSORS

BOOSTER DESIGN CONSIDERATION:

- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812, R5813, C5818 MODIFIED

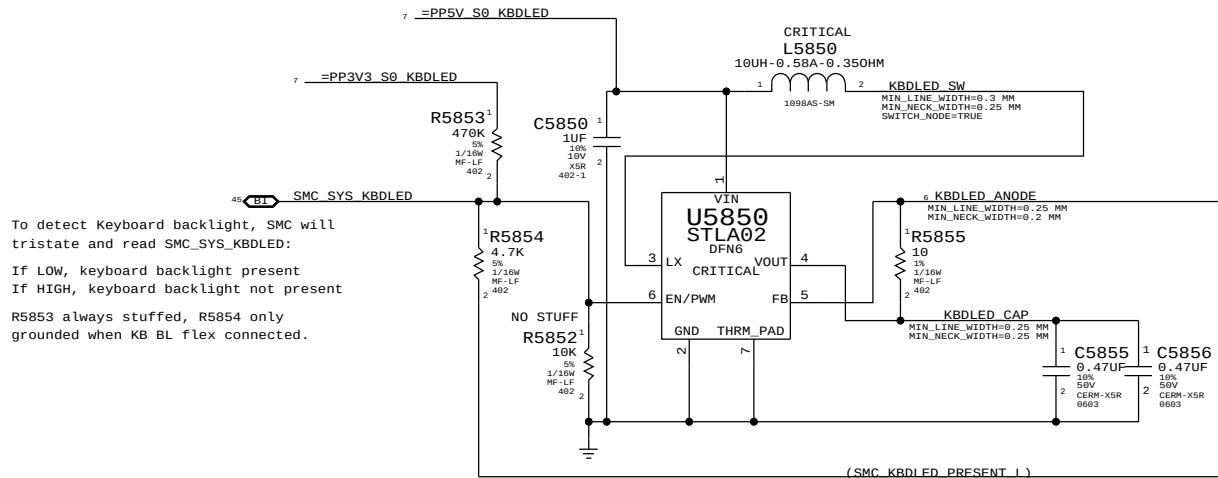


IPD Flex Connector



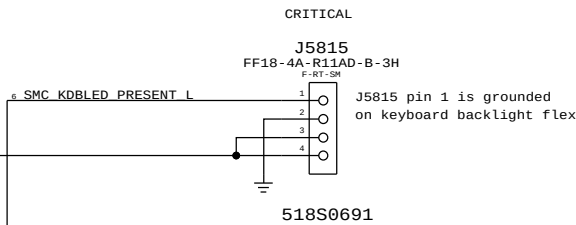
PIN 21 IS NC ON CUMULUS FLEX
PIN 18 IS NC ON Z2 FLEX

Keyboard Backlight Driver & Detection

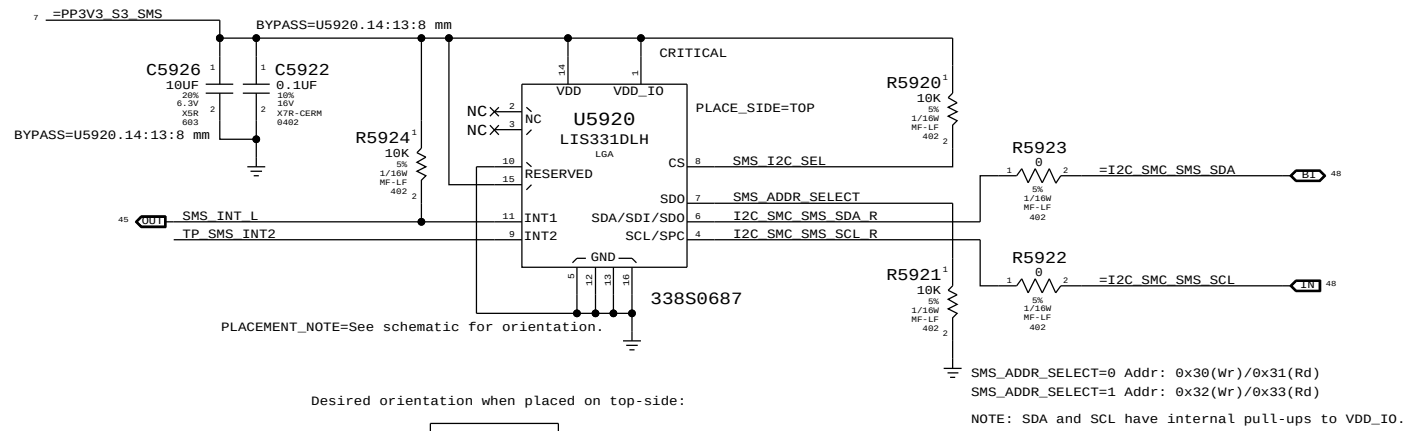


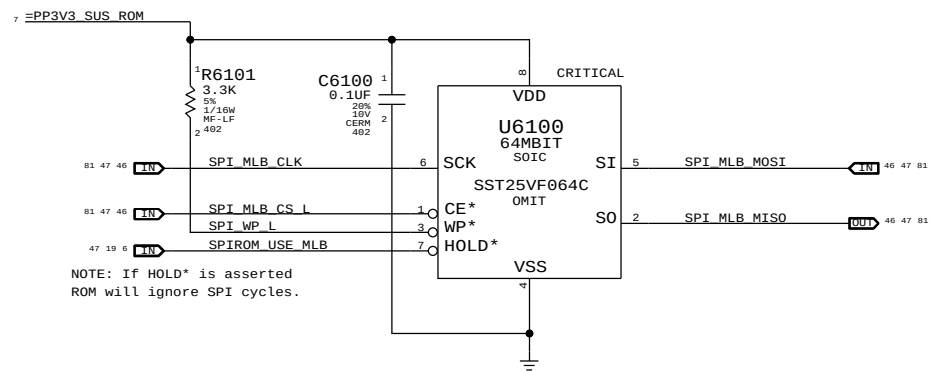
To detect Keyboard backlight, SMC will tristate and read SMC_SYS_KBDLED:
If LOW, keyboard backlight present
If HIGH, keyboard backlight not present
R5853 always stuffed, R5854 only grounded when KB BL flex connected.

Keyboard Backlight Connector

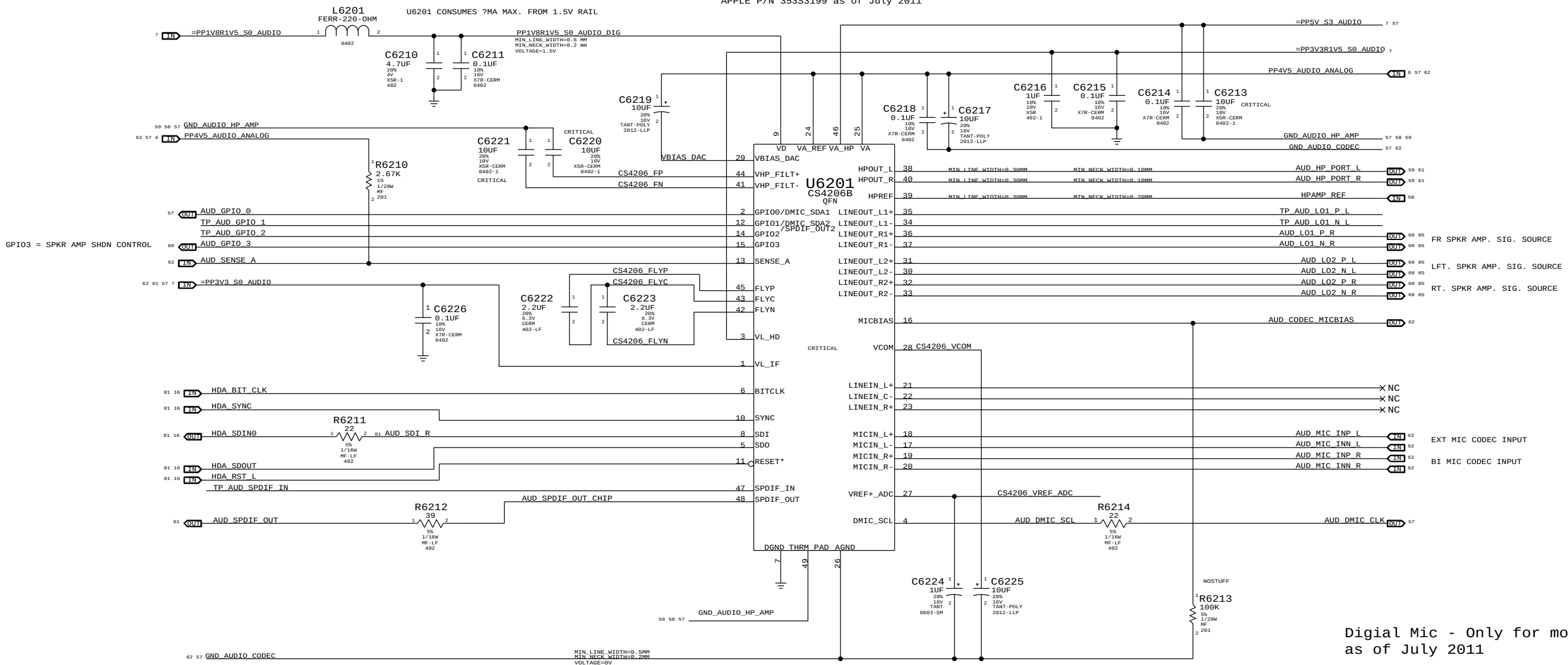


J5815 pin 1 is grounded on keyboard backlight flex



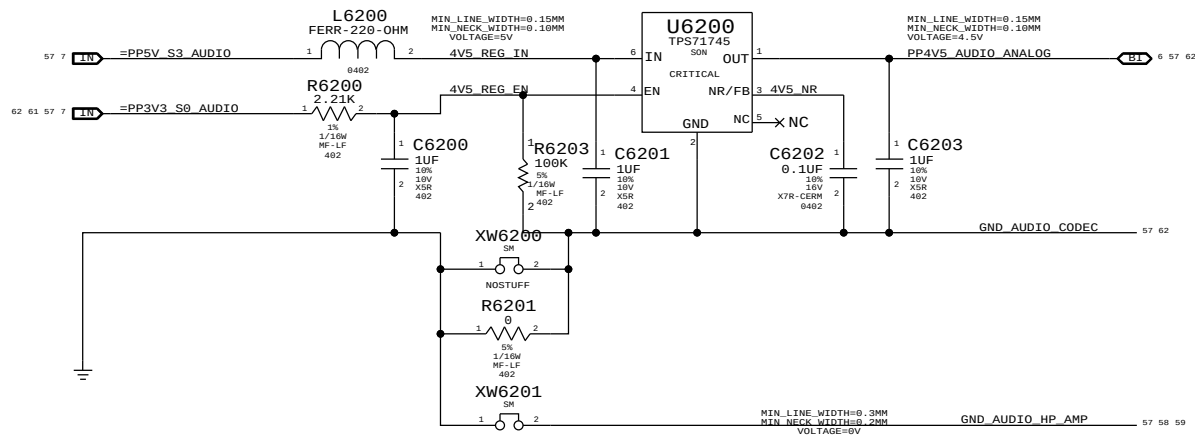


AUDIO CODEC
APPLE P/N 353S3199 as of July 2011



4.5V POWER SUPPLY FOR CODEC

APPLE P/N 353S2281 as of July 2011



NOTES ON J30 audio

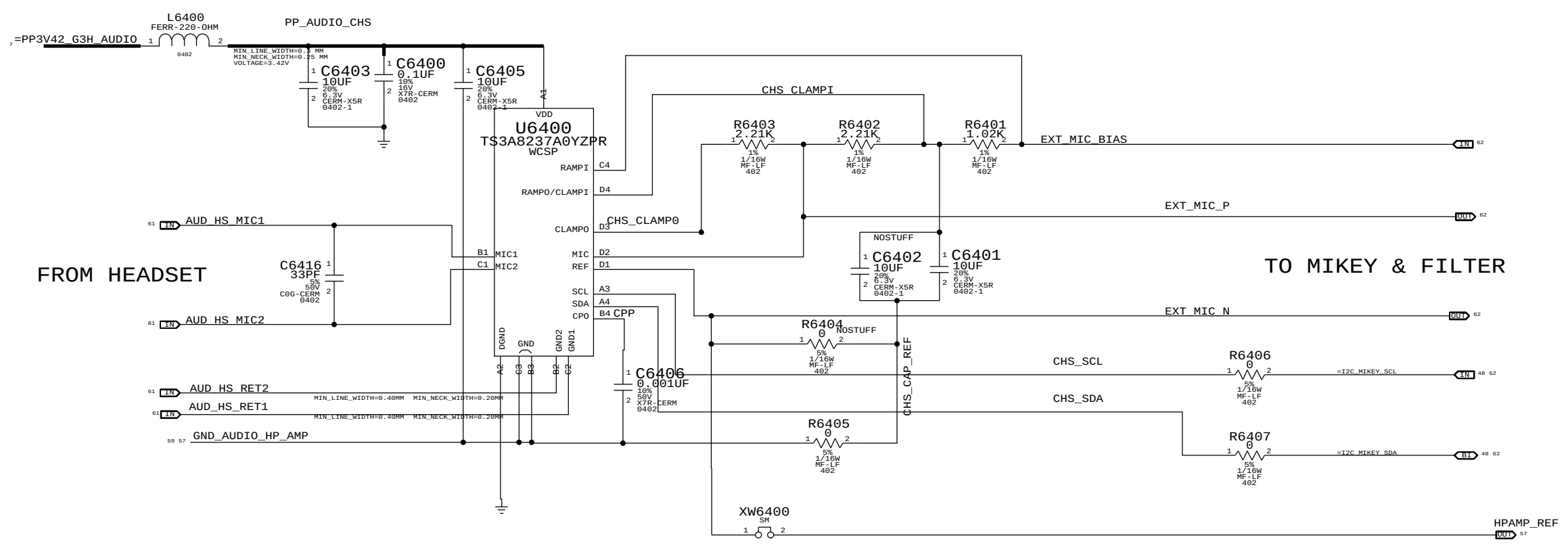
Codec HPamp used for Lineout/HPout. No external HPamp.
3 Spk amplifiers - 2 tweeters and a sub woofer
No line input capability
SPDIF out
China headset support

Digital Mic - Only for mock ups
as of July 2011

57 AUD_DMIC_CLK == TP_AUD_DMIC_CLK
MAKE_BASE=TRUE
57 AUD_GPIO_0 == TP_AUD_DMIC_SDATA
MAKE_BASE=TRUE

SYNC MASTER=KAVITHA J30		SYNC DATE=07/25/2011	
PAGE TITLE		AUDIO: CODEC/REGULATOR	
Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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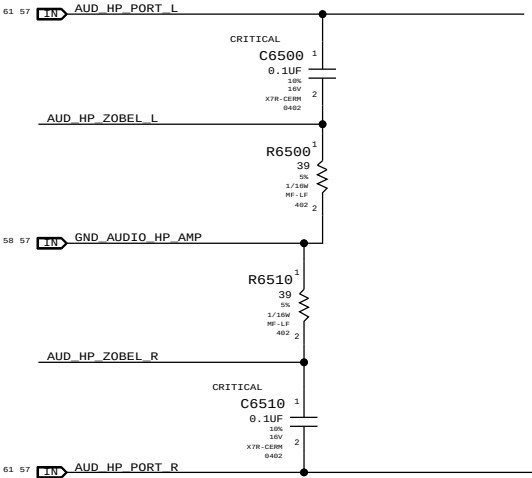
EXTERNAL (HEADSET) MIC INPUT CIRCUITRY APN:353S3066 as of July 2011



I2C ADDRESSES: CHS uses SMBus 0 connections

CHS	U6400	READ	0111	0111	0x77
CHS	U6400	WRITE	0111	0110	0x76

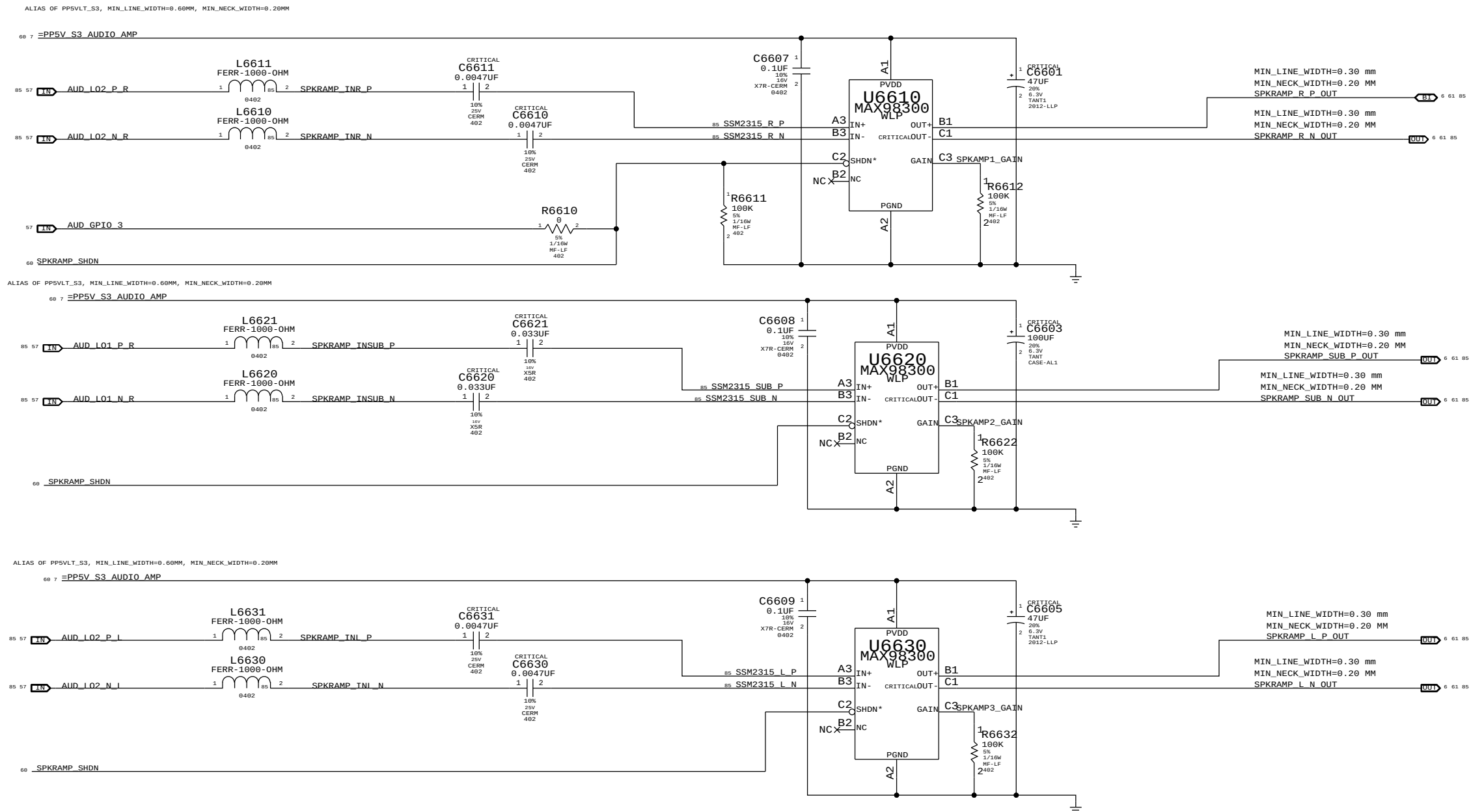
ZOBEL NETWORK & 1ST ORDER DAC FILTER PLACEHOLDER

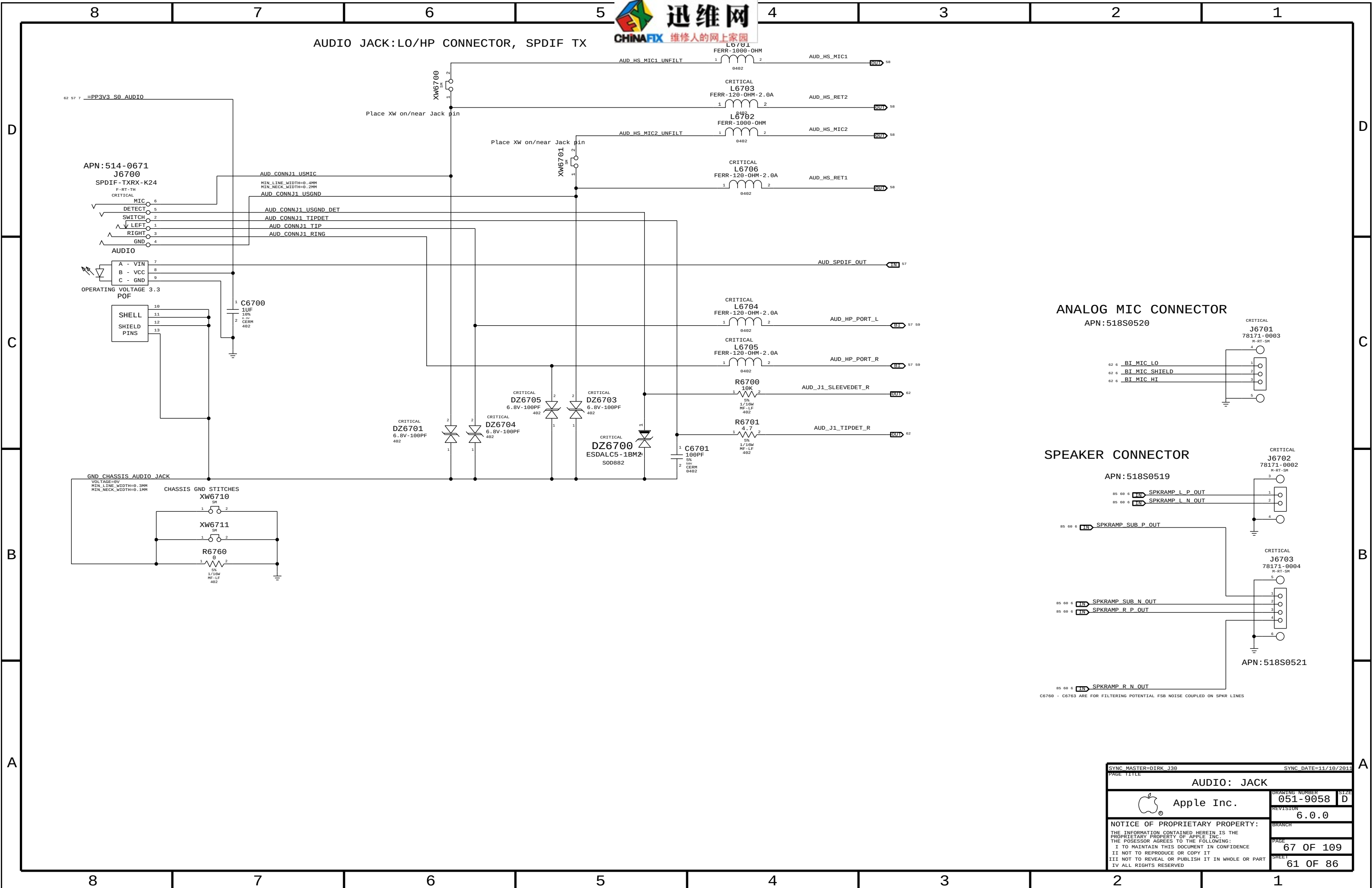


SATELLITE & SUB TWEETER AMPLIFIER
APN:353S2888 as of July 2011

SATELLITE	FC=1.2kHz typical
SUB	FC= 172 HZ typical
GAIN	3DB with Rin=28k typical

Gain Pin	Gain dB
Connect to VDD	12
Connect to VDD through 100k	9
Not connected	6
Connect to GND through 100k	3
Connect to GND	0





SYNC MASTER=DIRK J30		SYNC DATE=11/10/2011	
PAGE TITLE		AUDIO: JACK	
Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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CODEC OUTPUT SIGNAL PATHS

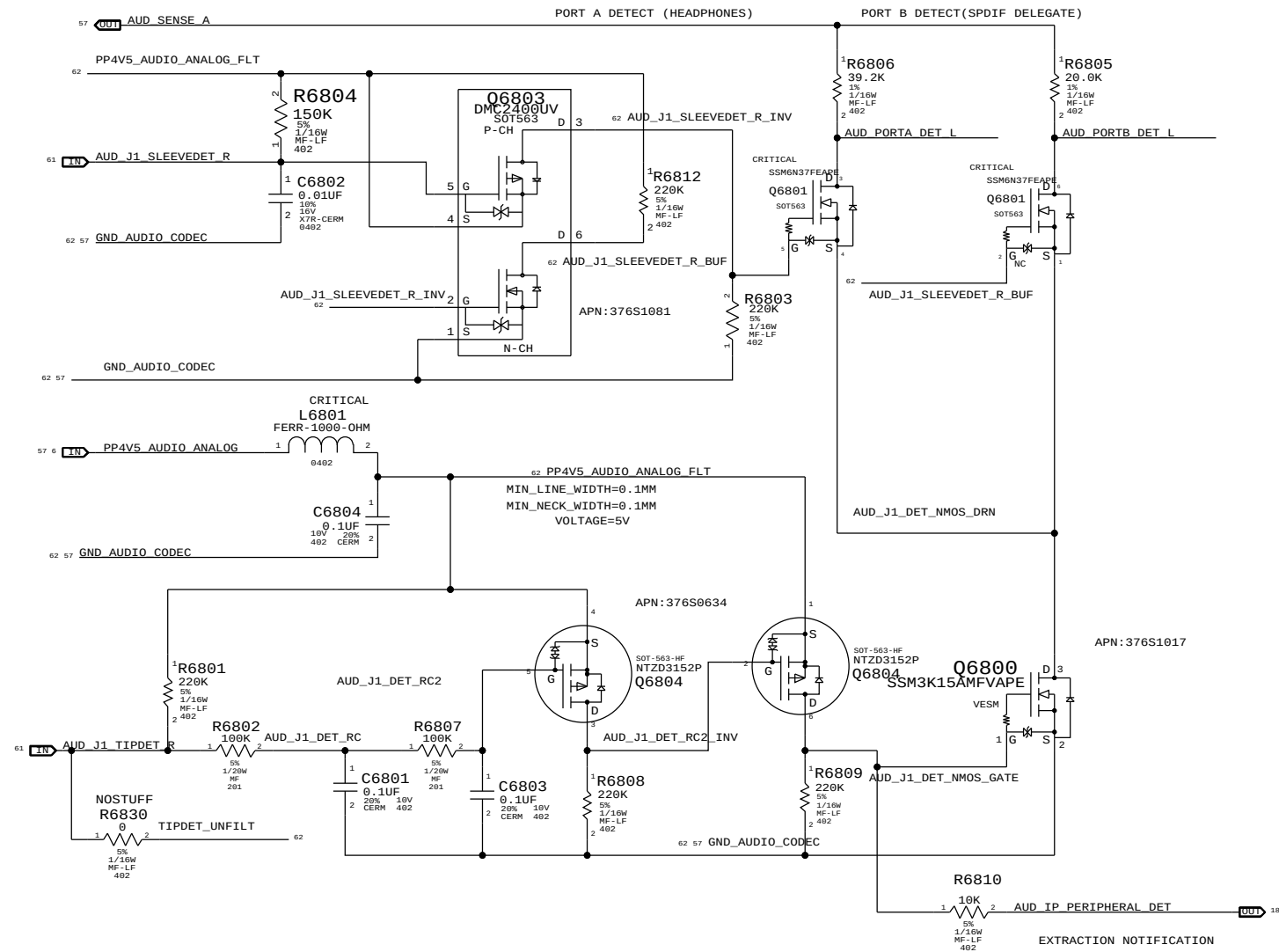
FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0x02 (2)	0x02 (2)	0x09 (9, A)	N/A	0x09 (A)
SATELLITES	0x04 (4)	0x04 (4)	0x0B (11)	GPIO_3	N/A
SUB	0x03 (3)	0x03 (03)	0x0A (10)	GPIO_3	N/A
SPDIF OUT	N/A	0x08 (8)	0x10 (16)	N/A	0x0D (B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
BUILT-IN MIC	0x06 (6)	0x0D (13, B, RIGHT)	MIC_BIAS (80%)	N/A
HEADSET MIC	0x06 (6)	0x0D (13, V22, B, LEFT)	MIKEY	MIKEY

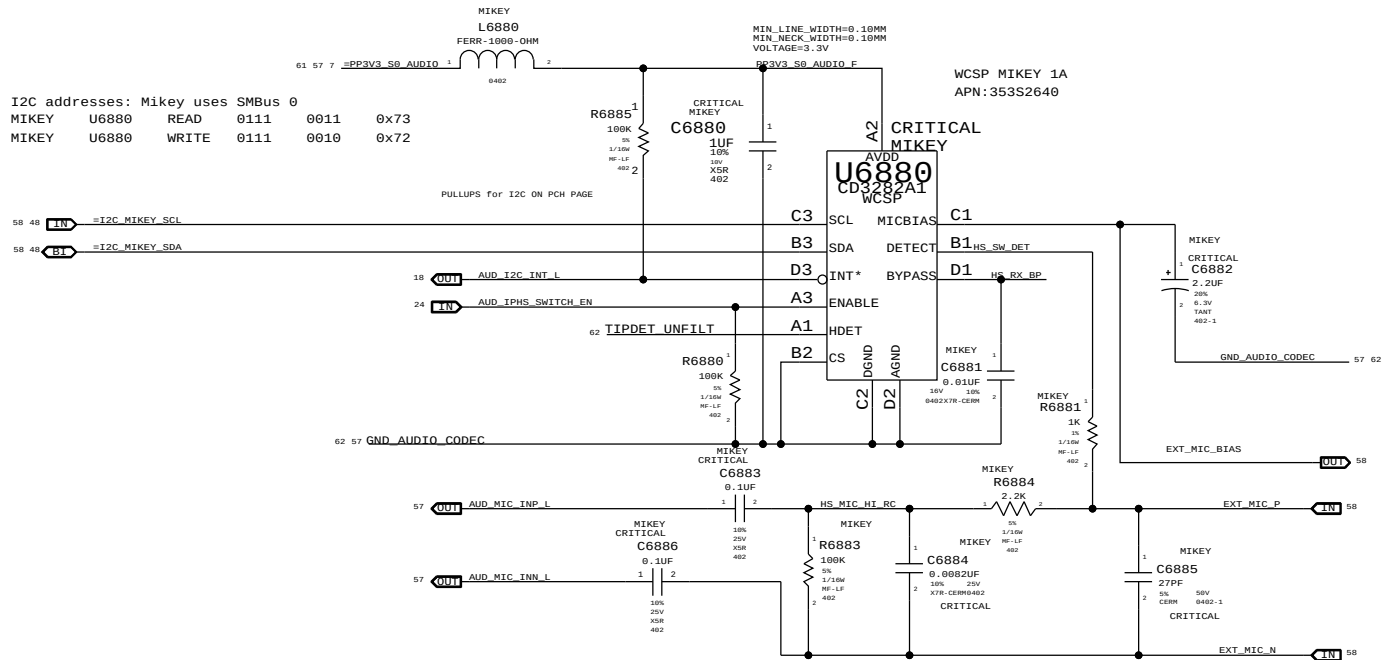
SOUTHBRIDGE RESOURCES

FUNCTION	SYSTEM GPIO	SYSTEM INTERRUPT
AUD_IPHS_SWITCH_EN	PANTHER_POINT GPIO16	N/A
AUD_I2C_INT_L	N/A	PANTHER_POINT GPIO5/PIRQH
AUD_IP_PERIPHERAL_DET	N/A	PANTHER_POINT GPIO3/PIRQH

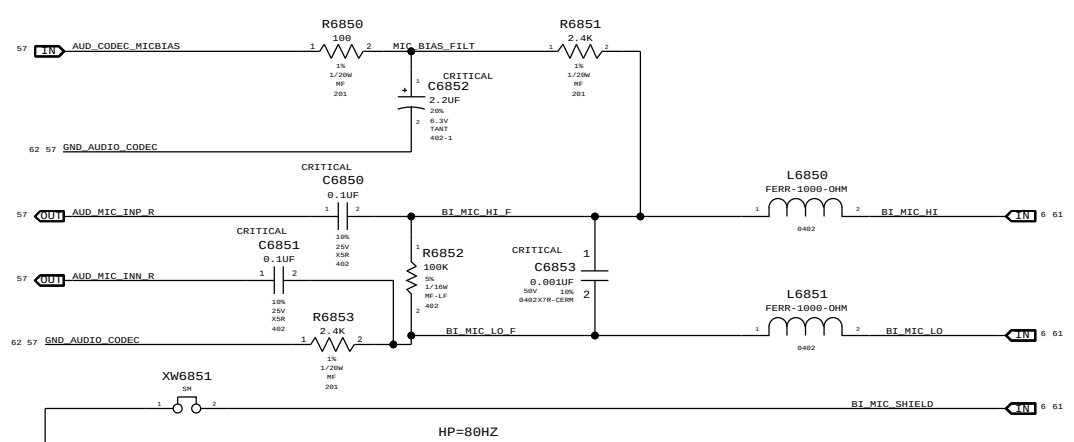


EXTRACTION NOTIFICATION

PORT B LEFT(HEADSET MIC)
HP=80HZ, LP=B.82KHZ

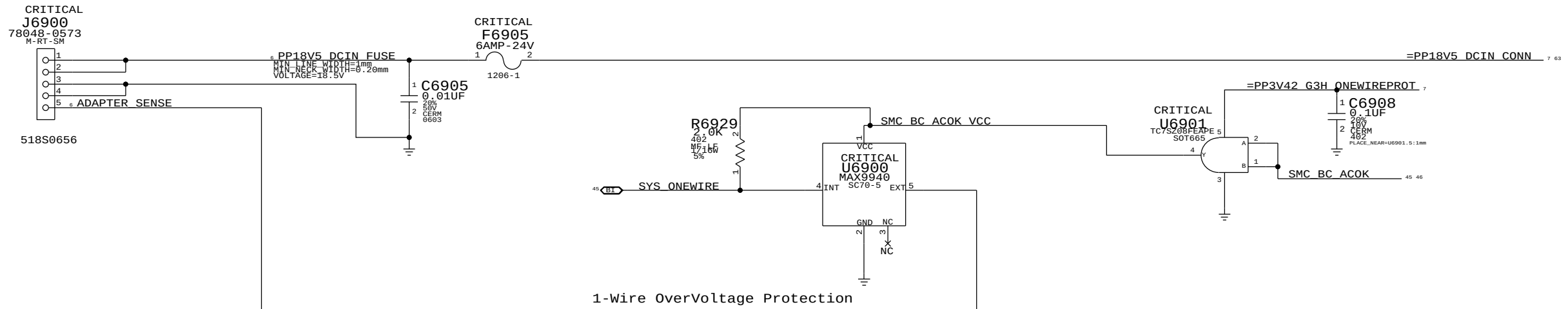


PORT B RIGHT(BUILT-IN MIC)

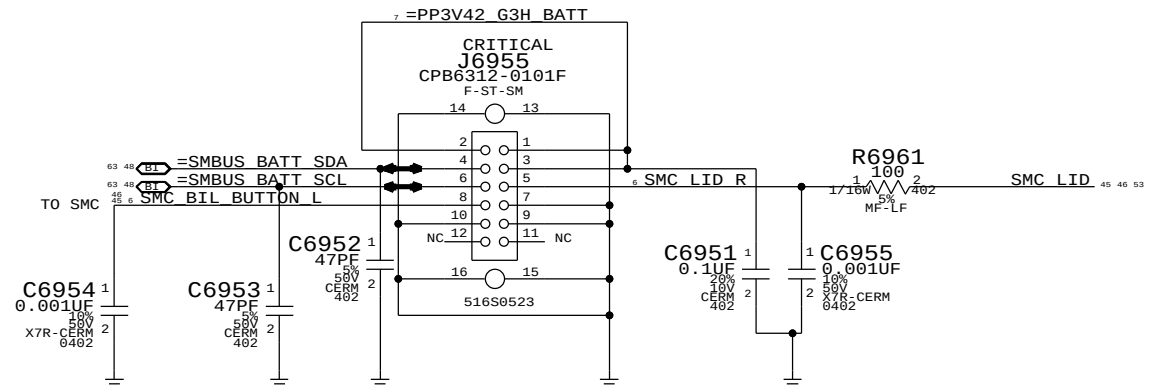


SYNC MASTER=DIRK J30		SYNC DATE=02/20/2012	
PAGE TITLE			
AUDIO:Jack Translators			
Apple Inc.		DRAWING NUMBER	051-9058
REVISION		6.0.0	
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MagSafe DC Power Jack

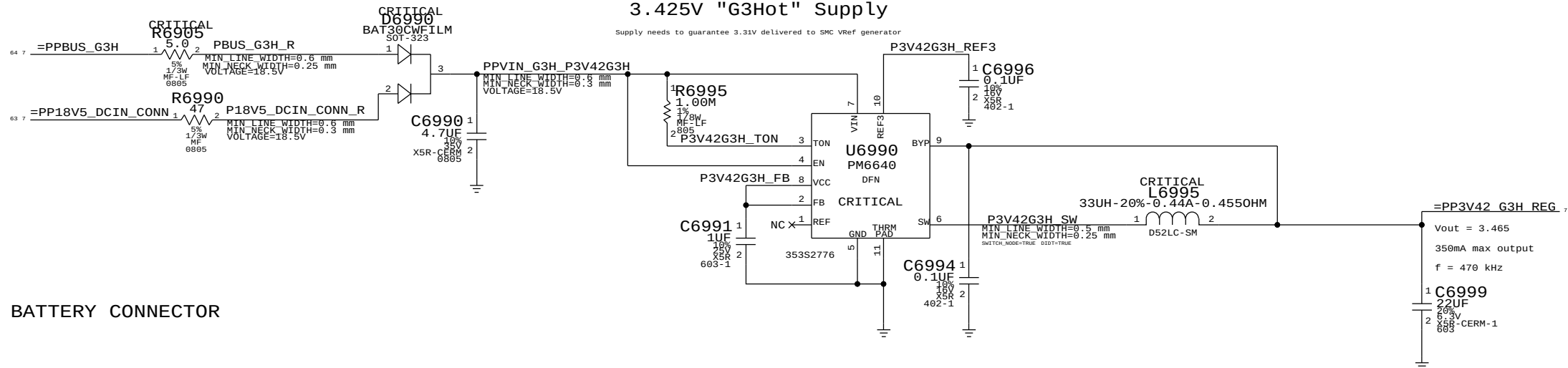


BIL CONNECTOR

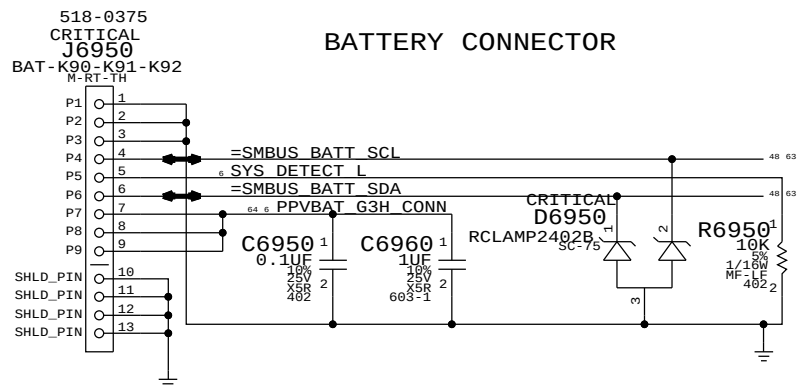


3.425V "G3Hot" Supply

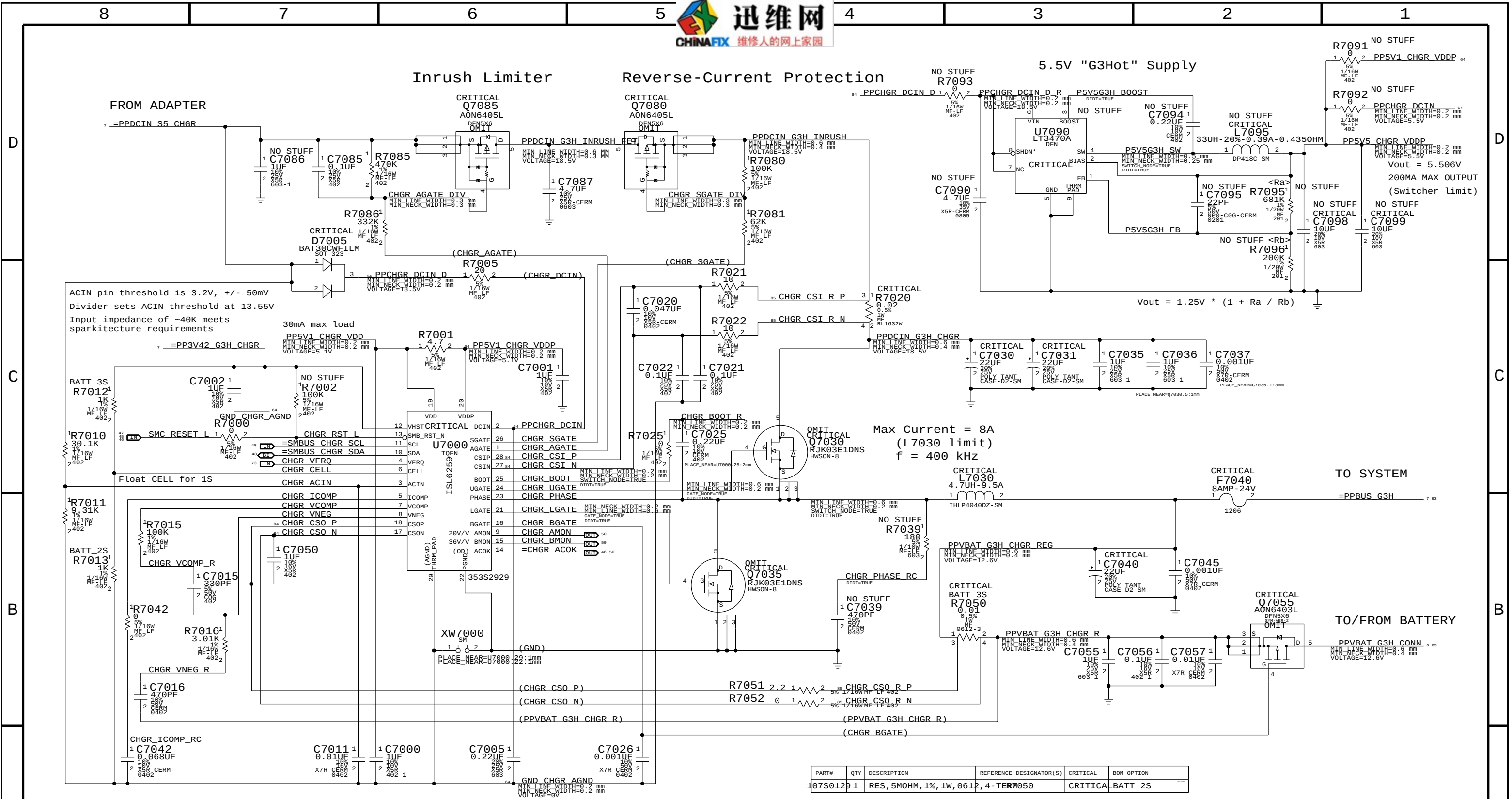
Supply needs to guarantee 3.31V delivered to SMC VRef generator



BATTERY CONNECTOR



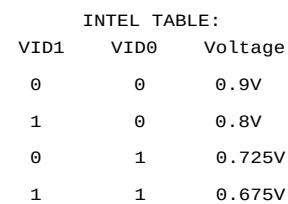
SYNC MASTER=JACK_338		SYNC DATE=07/29/2011	
PAGE TITLE		DC-In & Battery Connectors	
Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0927	2	FDMC3020DC	Q7030,Q7035	CRITICAL	CHARGER_POWER_FET:FAIR
376S0966	2	RJK03E1DNS	Q7030,Q7035	CRITICAL	CHARGER_POWER_FET:REN

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0761	1	SI7137DP	Q7055	CRITICAL	
376S0845	1	SI7149DP	Q7080	CRITICAL	
376S0845	1	SI7149DP	Q7085	CRITICAL	

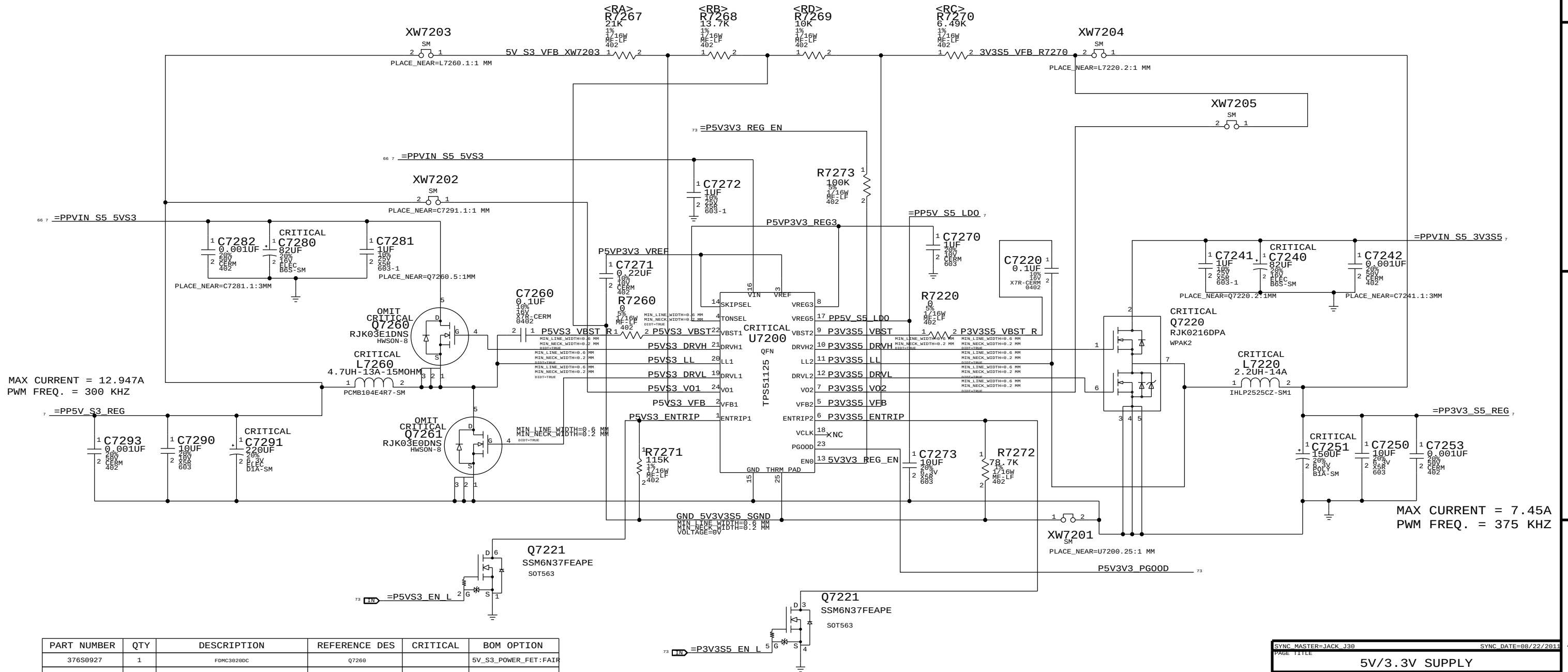
SYNC MASTER=JACK 330		SYNC DATE=09/27/2013	
PAGE TITLE			
PBus Supply & Battery Charger			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-9058	D
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		BRANCH	
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		SHEET	64 OF 86


$$\begin{aligned} OCP &= R7141 \times 8.5\mu A / R7140 \\ OCP &= 8.5A \end{aligned}$$

5V_S3/3.3V_S5 POWER SUPPLY

$$V_{OUT} = (2 * R_A / R_B) + 2$$

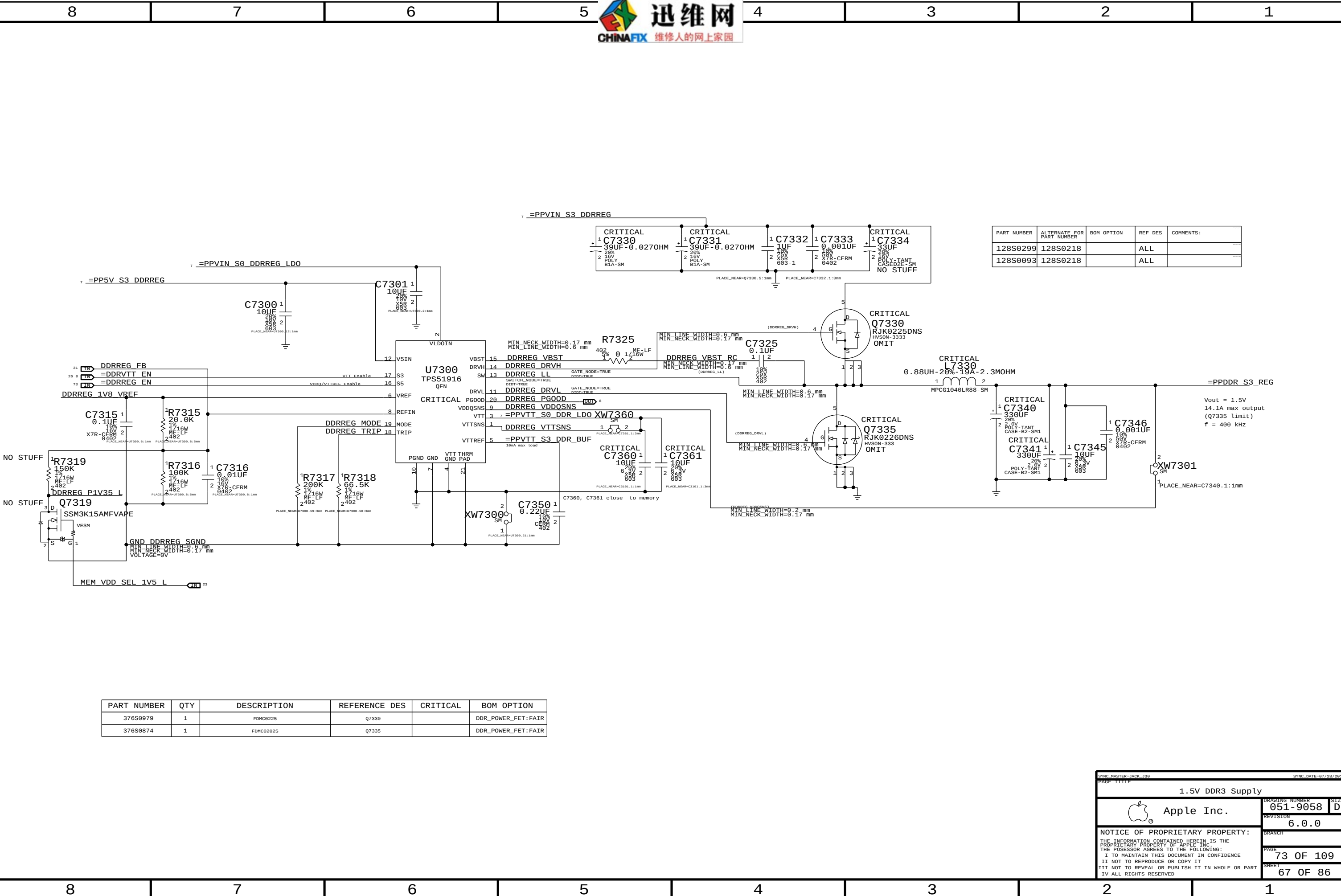
$$V_{OUT} = (2 * RC / RD) + 2$$



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0927	1	FDMC3020DC	Q7260		5V_S3_POWER_FET:FAIR
376S0928	1	FDMC25145DC	Q7261		5V_S3_POWER_FET:FAIR
376S0966	1	RJK03E1DNS	Q7260		5V_S3_POWER_FET:REN
376S0895	1	RJK03E0DNS	Q7261		5V_S3_POWER_FET:REN

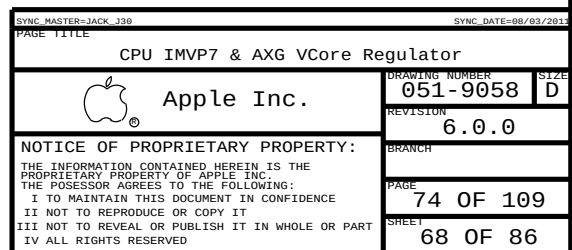
SEPERATED MASTER PG00D FOR BOTH 5V AND 3V3.

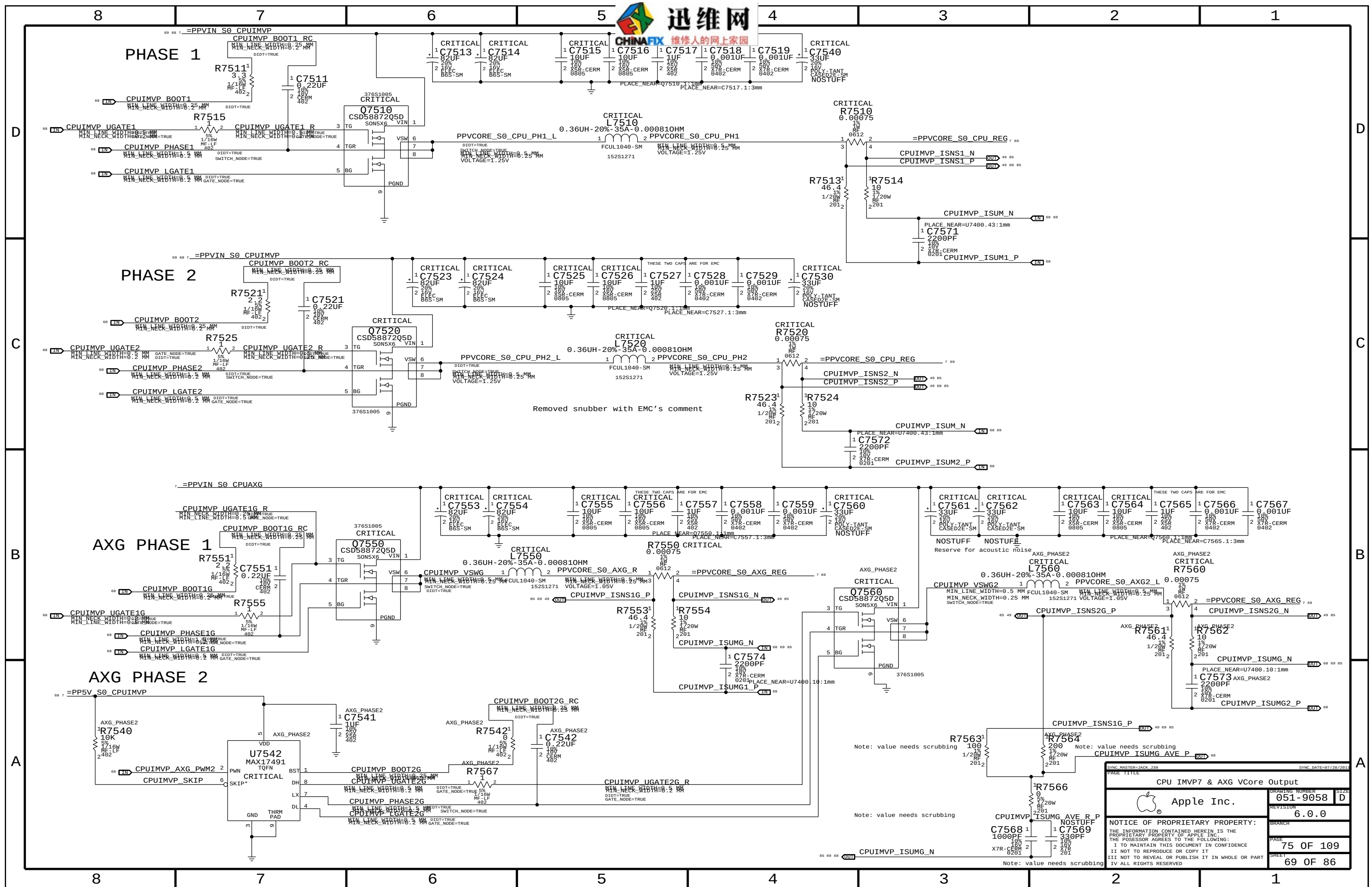
SYMC MASTER=JACK J36		SYMC DATE=08/22/2013	
PAGE TITLE			
5V/3.3V SUPPLY			
	Apple Inc.		DRAWING NUMBER 051-9058
			SIZE D
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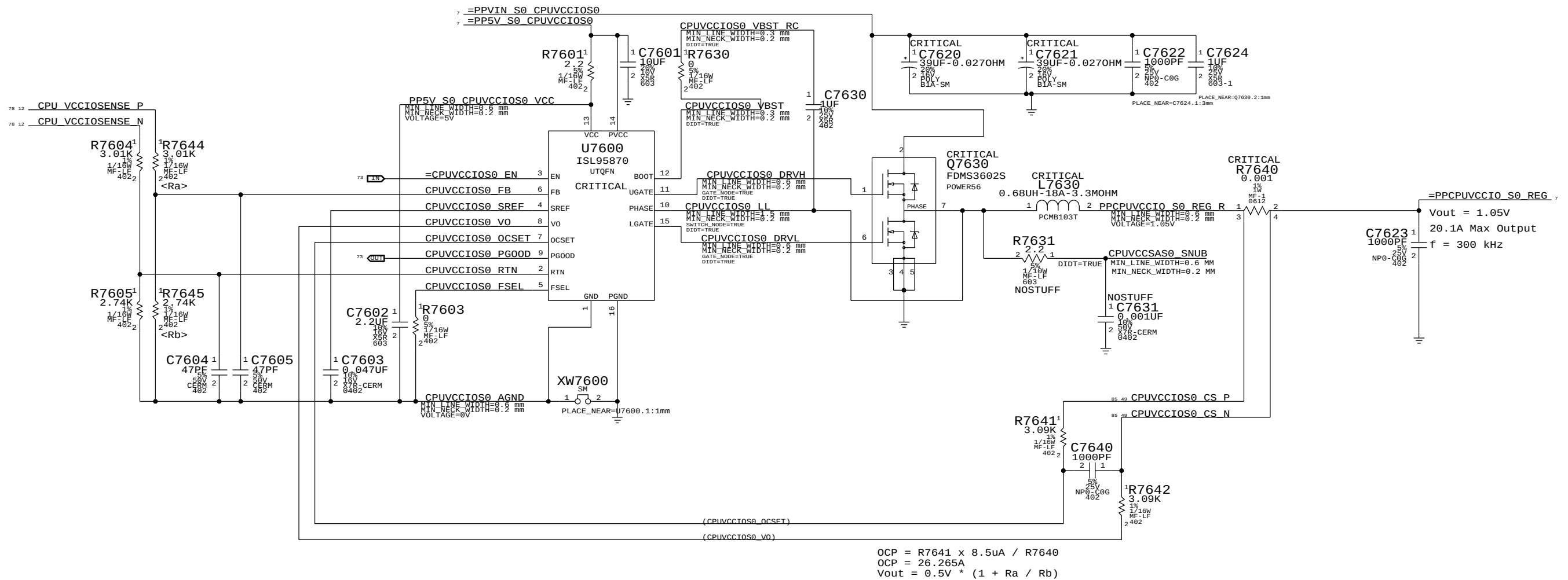
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
128S0299	128S0218		ALL	
128S0093	128S0218		ALL	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0979	1	FDMC0225	Q7330		DDR_POWER_FET:FAIR
376S0874	1	FDMC0202S	Q7335		DDR_POWER_FET:FAIR

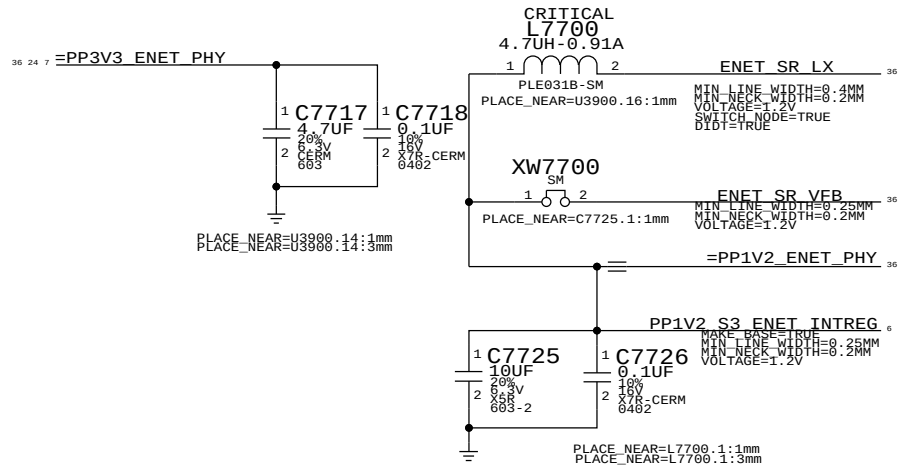




CPU VCCIO (1.05V S0) Regulator

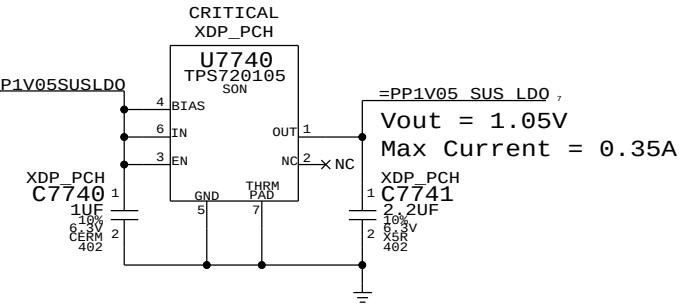


CAESAR IV 1.2V INT.VR CMPTS



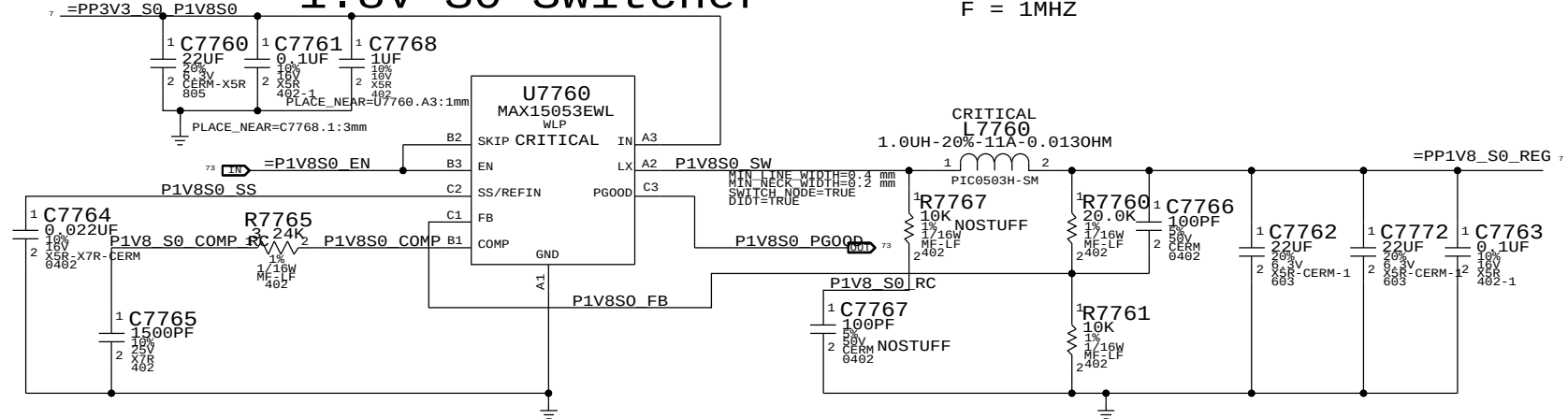
1.05V SUS LDO

Cougar Point requires JTAG pull-ups to be powered at 1.05V in SUS. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.

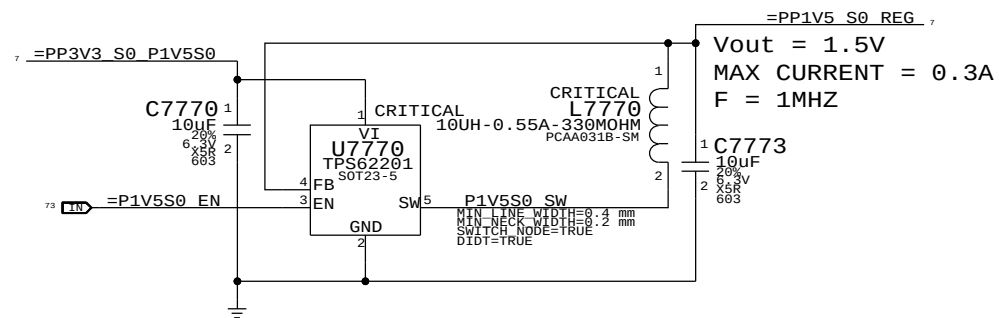


1.8V S0 Switcher

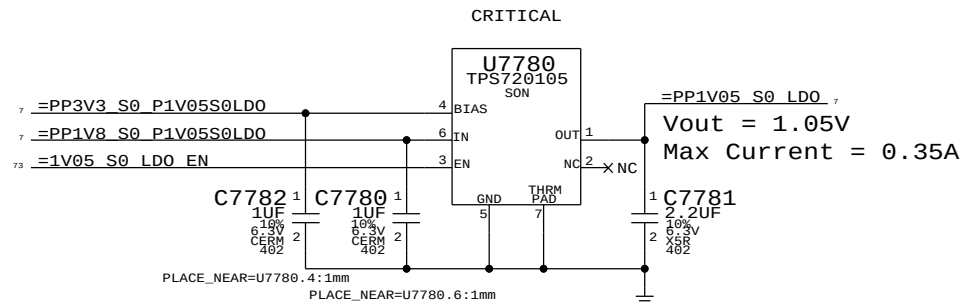
MAX Vout = 1.8V
CURRENT = 2A
F = 1MHZ



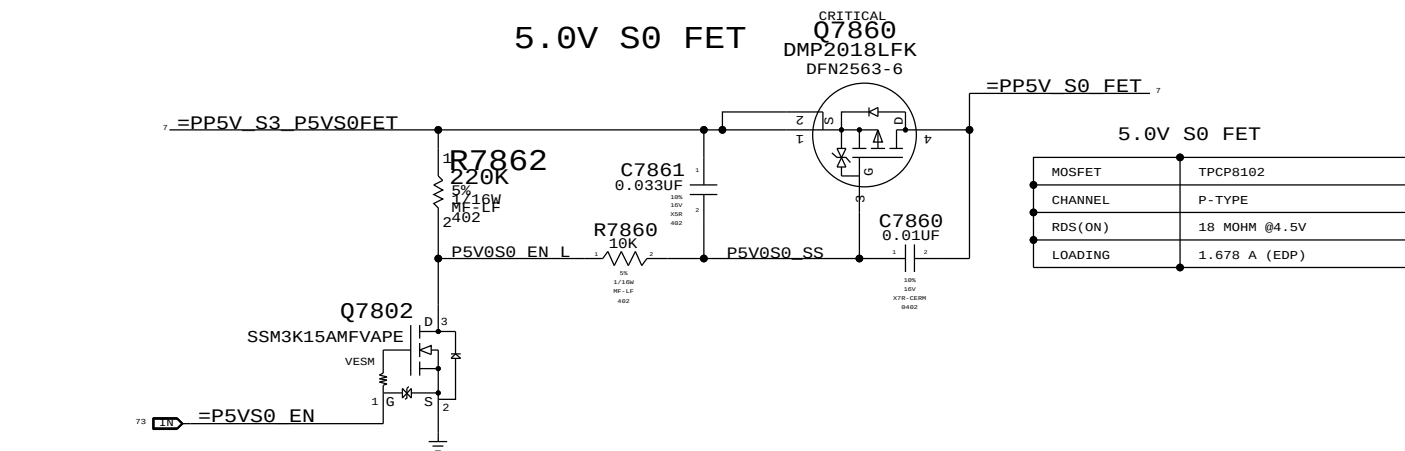
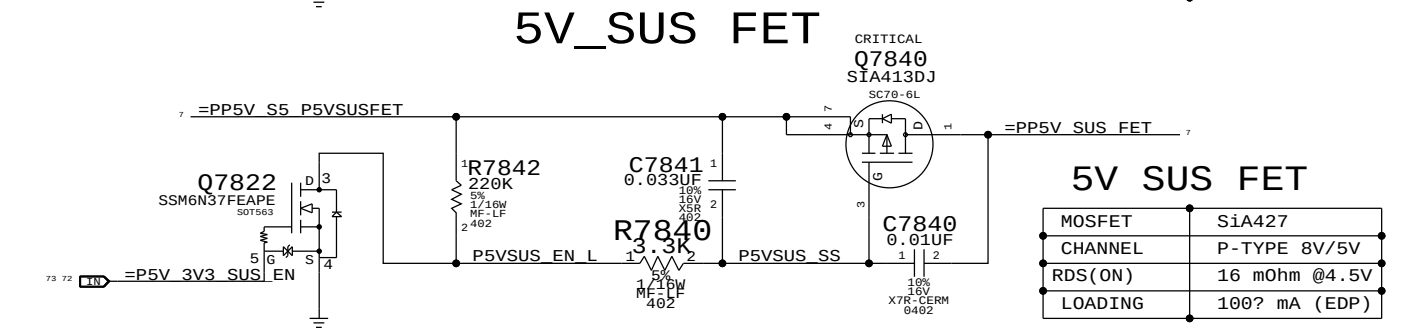
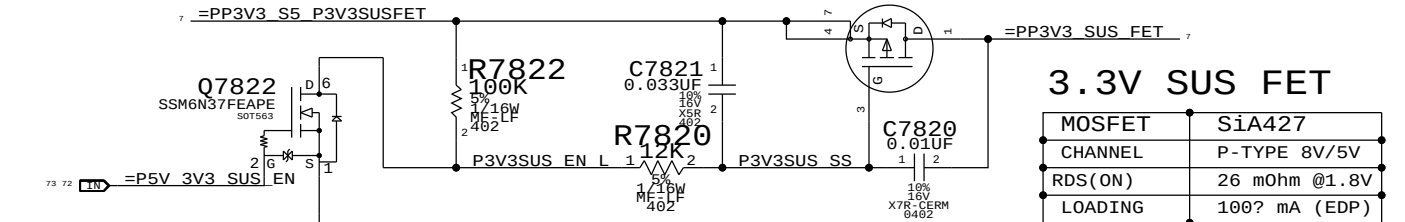
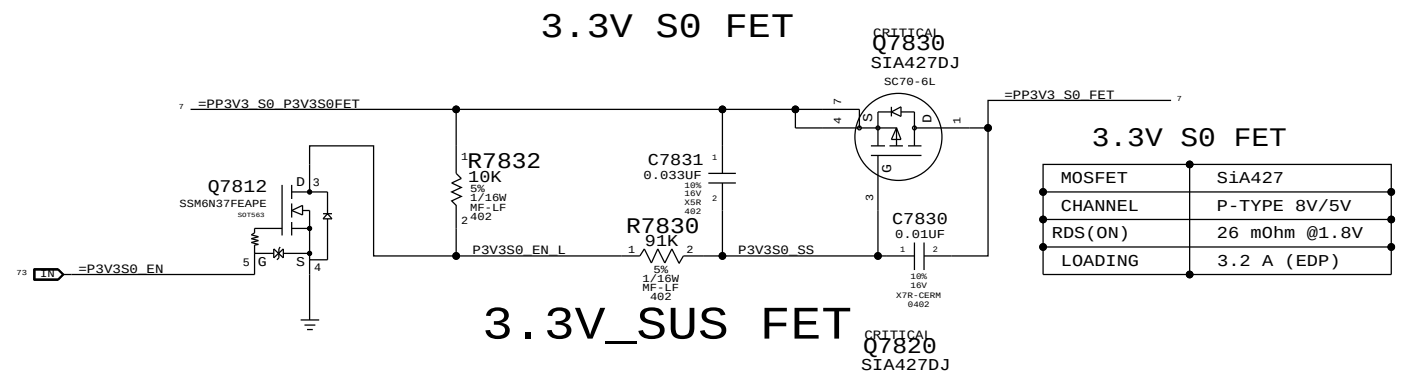
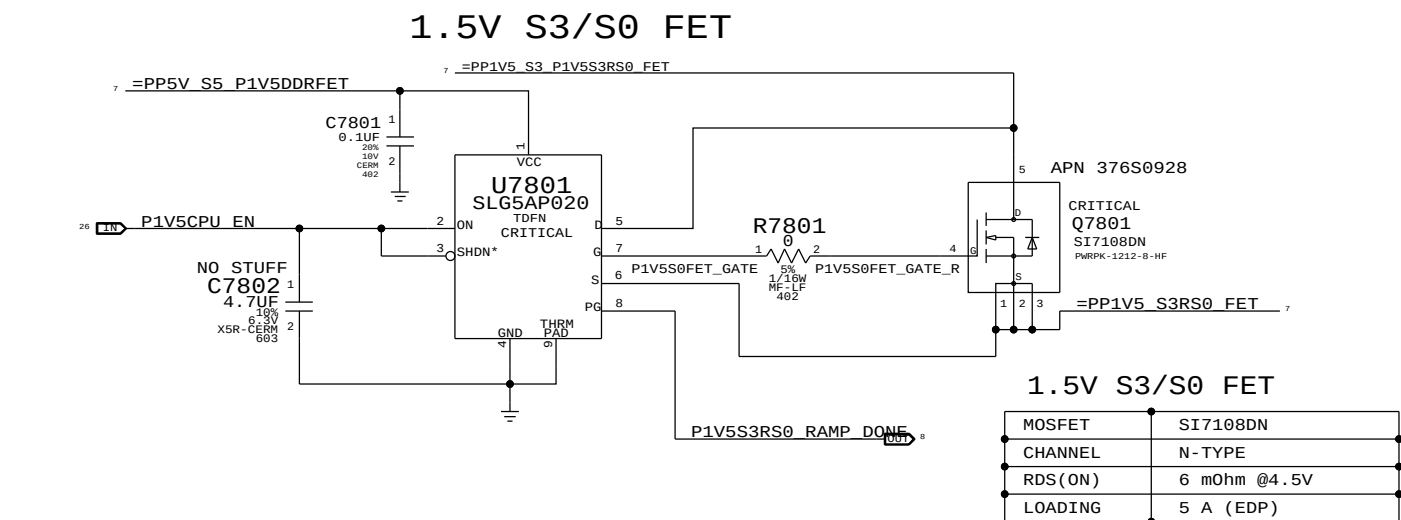
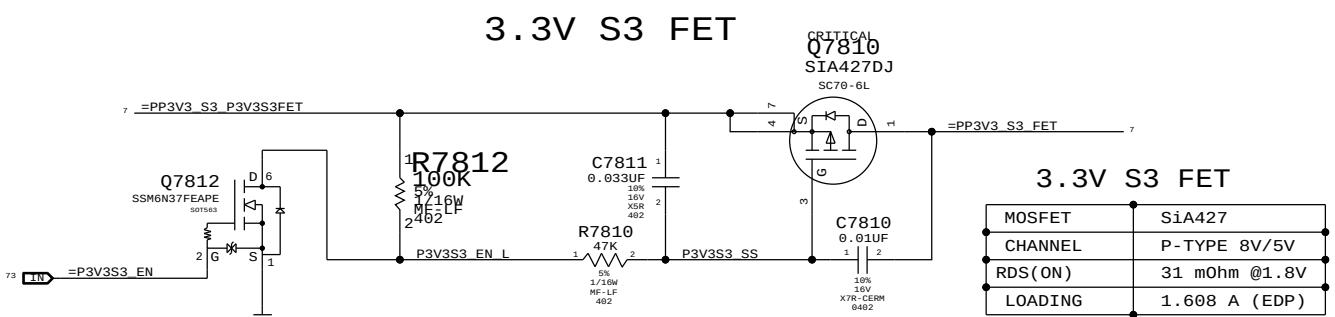
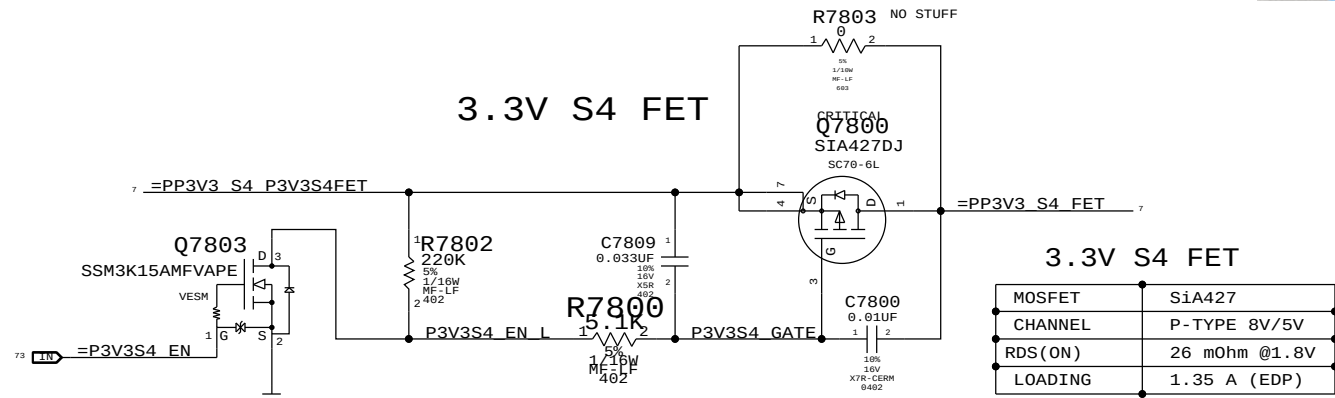
1.5V S0 Switcher



1.05V S0 LDO

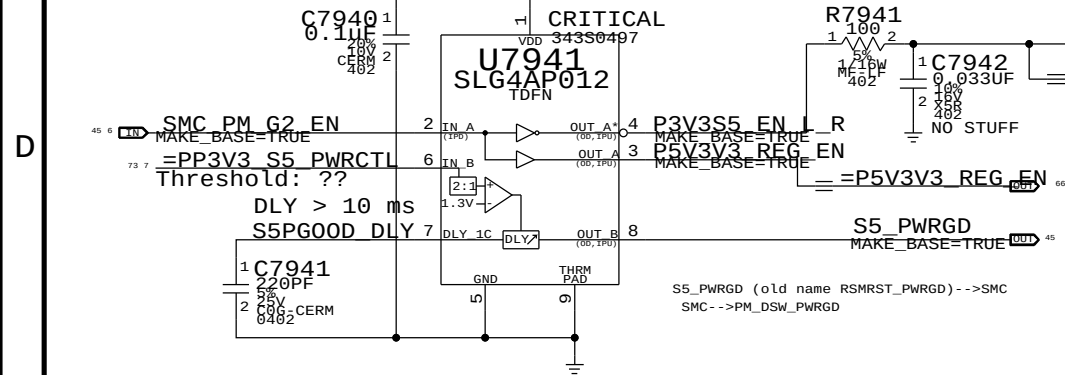


SYNC MASTER=JACK J30		SYNC DATE=07/28/2011	
PAGE TITLE			
Misc Power Supplies			
	Apple Inc.	DRAWING NUMBER	D
		051-9058	
		REVISION	
		6.0.0	
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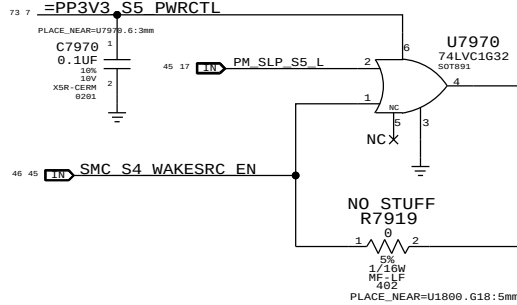
S5 Rail Enables & PGOOD

Internal pull-ups 100K +/- 20%

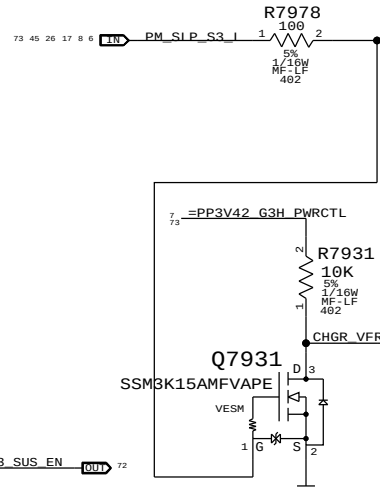


State	SMC_PM_G2_ENABLE	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1	1
Sleep (S3)	1	1	1	0
Deep Sleep (S4)	1	1	0	0
Deep Sleep (S5)	1	0	0	0
Battery Off (G3Hot)	0	0	0	0

3.3V S4 ENABLE



S0 ENABLE

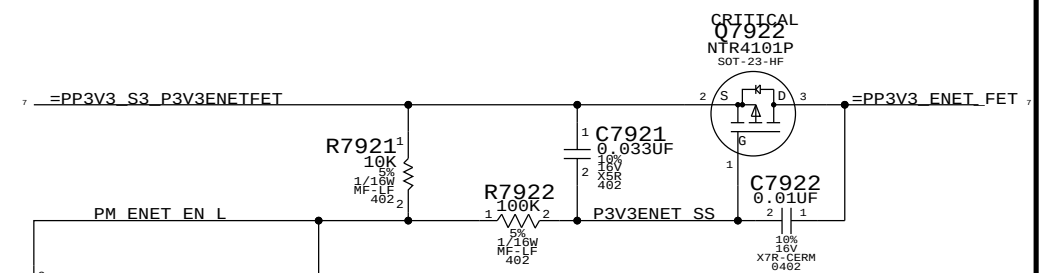


CHGR VFRQ Generation

ENET Enable Generation

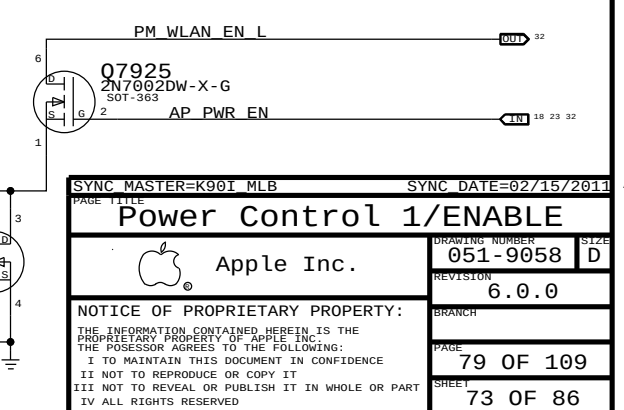
"ENET" = "S0" || ("S3" && "AC" && "WOL_EN")

3.3V ENET FET

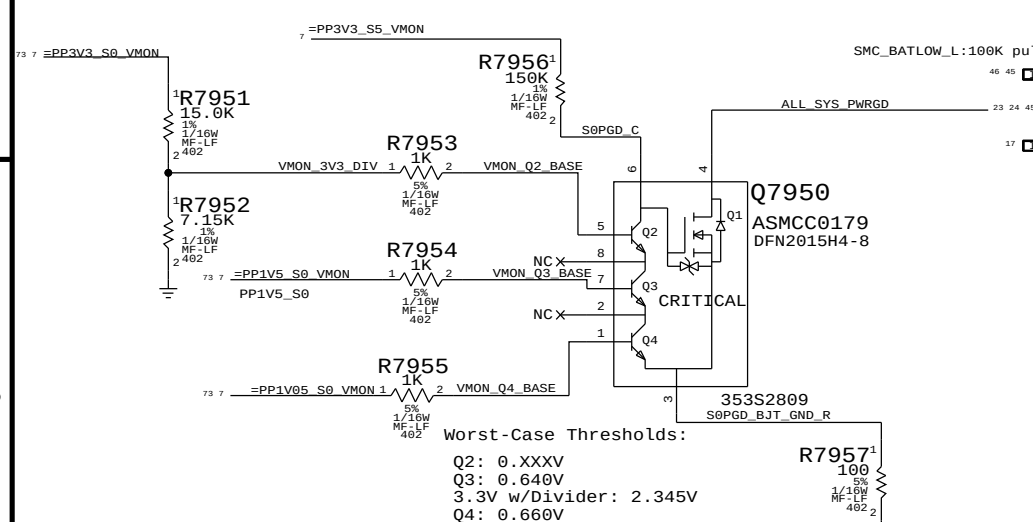


WLAN Enable Generation

"WLAN" = ("S3" && "AP_PWR_EN" && ("AC" || "S0"))
NOTE: S3 term is guaranteed by S3 pull-up on open-drain AP_PWR_EN signal.



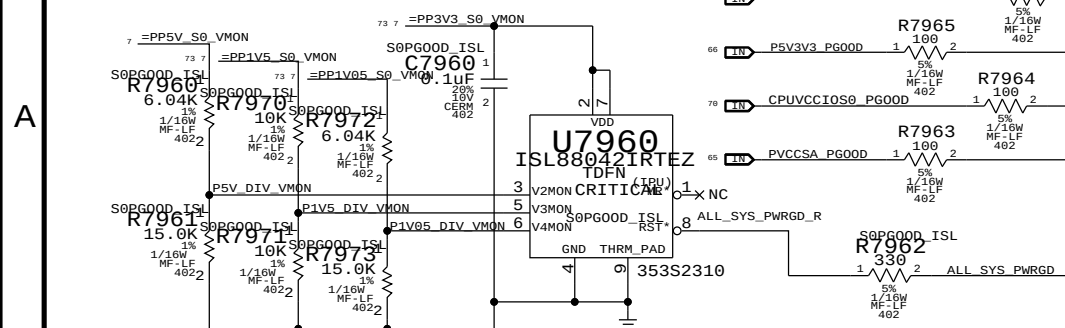
S0 Rail PGOOD (BJT Version)



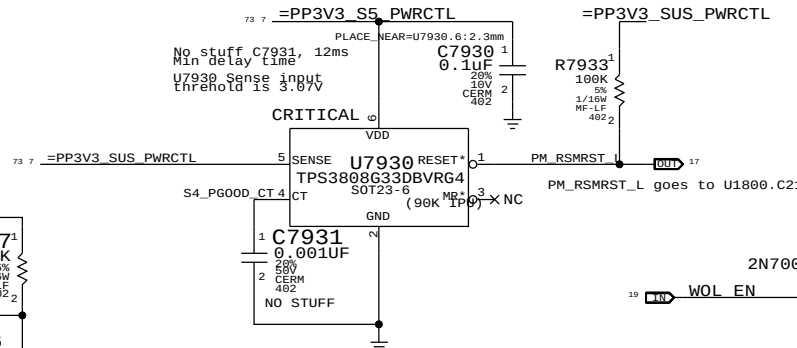
S0 Rail PGOOD Circuitry

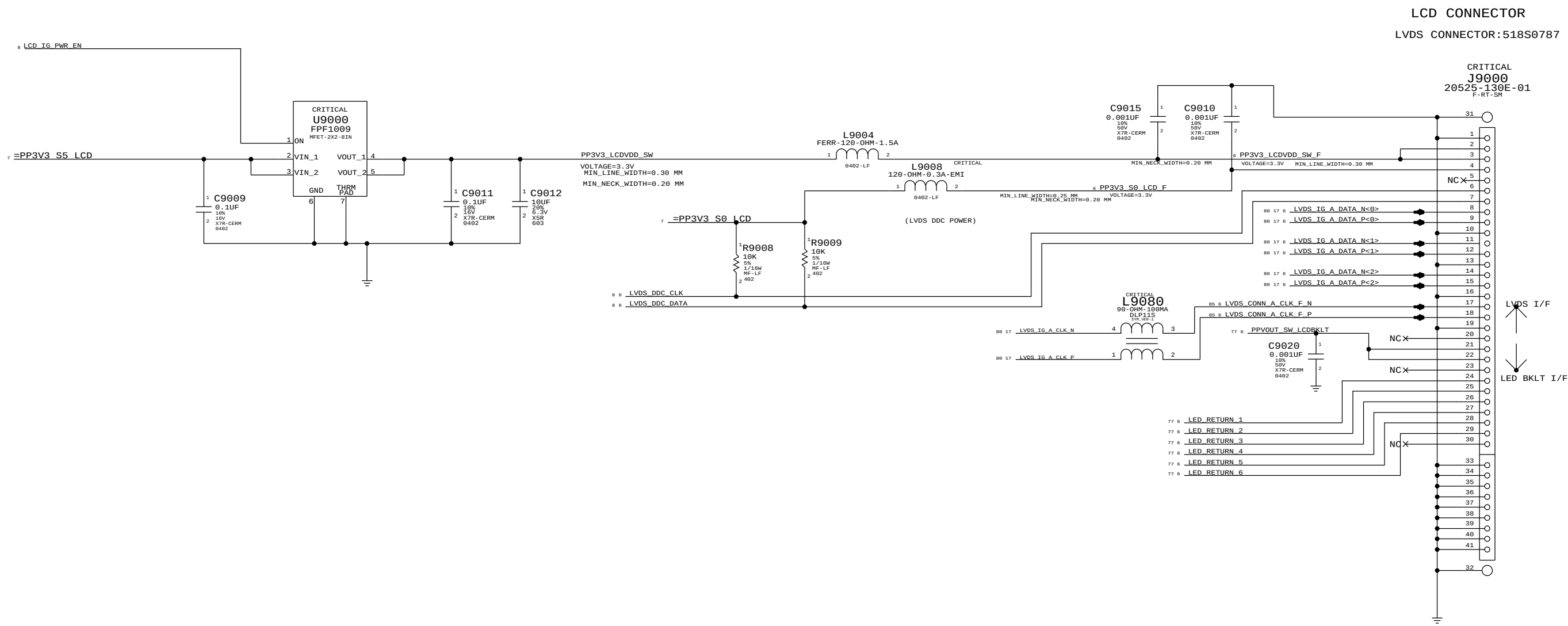
(ISL Version in development)

Thresholds:
VDD: 2.734V-3.010V
V2MON: 2.815V-3.099V
V3MON: 0.572V-0.630V
V4MON: 0.572V-0.630V

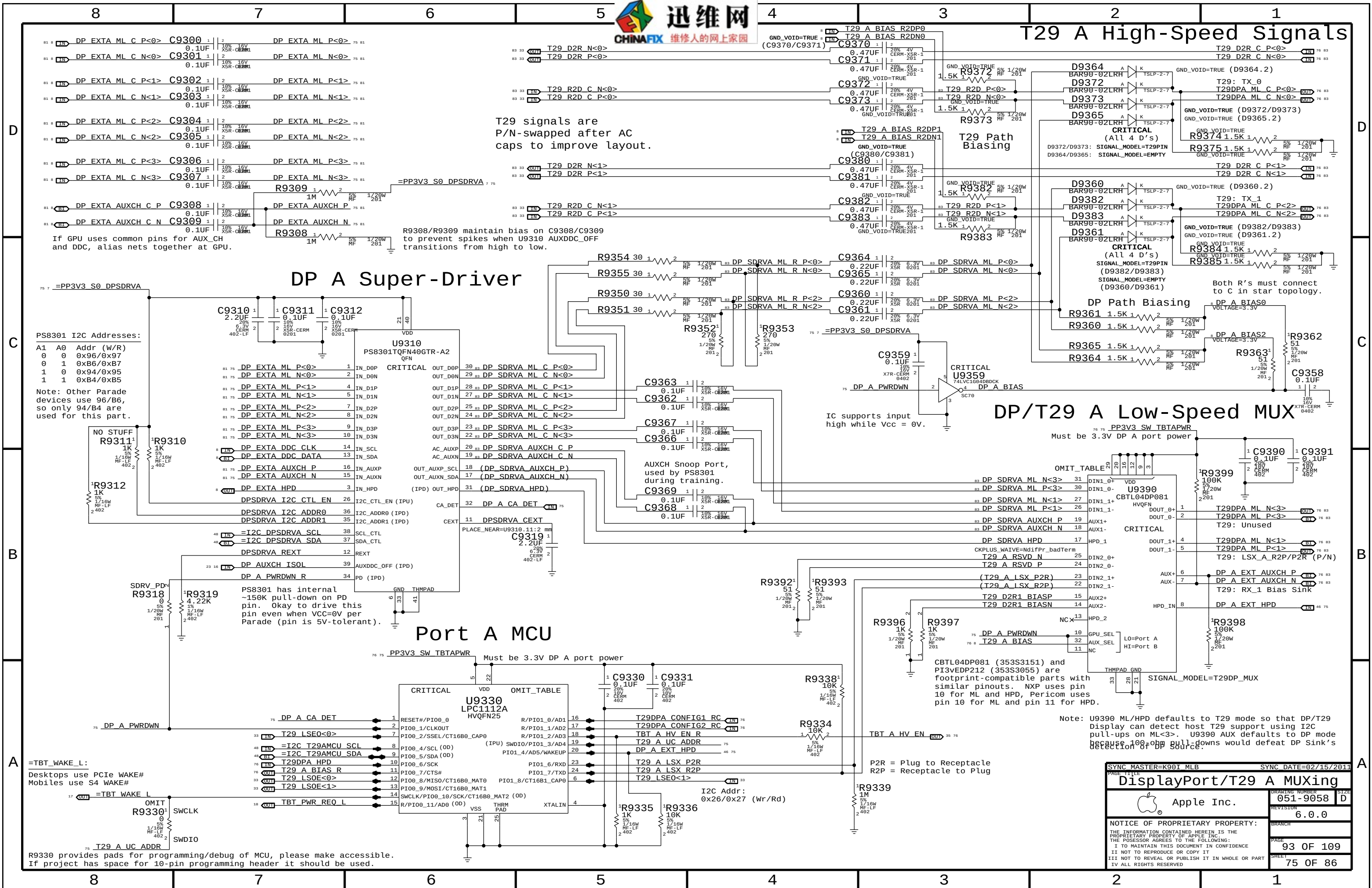


3.3V SUS Detect





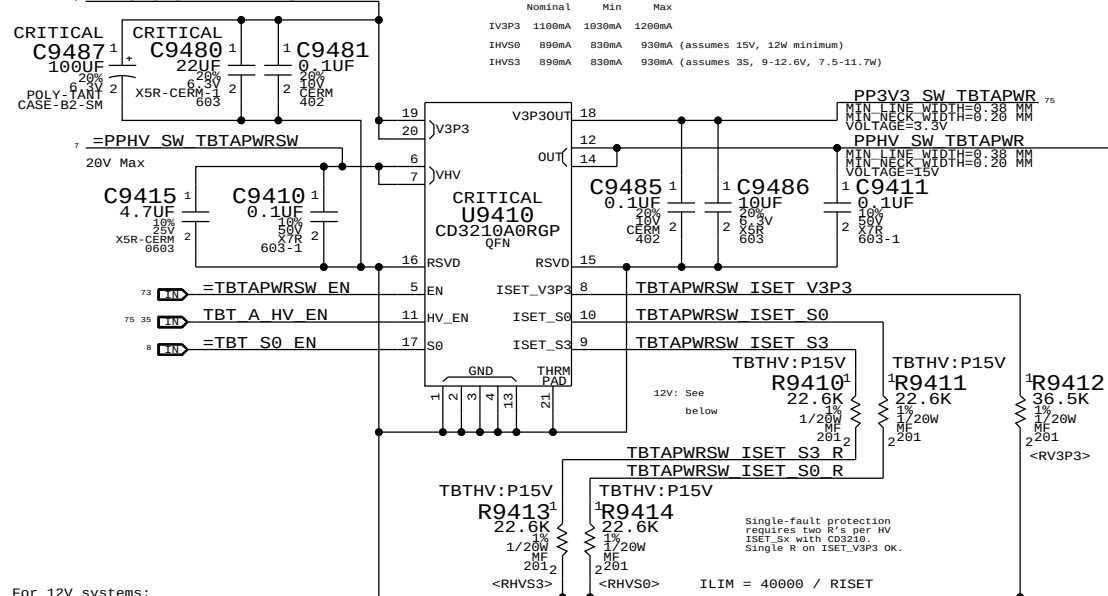
T29 A High-Speed Signals



3.3V/HV Power MUX

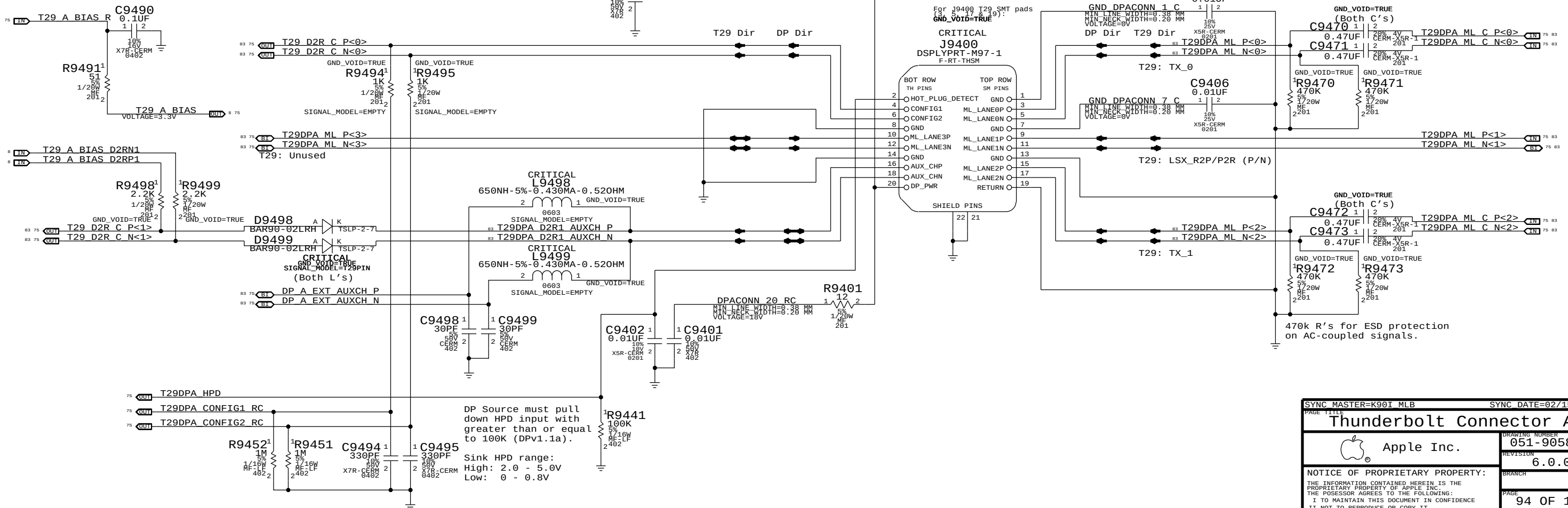
V3P3 must be S4 to support wake from Thunderbolt devices.

=PP3V3 S4 TBTAPWRSW



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9410,R9413		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9411,R9414		TBTHV:P12V

	Nominal	Min	Max
IHVS0/S3	1120mA	1090mA	1170mA (12W minimum)



Thunderbolt Connector A

SYNCH MASTER=K90I MLB		SYNCH DATE=02/15/2011	
PAGE TITLE			
Thunderbolt Connector A			
 Apple Inc.	DRAWING NUMBER	SIZE	
	051-9058	D	
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	BRANCH		
	PAGE	94 OF 109	
	SHEET	76 OF 86	

D

C

B

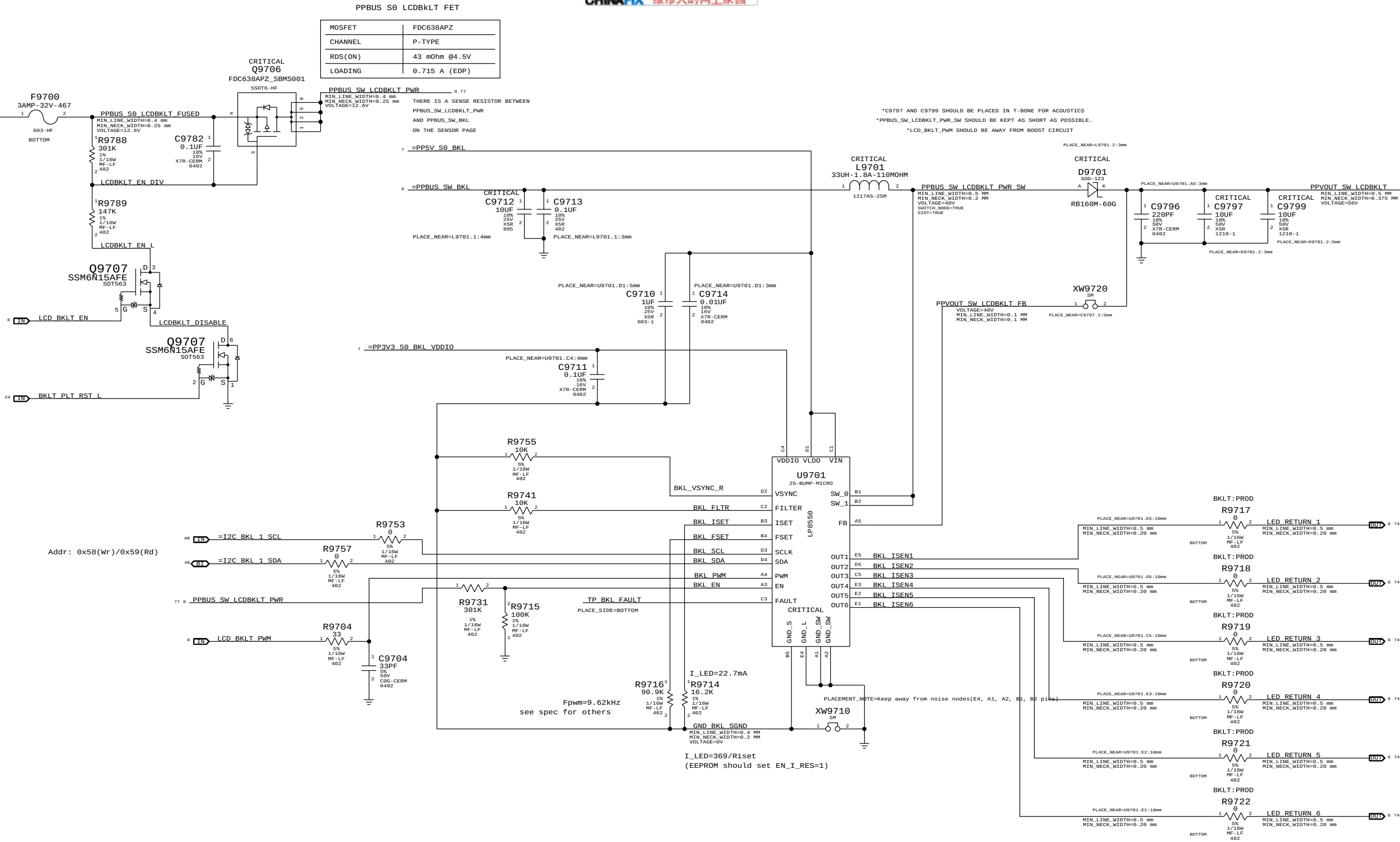
A

D

C

B

A



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3 RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402		SMR9717, R9718, R9719		BKLT:ENG
103S0198	3 RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402		SMR9720, R9721, R9722		BKLT:ENG

10.2 ohm resistors for current
measurement on LED strings.

SYNC MASTER=J31 MLB		SYNC DATE=07/08/2011	
PAGE TITLE			
LCD Backlight Driver			
DRAWING NUMBER		SIZE	
051-9058		D	
REVISION		6.0.0	
BRANCH			
PAGE		97 OF 109	
SHEET		77 OF 86	
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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?	CPU_AGTL	TOP, BOTTOM	=2X_DIELECTRIC	?
CPU_8MIL	*	8 MIL	?				
CPU_COMP	*	20 MIL	?				
CPU_ITP	*	=2:1_SPACING	?				
CPU_VCCSENSE	*	25 MIL	?				

Most CPU signals with impedance requirements are 50-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: Huron River SFF DG (DG-438297_v1.0), Section 4.18 and Huron River Platform Power Delivery DG v1.0 Section 2.7

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_E_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE	*	20 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCI_E_PCH_TX2TX	*	=3X_DIELECTRIC	?	PCI_E_PCH_TX2TX	TOP, BOTTOM	=4X_DIELECTRIC	?
PCI_E_PCH_TX2RX	*	=4X_DIELECTRIC	?	PCI_E_PCH_TX2RX	TOP, BOTTOM	=5X_DIELECTRIC	?
PCI_E_PCH_RX2RX	*	=3X_DIELECTRIC	?	PCI_E_PCH_RX2RX	TOP, BOTTOM	=4X_DIELECTRIC	?
PCI_E_PCH_RX2TX	*	=4X_DIELECTRIC	?	PCI_E_PCH_RX2TX	TOP, BOTTOM	=4X_DIELECTRIC	?
PCI_E_PCH_20THER	*	=3X_DIELECTRIC	?	PCI_E_PCH_20THER	TOP, BOTTOM	=4X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCI_E_PCH_TX	*_PCH_TX	*	PCI_E_PCH_TX2TX
PCI_E_PCH_TX	*_PCH_RX	*	PCI_E_PCH_TX2RX
PCI_E_PCH_RX	*_PCH_RX	*	PCI_E_PCH_RX2RX
PCI_E_PCH_RX	*_PCH_TX	*	PCI_E_PCH_RX2TX
PCI_E_PCH_TX	*	*	PCI_E_PCH_20THER
PCI_E_PCH_RX	*	*	PCI_E_PCH_20THER

SOURCE: 471984_Cheif_River_MS_PD6_1.0 and the spacing rule is adjusted per SI team feedback.

Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
DMI_S2N	PCIE_85D	PCIE_PCH_TX	DMI_S2N P<3:0>	9 17
DMI_S2N	PCIE_85D	PCIE_PCH_TX	DMI_S2N N<3:0>	9 17
DMI_N2S	PCIE_85D	PCIE_PCH_RX	DMI_N2S P<3:0>	9 17
DMI_N2S	PCIE_85D	PCIE_PCH_RX	DMI_N2S N<3:0>	9 17
FDI_DATA	PCIE_85D	PCIE_PCH_RX	FDI_DATA P<7:0>	9 17
FDI_DATA	PCIE_85D	PCIE_PCH_RX	FDI_DATA N<7:0>	9 17
	CPU_50S	CPU_AGTL	FDI_FSYNC<1..0>	9 17
	CPU_50S	CPU_AGTL	FDI_LSYNC<1..0>	9 17
	CPU_50S	CPU_AGTL	FDI_INT	9 17
CPU_PECI	CPU_50S	CPU_COMP	CPU_PECI	10 19 46
PM_SYNC	CPU_50S	CPU_AGTL	PM_SYNC	10 17
PM_MEM_PWRGD	CPU_50S	CPU_AGTL	PM_MEM_PWRGD	10 17 26
	CPU_50S	CPU_ITP	XDP_DBRESET_L	10 23 24
	CPU_50S	CPU_ITP	XDP_CPU_PRDY_L	10 23
	CPU_50S	CPU_ITP	XDP_CPU_PREQ_L	10 23
	CPU_50S	CPU_AGTL	PM_EXT_TS_L<0>	
	CPU_50S	CPU_AGTL	PM_EXT_TS_L<1>	
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<0>	10
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<1>	10
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<2>	10
	CPU_50S	CPU_ITP	CPU_CFG<11..0>	9 23
CPU_CATERR_L	CPU_50S	CPU_AGTL	CPU_CATERR_L	10 45
	CPU_50S	CPU_AGTL	CPU_VCCIO_SEL	8 12
CPU_PROCHOT_L	CPU_50S	CPU_AGTL	CPU_PROCHOT_L	10 45 46 68
CPU_PWRGD	CPU_50S	CPU_AGTL	CPU_PWRGD	10 19 23
PM_THRMTRIP_L	CPU_50S	CPU_8MIL	PM_THRMTRIP_L	10 19 46
DMI_CLK100M	CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M_CPU_P	10 16
DMI_CLK100M	CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M_CPU_N	10 16
ITPCPU_CLK100M	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M_P	10 16
ITPCPU_CLK100M	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M_N	10 16
ITPXDP_CLK100M	CLK_PCIE_90D	CLK_PCIE	ITPXDP_CLK100M_P	16 23
ITPXDP_CLK100M	CLK_PCIE_90D	CLK_PCIE	ITPXDP_CLK100M_N	16 23
XDP_CPU_CLK100M_P	CLK_PCIE_90D	CLK_PCIE	XDP_CPU_CLK100M_P	23
XDP_CPU_CLK100M_N	CLK_PCIE_90D	CLK_PCIE	XDP_CPU_CLK100M_N	23
EDP_COMP	CPU_27P4S	CPU_COMP	EDP_COMP	9
CPU_PEG_COMP	CPU_27P4S	CPU_COMP	CPU_PEG_COMP	9
XDP_TDI	CPU_50S	CPU_ITP	XDP_CPU_TDI	10 23
XDP_TDO	CPU_50S	CPU_ITP	XDP_CPU_TDO	10 23
XDP_TMS	CPU_50S	CPU_ITP	XDP_CPU_TMS	10 23
XDP_TCK	CPU_50S	CPU_ITP	XDP_CPU_TCK	10 23
XDP_TRST_L	CPU_50S	CPU_ITP	XDP_CPU_TRST_L	10 23
XDP_BPM_L	CPU_50S	CPU_ITP	XDP_BPM_L<3..0>	10 23
XDP_BPM_R_L	CPU_50S	CPU_ITP	CPU_CFG<15..12>	9 23
(FSB_CPURST_L)	CPU_50S	CPU_ITP	XDP_CPURST_L	23
CPU_VCCAXG_SENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	12 68
CPU_VCCAXG_SENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	12 68
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCIOSENSE_P	12 79
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCIOSENSE_N	12 79
CPU_VCCAXG_SENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_SENSE_P	12 68
CPU_VCCAXG_SENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_SENSE_N	12 68
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VDD0_SENSE_P	12
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VDD0_SENSE_N	12
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_P	9
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_N	9
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_P	9
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_N	9
CPU_VIDALERT_L	CPU_50S	CPU_COMP	CPU_VIDALERT_L	12 68
CPU_VIDSCLK	CPU_50S	CPU_COMP	CPU_VIDSCLK	12 68
CPU_VIDSOUT	CPU_50S	CPU_COMP	CPU_VIDSOUT	12 68

CPU_VCCSA_VID<0>
CPU_VCCSA_VID<1>

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_37S	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=STANDARD	=STANDARD
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_50S	TOP, BOTTOM	Y	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	TOP, BOTTOM	Y	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
MEM_50S	ISL10	N	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	ISL10	N	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
MEM_50S	ISL3, ISL4, ISL9	Y	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	ISL3, ISL4, ISL9	Y	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK P<5..0>	11 27
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK N<5..0>	11 27
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A CKE<3..0>	11 27
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A CS_L<3..0>	11 27
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A ODT<3..0>	11 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0>	11 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0>	11 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS_L	11 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS_L	11 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE_L	11 27
MEM_A_DQ_BYTE0	MEM_50S	MEM_DATA	MEM A DQ<7..0>	11 28
MEM_A_DQ_BYTE1	MEM_50S	MEM_DATA	MEM A DQ<15..8>	11 28
MEM_A_DQ_BYTE2	MEM_50S	MEM_DATA	MEM A DQ<23..16>	11 28
MEM_A_DQ_BYTE3	MEM_50S	MEM_DATA	MEM A DQ<31..24>	11 28
MEM_A_DQ_BYTE4	MEM_50S	MEM_DATA	MEM A DQ<39..32>	11 28
MEM_A_DQ_BYTE5	MEM_50S	MEM_DATA	MEM A DQ<47..40>	11 28
MEM_A_DQ_BYTE6	MEM_50S	MEM_DATA	MEM A DQ<55..48>	11 28
MEM_A_DQ_BYTE7	MEM_50S	MEM_DATA	MEM A DQ<63..56>	11 28

MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS P<0>	11 28
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS N<0>	11 28
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS P<1>	11 28
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS N<1>	11 28
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS P<2>	11 28
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS N<2>	11 28
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS P<3>	11 28
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS N<3>	11 28
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS P<4>	11 28
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS N<4>	11 28
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS P<5>	11 28
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS N<5>	11 28
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS P<6>	11 28
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS N<6>	11 28
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS P<7>	11 28
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS N<7>	11 28

MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK P<5..0>	11 29
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK N<5..0>	11 29
MEM_B_CNT1	MEM_37S	MEM_CTRL	MEM B CKE<3..0>	11 29
MEM_B_CNT1	MEM_37S	MEM_CTRL	MEM B CS_L<3..0>	11 29
MEM_B_CNT1	MEM_37S	MEM_CTRL	MEM B ODT<3..0>	11 29
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..0>	11 29
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0>	11 29
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS_L	11 29
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS_L	11 29
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE_L	11 29
MEM_B_DQ_BYTE0	MEM_50S	MEM_DATA	MEM B DQ<7..0>	11 28
MEM_B_DQ_BYTE1	MEM_50S	MEM_DATA	MEM B DQ<15..8>	11 28
MEM_B_DQ_BYTE2	MEM_50S	MEM_DATA	MEM B DQ<23..16>	11 28
MEM_B_DQ_BYTE3	MEM_50S	MEM_DATA	MEM B DQ<31..24>	11 28
MEM_B_DQ_BYTE4	MEM_50S	MEM_DATA	MEM B DQ<39..32>	11 28
MEM_B_DQ_BYTE5	MEM_50S	MEM_DATA	MEM B DQ<47..40>	11 28
MEM_B_DQ_BYTE6	MEM_50S	MEM_DATA	MEM B DQ<55..48>	11 28
MEM_B_DQ_BYTE7	MEM_50S	MEM_DATA	MEM B DQ<63..56>	11 28

MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS P<0>	11 28
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS N<0>	11 28
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS P<1>	11 28
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS N<1>	11 28
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS P<2>	11 28
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS N<2>	11 28
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS P<3>	11 28
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS N<3>	11 28
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS P<4>	11 28
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS N<4>	11 28
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS P<5>	11 28
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS N<5>	11 28
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS P<6>	11 28
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS N<6>	11 28
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS P<7>	11 28
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS N<7>	11 28

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=3:1_SPACING	?
MEM_CTRL2MEM	*	=2:5:1_SPACING	?
MEM_CMD2CMD	*	=1:5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1:5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	25 MILS	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_CLK	*	MEM_DATA2MEM
MEM_DATA	MEM_CTRL	*	MEM_DATA2MEM
MEM_DATA	MEM_CMD	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA
MEM_DATA	MEM_DQS	*	MEM_DATA2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_DATA	*	*	MEM_20THER
MEM_DQS	*	*	MEM_20THER

Need to support MEM*-style wildcards!

DDR3: Sandybridge SFF 2C when routed on Type-3 (Through hole) should follow rPGA guidelines

per Huron River SFF DG rev1.0 (#438297).

DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.

DQ to DQS matching per byte lane should be within 0.127mm.

DQS to clock matching should be within [CLK-63.5mm] and [CLK+38.1mm].

CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.0508mm.

CONTROL signals should be matched within [CLK-2.54mm] to [CLK+0.0mm] of CLK pairs.

A/BA/CMD signals should be matched within [CLK-12.7mm] to [CLK+12.7mm] of CLK pairs A/BA/CMD signals to each other should match within 5.08mm.

DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.

Maximum length of any signal from die pad to SODIMM pad is 119.83mm, from procesor ball to SODIMM pad is 88.9mm.

SOURCE: Huron River Platform DG, Rev 1.01 (#436735), Section 2.5

SYNC MASTER=K90T MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
Memory Constraints		DRAWING NUMBER	051-9058
 Apple Inc.		REVISION	6.0.0
		BRANCH	
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		SHEET	79 OF 86

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
LVDS_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_PCH	*	=3x_DIELECTRIC	?	DP_PCH	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_PCH_TX	*	=3x_DIELECTRIC	?	DP_PCH_TX	TOP,BOTTOM	=4x_DIELECTRIC	?
LVDS_PCH_TX	*	=3x_DIELECTRIC	?	LVDS_PCH_TX	TOP,BOTTOM	=4x_DIELECTRIC	?

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_PCH_TX	*	=3x_DIELECTRIC	?	SATA_PCH_TX	TOP,BOTTOM	=4x_DIELECTRIC	?
SATA_PCH_RX	*	=3x_DIELECTRIC	?	SATA_PCH_RX	TOP,BOTTOM	=4x_DIELECTRIC	?
SATA_ICOMP	*	8 MIL	?				

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA3_PCH_TX2TX	*	=4x_DIELECTRIC	?	SATA3_PCH_TX2TX	TOP,BOTTOM	=5x_DIELECTRIC	?
SATA3_PCH_TX2RX	*	=5x_DIELECTRIC	?	SATA3_PCH_TX2RX	TOP,BOTTOM	=6x_DIELECTRIC	?
SATA3_PCH_RX2RX	*	=4x_DIELECTRIC	?	SATA3_PCH_RX2RX	TOP,BOTTOM	=5x_DIELECTRIC	?
SATA3_PCH_RX2TX	*	=5x_DIELECTRIC	?	SATA3_PCH_RX2TX	TOP,BOTTOM	=6x_DIELECTRIC	?
SATA3_PCH_20THER	*	=4x_DIELECTRIC	?	SATA3_PCH_20THER	TOP,BOTTOM	=5x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA3_PCH_TX	*_PCH_TX	*	SATA3_PCH_TX2TX
SATA3_PCH_TX	*_PCH_RX	*	SATA3_PCH_TX2RX
SATA3_PCH_RX	*_PCH_RX	*	SATA3_PCH_RX2RX
SATA3_PCH_RX	*_PCH_TX	*	SATA3_PCH_RX2TX
SATA3_PCH_TX	*	*	SATA3_PCH_20THER
SATA3_PCH_RX	*	*	SATA3_PCH_20THER

SOURCE: 471984_Cheif_River_MS_PD6_1.0 and the spacing rule is adjusted per SI team feedback.

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.8

USB 3.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_PCH_TX2TX	*	=4x_DIELECTRIC	?	USB3_PCH_TX2TX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_PCH_TX2RX	*	=5x_DIELECTRIC	?	USB3_PCH_TX2RX	TOP,BOTTOM	=6x_DIELECTRIC	?
USB3_PCH_RX2RX	*	=4x_DIELECTRIC	?	USB3_PCH_RX2RX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_PCH_RX2TX	*	=5x_DIELECTRIC	?	USB3_PCH_RX2TX	TOP,BOTTOM	=6x_DIELECTRIC	?
USB3_PCH_20THER	*	=4x_DIELECTRIC	?	USB3_PCH_20THER	TOP,BOTTOM	=5x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_PCH_TX	*_PCH_TX	*	USB3_PCH_TX2TX
USB3_PCH_TX	*_PCH_RX	*	USB3_PCH_TX2RX
USB3_PCH_RX	*_PCH_RX	*	USB3_PCH_RX2RX
USB3_PCH_RX	*_PCH_TX	*	USB3_PCH_RX2TX
USB3_PCH_TX	*	*	USB3_PCH_20THER
USB3_PCH_RX	*	*	USB3_PCH_20THER

SOURCE: 471984_Cheif_River_MS_PD6_1.0 and the spacing rule is adjusted per SI team feedback.

Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET_TYPE
LVDS_IG_A_CLK	LVDS_90D	LVDS_PCH_TX	LVDS_IG_A_CLK_P
LVDS_IG_A_CLK	LVDS_90D	LVDS_PCH_TX	LVDS_IG_A_CLK_N
LVDS_IG_A_DATA	LVDS_90D	LVDS_PCH_TX	LVDS_IG_A_DATA_P<2..0>
LVDS_IG_A_DATA	LVDS_90D	LVDS_PCH_TX	LVDS_IG_A_DATA_N<2..0>
LVDS_IG_A_DATA	LVDS_90D	LVDS_PCH_TX	LVDS_IG_A_DATA_P<3>
LVDS_IG_A_DATA	LVDS_90D	LVDS_PCH_TX	LVDS_IG_A_DATA_N<3>
LVDS_IG_B_DATA	LVDS_90D	LVDS_PCH_TX	LVDS_IG_B_DATA_P<3..0>
LVDS_IG_B_DATA	LVDS_90D	LVDS_PCH_TX	LVDS_IG_B_DATA_N<3..0>
LVDS_IG_B_CLK	LVDS_90D	LVDS_PCH_TX	LVDS_IG_B_CLK_P
LVDS_IG_B_CLK	LVDS_90D	LVDS_PCH_TX	LVDS_IG_B_CLK_N
SATA_HDD_R2D	SATA_90D	SATA3_PCH_TX	SATA_HDD_R2D_C_P
SATA_HDD_R2D	SATA_90D	SATA3_PCH_TX	SATA_HDD_R2D_C_N
SATA_HDD_R2D_CONN	SATA_90D	SATA3_PCH_TX	SATA_HDD_R2D_P
SATA_HDD_R2D_CONN	SATA_90D	SATA3_PCH_TX	SATA_HDD_R2D_N
SATA_HDD_D2R	SATA_90D	SATA3_PCH_RX	SATA_HDD_D2R_P
SATA_HDD_D2R	SATA_90D	SATA3_PCH_RX	SATA_HDD_D2R_N
SATA_HDD_D2R_CONN	SATA_90D	SATA3_PCH_RX	SATA_HDD_D2R_C_P
SATA_HDD_D2R_CONN	SATA_90D	SATA3_PCH_RX	SATA_HDD_D2R_C_N
SATA_ODD_R2D	SATA_90D	SATA_PCH_TX	SATA_ODD_R2D_C_P
SATA_ODD_R2D	SATA_90D	SATA_PCH_TX	SATA_ODD_R2D_C_N
SATA_ODD_R2D	SATA_90D	SATA_PCH_TX	SATA_ODD_R2D_P
SATA_ODD_R2D	SATA_90D	SATA_PCH_TX	SATA_ODD_R2D_N
SATA_ODD_D2R	SATA_90D	SATA_PCH_RX	SATA_ODD_D2R_P
SATA_ODD_D2R	SATA_90D	SATA_PCH_RX	SATA_ODD_D2R_N
SATA_HDD_R2D_CONN	SATA_90D	SATA3_PCH_TX	SATA_HDD_R2D_RC_P
SATA_HDD_R2D_CONN	SATA_90D	SATA3_PCH_TX	SATA_HDD_R2D_RC_N
SATA_HDD_D2R_CONN	SATA_90D	SATA3_PCH_RX	SATA_HDD_D2R_RC_P
SATA_HDD_D2R_CONN	SATA_90D	SATA3_PCH_RX	SATA_HDD_D2R_RC_N
PCH_SATA_ICOMP	SATA_90D	SATA_ICOMP	PCH_SATAICOMP
USB_HUB1_UP	USB_85D	USB	USB_HUB_UP_P
USB_HUB1_UP	USB_85D	USB	USB_HUB_UP_N
USB_EXT_A	USB_85D	USB	USB_EXT_A_P
USB_EXT_A	USB_85D	USB	USB_EXT_A_N
USB_EXT_B	USB_85D	USB	USB_EXTB_MUX_P
USB_EXT_B	USB_85D	USB	USB_EXTB_MUX_N
USB_EXT_B	USB_85D	USB	USB_EXTB_MUXED_F_P
USB_EXT_B	USB_85D	USB	USB_EXTB_MUXED_F_N
USB_EXT_B	USB_85D	USB	USB_EXTB_F_P
USB_EXT_B	USB_85D	USB	USB_EXTB_F_N
USB_EXT_B	USB_85D	USB	USB_EXTB_MUXED_P
USB_EXT_B	USB_85D	USB	USB_EXTB_MUXED_N
USB_EXT_D_XHCI	USB_85D	USB	USB_EXTD_XHCI_P
USB_EXT_D_XHCI	USB_85D	USB	USB_EXTD_XHCI_N
USB_EXT_B_EHCI	USB_85D	USB	USB_EXTB_EHCI_P
USB_EXT_B_EHCI	USB_85D	USB	USB_EXTB_EHCI_N
USB_EXT_B_XHCI	USB_85D	USB	USB_EXTB_XHCI_P
USB_EXT_B_XHCI	USB_85D	USB	USB_EXTB_XHCI_N
USB3_EXT_RX	USB_85D	USB3_PCH_RX	USB3_EXT_A_RX_P
USB3_EXT_RX	USB_85D	USB3_PCH_RX	USB3_EXT_A_RX_N
USB3_EXT_TX	USB_85D	USB3_PCH_TX	USB3_EXT_A_TX_P
USB3_EXT_TX	USB_85D	USB3_PCH_TX	USB3_EXT_A_TX_N
USB3_EXT_RX	USB_85D	USB3_PCH_RX	USB3_EXTB_RX_P
USB3_EXT_RX	USB_85D	USB3_PCH_RX	USB3_EXTB_RX_N
USB3_EXT_TX	USB_85D	USB3_PCH_TX	USB3_EXTB_TX_P
USB3_EXT_TX	USB_85D	USB3_PCH_TX	USB3_EXTB_TX_N
USB3_EXT_A_RX	USB_85D	USB3_PCH_RX	USB3_EXT_A_RX_F_P
USB3_EXT_A_RX	USB_85D	USB3_PCH_RX	USB3_EXT_A_RX_F_N
USB3_EXT_A_TX	USB_85D	USB3_PCH_TX	USB3_EXT_A_TX_F_P
USB3_EXT_A_TX	USB_85D	USB3_PCH_TX	USB3_EXT_A_TX_F_N
USB3_EXTB_RX	USB_85D	USB3_PCH_RX	USB3_EXTB_RX_F_P
USB3_EXTB_RX	USB_85D	USB3_PCH_RX	USB3_EXTB_RX_F_N
USB3_EXTB_TX	USB_85D	USB3_PCH_TX	USB3_EXTB_TX_F_P
USB3_EXTB_TX	USB_85D	USB3_PCH_TX	USB3_EXTB_TX_F_N
USB3_EXT_A_TX	USB_85D	USB3_PCH_TX	USB3_EXT_A_TX_C_P
USB3_EXT_A_TX	USB_85D	USB3_PCH_TX	USB3_EXT_A_TX_C_N
USB3_EXTB_TX	USB_85D	USB3_PCH_TX	USB3_EXTB_TX_C_P
USB3_EXTB_TX	USB_85D	USB3_PCH_TX	USB3_EXTB_TX_C_N
USB_EXT_A	USB_85D	USB	USB_SMC_P
USB_EXT_A	USB_85D	USB	USB_SMC_N
USB_EXT_C	USB_85D	USB	USB_EXTC_P
USB_EXT_C	USB_85D	USB	USB_EXTC_N
USB_CAMERA	USB_85D	USB	USB_CAMERA_P
USB_CAMERA	USB_85D	USB	USB_CAMERA_N
USB_CAMERA	USB_85D	USB	USB_CAMERA_CONN_P
USB_CAMERA	USB_85D	USB	USB_CAMERA_CONN_N
USB_BT	USB_85D	USB	USB_BT_P
USB_BT	USB_85D	USB	USB_BT_N
USB_BT	USB_85D	USB	USB_BT_CONN_P
USB_BT	USB_85D	USB	USB_BT_CONN_N
USB_TPAD	USB_85D	USB	USB_TPAD_P
USB_TPAD	USB_85D	USB	USB_TPAD_N
USB_IR	USB_85D	USB	USB_IR_P
USB_IR	USB_85D	USB	USB_IR_N
PCH_USB_RBIAS	PCH_USB_RBIAS		PCH_USB_RBIAS
PCH_DIEFCLK_UNUSED	CLK_PCTE_90D	CLK_PCTE	PCIE_CLK100M_PCH_P
PCH_DIEFCLK_UNUSED	CLK_PCTE_90D	CLK_PCTE	PCIE_CLK100M_PCH_N
PCH_DIEFCLK_UNUSED	CLK_PCTE_90D	CLK_PCTE	PCH_CLK96M_DOT_P
PCH_DIEFCLK_UNUSED	CLK_PCTE_90D	CLK_PCTE	PCH_CLK96M_DOT_N
PCH_DIEFCLK_UNUSED	CLK_PCTE_90D	CLK_PCTE	PCH_CLK100M_SATA_P
PCH_DIEFCLK_UNUSED	CLK_PCTE_90D	CLK_PCTE	PCH_CLK100M_SATA_N
CPU_59S	CLK_PCTE	CLK_PCTE	PCH_CLK143M_REFCLK
LPC_CLK33M	CLK_PCTE	CLK_PCTE	PCH_CLK33M_PCIIN

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SOURCE: Calpella Platform Design Guide for Ibox Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2X_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2X_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibox Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCI-Express Signal Constraints

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_T29_TX2TX	*	=3X_DIELECTRIC	?
PCIE_T29_TX2RX	*	=4X_DIELECTRIC	?
PCIE_T29_RX2RX	*	=3X_DIELECTRIC	?
PCIE_T29_RX2TX	*	=4X_DIELECTRIC	?
PCIE_T29_20THER	*	=3X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_T29_TX2TX	TOP,BOTTOM	=4X_DIELECTRIC	?
PCIE_T29_TX2RX	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_T29_RX2RX	TOP,BOTTOM	=4X_DIELECTRIC	?
PCIE_T29_RX2TX	TOP,BOTTOM	=4X_DIELECTRIC	?
PCIE_T29_20THER	TOP,BOTTOM	=4X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_T29_TX	*_TX	*	PCIE_T29_TX2TX
PCIE_T29_TX	*_RX	*	PCIE_T29_TX2RX
PCIE_T29_RX	*_RX	*	PCIE_T29_RX2RX
PCIE_T29_RX	*_TX	*	PCIE_T29_RX2TX
PCIE_T29_TX	*	*	PCIE_T29_20THER
PCIE_T29_RX	*	*	PCIE_T29_20THER

SOURCE: 471984_Cheif_River_MS_PD6_1.0 and the spacing rule is adjusted per SI team feedback.

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_25M_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2X_DIELECTRIC	?
CLK_25M	*	=5X_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET_TYPE
LPC_AD	LPC_50S	LPC	LPC_AD<3..0>
LPC_FRAME_L	LPC_50S	LPC	LPC_FRAME_L
LPC_RESET_L	LPC_50S	LPC	LPC_RESET_L
LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC_R
LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC
LPC_CLK33M	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS
SMBUS_PCH_CLK	SMB_50S	SMB	SMBUS_PCH_CLK
SMBUS_PCH_DATA	SMB_50S	SMB	SMBUS_PCH_DATA
SMBUS_PCH_0_CLK	SMB_50S	SMB	SML_PCH_0_CLK
SMBUS_PCH_0_DATA	SMB_50S	SMB	SML_PCH_0_DATA
SMBUS_SMC_0_50_SCL	SMB_50S	SMB	SML_PCH_1_CLK
SMBUS_SMC_0_50_SDA	SMB_50S	SMB	SML_PCH_1_DATA
HDA_BIT_CLK	HDA_50S	HDA	HDA_BIT_CLK
HDA_BIT_CLK_R	HDA_50S	HDA	HDA_BIT_CLK_R
HDA_SYNC	HDA_50S	HDA	HDA_SYNC
HDA_SYNC_R	HDA_50S	HDA	HDA_SYNC_R
HDA_RST_L	HDA_50S	HDA	HDA_RST_R_L
HDA_RST_L	HDA_50S	HDA	HDA_RST_L
HDA_SDIN0	HDA_50S	HDA	HDA_SDIN0
HDA_SDI_R	HDA_50S	HDA	AUD_SDI_R
HDA_SDO0T	HDA_50S	HDA	HDA_SDO0T
HDA_SDO0T_R	HDA_50S	HDA	HDA_SDO0T_R
PM_SUS_CLK	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK
SPT_CLK	SPT_50S	SPT	SPT_CLK_R
SPT_CLK	SPT_50S	SPT	SPT_CLK
SPT_M0SI	SPT_50S	SPT	SPT_M0SI_R
SPT_M0SI	SPT_50S	SPT	SPT_M0SI
SPT_MISO	SPT_50S	SPT	SPT_MISO
SPT_CS0	SPT_50S	SPT	SPT_CS0_R_L
SPT_CS0	SPT_50S	SPT	SPT_CS0_L
SPT_MLB_CLK	SPT_50S	SPT	SPT_MLB_CLK
SPT_MLB_CS_L	SPT_50S	SPT	SPT_MLB_CS_L
SPT_MLB_M0SI	SPT_50S	SPT	SPT_MLB_M0SI
SPT_MLB_MISO	SPT_50S	SPT	SPT_MLB_MISO
SPT_SMC_MISO	SPT_50S	SPT	SPT_SMC_MISO
SPT_SMC_M0SI	SPT_50S	SPT	SPT_SMC_M0SI
SPT_SMC_CLK	SPT_50S	SPT	SPT_SMC_CLK
SPT_SMC_CS_L	SPT_50S	SPT	SPT_SMC_CS_L
PCIE_ENET_R2D	PCIE_85D	PCIE_PCH_TX	PCIE_ENET_R2D_P
PCIE_ENET_R2D	PCIE_85D	PCIE_PCH_TX	PCIE_ENET_R2D_N
PCIE_ENET_R2D	PCIE_85D	PCIE_PCH_TX	PCIE_ENET_R2D_C_P
PCIE_ENET_R2D_C_N	PCIE_85D	PCIE_PCH_TX	PCIE_ENET_R2D_C_N
PCIE_ENET_D2R	PCIE_85D	PCIE_PCH_RX	PCIE_ENET_D2R_P
PCIE_ENET_D2R	PCIE_85D	PCIE_PCH_RX	PCIE_ENET_D2R_N
PCIE_ENET_D2R_C_P	PCIE_85D	PCIE_PCH_RX	PCIE_ENET_D2R_C_P
PCIE_ENET_D2R_C_N	PCIE_85D	PCIE_PCH_RX	PCIE_ENET_D2R_C_N
PCIE_AP_R2D_P	PCIE_85D	PCIE_PCH_TX	PCIE_AP_R2D_P
PCIE_AP_R2D_N	PCIE_85D	PCIE_PCH_TX	PCIE_AP_R2D_N
PCIE_AP_R2D_C_P	PCIE_85D	PCIE_PCH_TX	PCIE_AP_R2D_C_P
PCIE_AP_R2D_C_N	PCIE_85D	PCIE_PCH_TX	PCIE_AP_R2D_C_N
PCIE_AP_D2R_P	PCIE_85D	PCIE_PCH_RX	PCIE_AP_D2R_P
PCIE_AP_D2R_N	PCIE_85D	PCIE_PCH_RX	PCIE_AP_D2R_N
PCIE_FW_R2D_P	PCIE_85D	PCIE_PCH_TX	PCIE_FW_R2D_P
PCIE_FW_R2D_N	PCIE_85D	PCIE_PCH_TX	PCIE_FW_R2D_N
PCIE_FW_R2D_C_P	PCIE_85D	PCIE_PCH_TX	PCIE_FW_R2D_C_P
PCIE_FW_R2D_C_N	PCIE_85D	PCIE_PCH_TX	PCIE_FW_R2D_C_N
PCIE_FW_D2R_P	PCIE_85D	PCIE_PCH_RX	PCIE_FW_D2R_P
PCIE_FW_D2R_N	PCIE_85D	PCIE_PCH_RX	PCIE_FW_D2R_N
PCIE_FW_D2R_C_P	PCIE_85D	PCIE_PCH_RX	PCIE_FW_D2R_C_P
PCIE_FW_D2R_C_N	PCIE_85D	PCIE_PCH_RX	PCIE_FW_D2R_C_N
PCIE_AP_D2R_PI_P	PCIE_85D	PCIE_PCH_RX	PCIE_AP_D2R_PI_P
PCIE_AP_D2R_PI_N	PCIE_85D	PCIE_PCH_RX	PCIE_AP_D2R_PI_N
PCIE_AP_R2D_PI_P	PCIE_85D	PCIE_PCH_RX	PCIE_AP_R2D_PI_P
PCIE_AP_R2D_PI_N	PCIE_85D	PCIE_PCH_RX	PCIE_AP_R2D_PI_N
PEG_CLK100M_P	CLK_PCTE_98D	CLK_PCTE	PEG_CLK100M_P
PEG_CLK100M_N	CLK_PCTE_98D	CLK_PCTE	PEG_CLK100M_N
PCIE_CLK100M_ENET_P	CLK_PCTE_98D	CLK_PCTE	PCIE_CLK100M_ENET_P
PCIE_CLK100M_ENET_N	CLK_PCTE_98D	CLK_PCTE	PCIE_CLK100M_ENET_N
PCIE_CLK100M_AP_P	CLK_PCTE_98D	CLK_PCTE	PCIE_CLK100M_AP_P
PCIE_CLK100M_AP_N	CLK_PCTE_98D	CLK_PCTE	PCIE_CLK100M_AP_N
PCIE_CLK100M_FW_P	CLK_PCTE_98D	CLK_PCTE	PCIE_CLK100M_FW_P
PCIE_CLK100M_FW_N	CLK_PCTE_98D	CLK_PCTE	PCIE_CLK100M_FW_N
PCIE_CLK100M_EXCARD_P	CLK_PCTE_98D	CLK_PCTE	PCIE_CLK100M_EXCARD_P
PCIE_CLK100M_EXCARD_N	CLK_PCTE_98D	CLK_PCTE	PCIE_CLK100M_EXCARD_N
PCH_VSS_NCTF<1>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<1>
PCH_VSS_NCTF<2>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<2>
PCH_VSS_NCTF<5>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<5>
TP_PCH_VSS_NCTF<7>	CPU_27B4S	CPU_COMP	TP_PCH_VSS_NCTF<7>
PCH_VSS_NCTF<9>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<9>
PCH_VSS_NCTF<9>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<9>
PCH_VSS_NCTF<11>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<11>
PCH_VSS_NCTF<12>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<12>
PCH_VSS_NCTF<15>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<15>
PCH_VSS_NCTF<17>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<17>
PCH_VSS_NCTF<19>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<19>
PCH_VSS_NCTF<21>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<21>
PCH_VSS_NCTF<22>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<22>
PCH_VSS_NCTF<25>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<25>
PCH_VSS_NCTF<27>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<27>
PCH_VSS_NCTF<29>	CPU_27B4S	CPU_COMP	PCH_VSS_NCTF<29>

Chipset Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET_TYPE
DP_EXT_A_ML	DP_85D	DP_PCH_TX	DP_EXT_A_ML_C_P<3..0>
DP_EXT_A_ML	DP_85D	DP_PCH_TX	DP_EXT_A_ML_C_N<3..0>
DP_EXT_A_ML	DP_85D	DP_PCH_TX	DP_EXT_A_ML_P<3..0>
DP_EXT_A_AUXCH	DP_85D	DP_PCH_TX	DP_EXT_A_AUXCH_C_P
DP_EXT_A_AUXCH	DP_85D	DP_PCH_TX	DP_EXT_A_AUXCH_C_N
DP_EXT_A_AUXCH	DP_85D	DP_PCH_TX	DP_EXT_A_AUXCH_P
DP_EXT_A_AUXCH	DP_85D	DP_PCH_TX	DP_EXT_A_AUXCH_N
PCIE_T29_R2D	PCIE_85D	PCIE_T29_RX	PCIE_T29_R2D_C_P<3..0>
PCIE_T29_R2D	PCIE_85D	PCIE_T29_RX	PCIE_T29_R2D_C_N<3..0>
PCIE_T29_R2D	PCIE_85D	PCIE_T29_RX	PCIE_T29_R2D_P<3..0>
PCIE_T29_R2D	PCIE_85D	PCIE_T29_RX	PCIE_T29_R2D_N<3..0>
PCIE_T29_D2R	PCIE_85D	PCIE_T29_TX	PCIE_T29_D2R_P<3..0>
PCIE_T29_D2R	PCIE_85D	PCIE_T29_TX	PCIE_T29_D2R_N<3..0>
PCIE_T29_D2R	PCIE_85D	PCIE_T29_TX	PCIE_T29_D2R_C_P<3..0>
PCIE_T29_D2R	PCIE_85D	PCIE_T29_TX	PCIE_T29_D2R_C_N<3..0>
PCIE_CLK100M_T29	CLK_PCTE_98D	CLK_PCTE	PCIE_CLK100M_T29_P
PCIE_CLK100M_T29	CLK_PCTE_98D	CLK_PCTE	PCIE_CLK100M_T29_N

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET_TYPE
SYSCLK_CLK32K_RTC	CLK_SLOW_55S	CLK_SLOW	SYSCLK_CLK32K_RTC
SYSCLK_CLK25M_SB	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_SB
SYSCLK_CLK25M_SB_R	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_SB_R
SYSCLK_CLK25M_ENET	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_ENET
SYSCLK_CLK25M_ENET_R	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_ENET_R
SYSCLK_CLK25M_T29	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29
SYSCLK_CLK25M_T29_R	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_R

CAESAR IV (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_3X	*	=3:1_SPACING	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_CR_DATA	*	8MIL	?

CAESAR IV (Ethernet PHY) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	0.6 MM	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FW_110D	*	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

et Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	ENET_50S	ENET_3X	BCM5764_CLK25M_XTALI
	ENET_50S	ENET_3X	BCM5764_CLK25M_XTALO
	ENET_50S	ENET_3X	ENET_RESET_L
	ENET_100D	ENET_MDI	ENET_MDI_P<3..0>
	ENET_100D	ENET_MDI	ENET_MDI_N<3..0>
	ENET_50S	ENET_CR_DATA	ENET_CR_DATA<7..0>
	ENET_50S	ENET_CR_DATA	ENET_CR_CMD
	ENET_50S	ENET_CR_DATA	ENET_CR_CLK
	ENET_50S	ENET_CR_DATA	SDCONN_DATA<7..0>
	ENET_50S	ENET_CR_DATA	SDCONN_CMD
	ENET_50S	ENET_CR_DATA	SDCONN_CLK
	ENET_50S	ENET_CR_DATA	SDCONN_CLK_L

FireWire Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	FW_110D	FW_TP	FW_P0_TPA_P
	FW_110D	FW_TP	FW_P0_TPA_N
	FW_110D	FW_TP	FW_P0_TPB_P
	FW_110D	FW_TP	FW_P0_TPB_N
	FW_110D	FW_TP	FW_P1_TPA_P
	FW_110D	FW_TP	FW_P1_TPA_N
	FW_110D	FW_TP	FW_P1_TPB_P
	FW_110D	FW_TP	FW_P1_TPB_N
Port 2 Not Used			

DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

T29 I2C Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_I2C	*	=2x_DIELECTRIC	?

T29 SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SPI	*	=2x_DIELECTRIC	?

T29/DP Connector Signal Constraints

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29DP	*	=5x_DIELECTRIC	?	T29DP	TOP, BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius's T29 Routing Notes

T29 IC Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
REG1		DP_85D	DP_PCH_TX	DP T29SNK0 ML C P<3..0>	0 33
REG1		DP_85D	DP_PCH_TX	DP T29SNK0 ML C N<3..0>	0 33
REG1	DP T29SNK0_MI	DP_85D	DP_PCH_TX	DP T29SNK0 ML P<3..0>	0 33
REG1	DP T29SNK0_MI	DP_85D	DP_PCH_TX	DP T29SNK0 ML N<3..0>	0 33
REG1		DP_85D	DP_PCH	DP T29SNK0 AUXCH C P	0 33
REG1		DP_85D	DP_PCH	DP T29SNK0 AUXCH C N	0 33
REG1	DP T29SNK0_AUXCH	DP_85D	DP_PCH	DP T29SNK0 AUXCH P	0 33
REG1	DP T29SNK0_AUXCH	DP_85D	DP_PCH	DP T29SNK0 AUXCH N	0 33
REG1		DP_85D	DP_PCH_TX	DP T29SNK1 ML C P<3..0>	0 33
REG1		DP_85D	DP_PCH_TX	DP T29SNK1 ML C N<3..0>	0 33
REG1	DP T29SNK1_MI	DP_85D	DP_PCH_TX	DP T29SNK1 ML P<3..0>	0 33
REG1	DP T29SNK1_MI	DP_85D	DP_PCH_TX	DP T29SNK1 ML N<3..0>	0 33
REG1		DP_85D	DP_PCH	DP T29SNK1 AUXCH C P	0 33
REG1		DP_85D	DP_PCH	DP T29SNK1 AUXCH C N	0 33
REG1	DP T29SNK1_AUXCH	DP_85D	DP_PCH	DP T29SNK1 AUXCH P	0 33
REG1	DP T29SNK1_AUXCH	DP_85D	DP_PCH	DP T29SNK1 AUXCH N	0 33
REG1		DP_85D	DTSPAYPORT	DP T29SRC ML C P<3..0>	0 33
REG1		DP_85D	DTSPAYPORT	DP T29SRC ML C N<3..0>	0 33
REG1		DP_85D	DTSPAYPORT	DP T29SRC AUXCH C P	0 33
REG1		DP_85D	DTSPAYPORT	DP T29SRC AUXCH C N	0 33
REG1		T29_T2C_55S	T29_T2C	I2C T29_SCL	0 33 48
REG1		T29_T2C_55S	T29_T2C	I2C T29_SDA	0 33 48
REG1	T29_SPT_CLK	T29_SPT_55S	T29_SPT	T29 SPI_CLK	0 33
REG1	T29_SPT_MOST	T29_SPT_55S	T29_SPT	T29 SPI_MOST	0 33
REG1	T29_SPT_MISO	T29_SPT_55S	T29_SPT	T29 SPI_MISO	0 33
REG1	T29_SPT_CS_L	T29_SPT_55S	T29_SPT	T29 SPI_CS_L	0 33
REG1		T29DP_80D	T29DP	T29 R2D C P<3..0>	0 33 75
REG1		T29DP_80D	T29DP	T29 R2D C N<3..0>	0 33 75
REG1		T29DP_100D	T29DP	T29 D2R P<3..0>	0 33 75
REG1		T29DP_100D	T29DP	T29 D2R N<3..0>	0 33 75

Only used on hosts supporting T29 video-in

Net Properties

ELECTRICAL CONSTRAINT SET	NET TYPE			
	PHYSICAL	SPACING		
T29_R2D0	T29DP_80D	T29DP	T29_R2D P<0>	75
T29_R2D0	T29DP_80D	T29DP	T29_R2D N<0>	75
T29_R2D1	T29DP_80D	T29DP	T29_R2D P<1>	75
T29_R2D1	T29DP_80D	T29DP	T29_R2D N<1>	75
T29_R2D0	T29DP_80D	T29DP	T29_R2D C F P<1..0>	75
T29_R2D0	T29DP_80D	T29DP	T29_R2D C F N<1..0>	75
T29_R2R0	T29DP_100D	T29DP	T29_R2R C P<0>	75 76
T29_R2R0	T29DP_100D	T29DP	T29_R2R C N<0>	75 76
T29_R2R1	T29DP_100D	T29DP	T29_R2R C P<1>	75 76
T29_R2R1	T29DP_100D	T29DP	T29_R2R C N<1>	75 76
T29DP_100D	T29DP_100D	T29DP	T29DPA_D2R1_AUXCH_P	75 76
T29DP_100D	T29DP_100D	T29DP	T29DPA_D2R1_AUXCH_N	76
T29DP_80D	T29DP_80D	T29DP	DP_SDRVA_ML_C_P<3..0>	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVA_ML_C_N<3..0>	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVA_ML_R_P<3..0>	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVA_ML_R_N<3..0>	75
DP_SDRVA_MI_EVEN	T29DP_80D	T29DP	DP_SDRVA_ML_P<2..0:2>	75 83
DP_SDRVA_MI_EVEN	T29DP_80D	T29DP	DP_SDRVA_ML_N<2..0:2>	75 83
DP_SDRVA_MI_ODD	T29DP_80D	T29DP	DP_SDRVA_ML_P<3..1:2>	75
DP_SDRVA_MI_ODD	T29DP_80D	T29DP	DP_SDRVA_ML_N<3..1:2>	75
DP_SDRVA_AUXCH	T29DP_80D	T29DP	DP_SDRVA_AUXCH_P	75
DP_SDRVA_AUXCH	T29DP_80D	T29DP	DP_SDRVA_AUXCH_N	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVA_AUXCH_C_P	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVA_AUXCH_C_N	75
T29DP_80D	T29DP_80D	T29DP	T29DPA_ML_P<3..0>	75 76
T29DP_80D	T29DP_80D	T29DP	T29DPA_ML_N<3..0>	75 76
T29DP_80D	T29DP_80D	T29DP	T29DPA_ML_C_P<3..0>	75 76
T29DP_80D	T29DP_80D	T29DP	T29DPA_ML_C_N<3..0>	75 76
T29DP_80D	T29DP_80D	T29DP	DP_A_EXT_AUXCH_P	75 76
T29DP_80D	T29DP_80D	T29DP	DP_A_EXT_AUXCH_N	75 76
T29_R2D2	T29DP_80D	T29DP	T29_R2D P<2>	75
T29_R2D2	T29DP_80D	T29DP	T29_R2D N<2>	75
T29_R2D3	T29DP_80D	T29DP	T29_R2D P<3>	75
T29_R2D3	T29DP_80D	T29DP	T29_R2D N<3>	75
T29_R2D0	T29DP_80D	T29DP	T29_R2D C F P<3..2>	75
T29_R2D0	T29DP_80D	T29DP	T29_R2D C F N<3..2>	75
T29_R2R2	T29DP_100D	T29DP	T29_R2R C P<2>	75
T29_R2R2	T29DP_100D	T29DP	T29_R2R C N<2>	75
T29_R2R3	T29DP_100D	T29DP	T29_R2R C P<3>	75
T29_R2R3	T29DP_100D	T29DP	T29_R2R C N<3>	75
T29DP_100D	T29DP_100D	T29DP	T29DPB_D2R3_AUXCH_P	75
T29DP_100D	T29DP_100D	T29DP	T29DPB_D2R3_AUXCH_N	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVB_ML_C_P<3..0>	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVB_ML_C_N<3..0>	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVB_ML_R_P<3..0>	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVB_ML_R_N<3..0>	75
DP_SDRVB_MI_EVEN	T29DP_80D	T29DP	DP_SDRVB_ML_P<2..0:2>	83
DP_SDRVB_MI_EVEN	T29DP_80D	T29DP	DP_SDRVB_ML_N<2..0:2>	83
DP_SDRVB_MI_ODD	T29DP_80D	T29DP	DP_SDRVB_ML_P<3..1:2>	75
DP_SDRVB_MI_ODD	T29DP_80D	T29DP	DP_SDRVB_ML_N<3..1:2>	75
DP_SDRVB_AUXCH	T29DP_80D	T29DP	DP_SDRVB_AUXCH_P	75
DP_SDRVB_AUXCH	T29DP_80D	T29DP	DP_SDRVB_AUXCH_N	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVB_AUXCH_C_P	75
T29DP_80D	T29DP_80D	T29DP	DP_SDRVB_AUXCH_C_N	75
T29DP_80D	T29DP_80D	T29DP	T29DPB_ML_P<3..0>	75
T29DP_80D	T29DP_80D	T29DP	T29DPB_ML_N<3..0>	75
T29DP_80D	T29DP_80D	T29DP	T29DPB_ML_C_P<3..0>	75
T29DP_80D	T29DP_80D	T29DP	T29DPB_ML_C_N<3..0>	75
T29DP_80D	T29DP_80D	T29DP	DP_B_EXT_AUXCH_P	75
T29DP_80D	T29DP_80D	T29DP	DP_B_EXT_AUXCH_N	75

Only used on dual-port hosts.

Net Properties

SMBus Charger Net Properties

SYNC MASTER=K907 MLB		SYNC DATE=02/15/2013	
PAGE TITLE			
SMC Constraints			
	Apple Inc.		DRAWING NUMBER 051-9058
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		SHEET 84 OF 86	

8

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K90i Board-Specific Spacing & Physical Constraints

BOARD LAYERS			BOARD AREAS			BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM			NO_TYPE, BGA			MM	16.2

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP, BOTTOM	Y	0.110 MM	0.090 MM			
50_OHM_SE	*	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP, BOTTOM	Y	0.105 MM	0.105 MM			
40_OHM_SE	ISL10	N	0.126 MM	0.126 MM	=STANDARD	=STANDARD	=STANDARD
40_OHM_SE	ISL3, ISL4, ISL9	Y	0.126 MM	0.126 MM	=STANDARD	=STANDARD	=STANDARD
40_OHM_SE	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP, BOTTOM	Y	0.100 MM	0.1 MM			
37_OHM_SE	ISL10	N	0.145 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD
37_OHM_SE	ISL3, ISL4, ISL9	Y	0.145 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD
37_OHM_SE	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP, BOTTOM	Y	0.310 MM	0.2 MM			
27P4_OHM_SE	*	Y	0.235 MM	0.2 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP, BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.070 MM	0.070 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3, ISL4, ISL9	Y	0.140 MM	0.140 MM		0.190 MM	0.190 MM
72_OHM_DIFF	ISL10	N	0.140MM	0.140 MM		0.190 MM	0.190 MM
72_OHM_DIFF	TOP, BOTTOM	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL4	Y	0.101 MM	0.1 MM		0.170 MM	0.170 MM
85_OHM_DIFF	ISL9, ISL10	Y	0.101 MM	0.1 MM		0.170 MM	0.170 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.1 MM		0.190 MM	0.190 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL4	Y	0.091 MM	0.091 MM		0.180 MM	0.180 MM
90_OHM_DIFF	ISL9, ISL10	Y	0.091 MM	0.091 MM		0.180 MM	0.180 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.111 MM	0.111 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3, ISL4	Y	0.076 MM	0.076 MM		0.250 MM	0.250 MM
100_OHM_DIFF	ISL9, ISL10	Y	0.076 MM	0.076 MM		0.250 MM	0.250 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.085 MM	0.085 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3, ISL4	Y	0.068 MM	0.068 MM		0.250 MM	0.250 MM
110_OHM_DIFF	ISL9, ISL10	Y	0.068 MM	0.068 MM		0.250 MM	0.250 MM
110_OHM_DIFF	TOP, BOTTOM	Y	0.081 MM	0.081 MM		0.250 MM	0.250 MM

NOTE: These are Intel recommended impedances for PEG, unused on K90i.

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP, BOTTOM	Y	0.105 MM	0.105 MM			
40_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3, ISL4	Y	0.115 MM	0.115 MM		0.180 MM	0.180 MM
80_OHM_DIFF	ISL9, ISL10	Y	0.115 MM	0.115 MM		0.180 MM	0.180 MM
80_OHM_DIFF	TOP, BOTTOM	Y	0.140 MM	0.140 MM		0.190 MM	0.190 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P3MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P3MM
MEM_CLK	*	BGA	BGA_P2MM
CLK_PCIE	*	BGA	BGA_P2MM
CLK_SLOW	*	BGA	BGA_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.140 MM	?
3X_DIELECTRIC	*	0.210 MM	?
4X_DIELECTRIC	*	0.280 MM	?
5X_DIELECTRIC	*	0.350 MM	?
6X_DIELECTRIC	*	0.420 MM	?
7X_DIELECTRIC	*	0.490 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_DIFF_BGA	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
85_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
85_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 85_DIFF_BGA is 85-ohms differential impedance on outer layers and 80-ohms on inner layers.

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_DIFF_BGA	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
90_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
90_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 90_DIFF_BGA is 90-ohms differential impedance on outer layers and 85-ohms on inner layers.

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.

NOTE: 110_DIFF is 110-ohms differential impedance on outer layers and 105-ohms on inner layers.

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SYNC_MASTER=K90I_MLB

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PAGE TITLE

PCB Rule Definitions



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PAGE

109 OF 109

SHEET

86 OF 86

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